

Associator Field Theory

Consolidated Findings Report

Laws, equations, proof sketches, computations, and applications

Compiled from the Master Document v2 and recovered technical supplement.

Status note. This report consolidates exploratory mathematics.

It separates proof-level derivations, machine-checked evidence, and conjectures. Many results are new to this research program and have not been peer-reviewed. The finite magma certificates are machine-certified by the recovered verifier. The Cayley–Dickson formulas use the exact sign convention fixed in the recovered technical supplement.

April 29, 2026

Contents

1	Executive summary	3
2	Sources, status labels, and notation	4
2.1	Source base	4
2.2	Status labels	4
2.3	Notation	4
3	Finite one-bug magmas and hidden-state curvature	5
3.1	Associator field and semigroup distance	5
3.2	Compact fan family	5
3.3	Recovered small hard examples	5
3.4	Projection Associator Equation	6
4	Cayley–Dickson convention and Boolean curvature	6
5	Rank-Birth Theorem and 16-state automaton	6
5.1	Triangular Rank Recurrence	6
5.2	Rank-Birth closed form	7
5.3	Minimal 16-state column-span automaton	7
5.4	Automaton state distribution	7
5.5	Birth-time distribution	8
6	Local potentials and finite sheaf structure	9
6.1	Three-plane potential law	9
6.2	Vertical and graph laws	9
7	Holonomy, Fourier scars, and sensitivity	9
7.1	One-gauge holonomy spectrum	9
7.2	Holonomy–Walsh duality	10
7.3	Terminal-color ANF law	10
7.4	Influence and noise laws	11
8	Atlas theory: exact coherence, compression, and optimality	12
8.1	Exact atlas theorem	12
8.2	Schubert and suspension laws	12
8.3	Birth-CDF interpretation	13
8.4	All-gauge suffix optimality	13
8.4.1	Affine-rank coverage law	13
8.4.2	Proof chain for the affine-rank law	14
9	Redundancy, reliability, and atlas weight enumerators	14
9.1	Random-walk redundancy law	14
9.2	Multivariate atlas weight enumerator	15
9.3	Redundant coverage thresholds	15
10	Breadth versus depth: affine closures and Pareto structure	16
10.1	Affine closure multiplicity law	16
10.2	Consensus and Pareto frontier	17
11	Rigidity and minimal hitting skeletons	17

11.1	Flag-jump fiber law	17
11.2	Suffix Simplex Rigidity conjectures	17
12	Algorithms and diagnostics	18
12.1	Streaming associator evaluation	18
12.2	Spectral chart discovery	18
12.3	Source-separation pipeline	19
13	Applications	19
13.1	Finite geometry	19
13.2	Coding theory	19
13.3	Fault-tolerant algebraic repair	19
13.4	Scientific computing	19
13.5	AI and tool composition	19
14	Corroborating data tables	20
14.1	Cayley–Dickson failure counts	20
14.2	Suffix coverage versus random coverage	20
14.3	Multiplicity counts in exact atlas	20
14.4	Influence check values	21
15	Open problems and next targets	21
16	Final synthesis	21

1 Executive summary

Associator Field Theory studies nonassociativity as a field, not merely as a scalar defect count. The guiding question is:

$$(xy)z \neq x(yz) \quad \text{but why?}$$

A failure may come from a wrong multiplication-table entry, a hidden state that was quotient-collapsed, a cocycle curvature such as the Cayley–Dickson sign twist, or a sheaf-like obstruction where local corrections exist but do not glue globally.

The earlier source documents established one-bug finite magmas with large semigroup repair distance, a projection associator equation for hidden-state curvature, Cayley–Dickson local potentials, holonomy, exact atlas number $\tau_d = d - 2$, and structured suffix compression laws. This continuation adds a large set of new results, the most important of which are:

1. **Rank-Birth Theorem.** The Cayley–Dickson Boolean associator is generated by the moment when the coordinate columns of (u, v, w) first span $\mathbb{F}_2^3$. After birth, the curvature is transported by the parity $u_i + w_i$.
2. **Minimal 16-state automaton.** The full associator Φ_n is computed by a minimal 16-state column-span automaton. Its state distribution gives the Cayley–Dickson bad-triple count exactly.
3. **Birth-CDF compression.** The suffix atlas compression curve is exactly the cumulative distribution function of rank-birth time:

$$F_r = \mathbb{P}(\beta \leq r + 2).$$

4. **All-gauge suffix optimality.** For any r -gauge family,

$$\text{Opt}_{d,r} = 8^{d-r-2} \begin{bmatrix} r+2 \\ 3 \end{bmatrix}_2.$$

Equivalently, no r gauges cover beyond an $(r+2)$ -dimensional coherent birth core. The proof proceeds through affine closure, kernel stratification, and two local shadow inequalities.

5. **Affine-rank law.** First-coverage breadth is controlled by affine rank, not raw chart count:

$$\text{Cov}_d(\Gamma) \leq 8^{d-a(\Gamma)-3} \begin{bmatrix} a(\Gamma) + 3 \\ 3 \end{bmatrix}_2.$$

6. **Atlas reliability and redundancy.** The exact atlas is a redundant 8-state return system. A plane born at m is covered by

$$1 + \text{Binomial}(d - m, 1/8)$$

canonical charts.

7. **Breadth-depth split.** Suffix charts maximize breadth. Affine closures create depth/redundancy without increasing first coverage. This yields a Pareto theory of coverage versus consensus.
8. **Spectral theory.** Holonomy is encoded in diagonal Walsh coefficients of Φ_n ; the strongest Fourier scars are the best one-chart coherence scars.
9. **Sensitivity law.** The total average sensitivity is

$$I_{\text{tot}}(n) = 2n - \frac{94}{21} + O(n2^{-n}),$$

reflecting two post-birth transport channels and mean birth delay $94/21$.

The most compact conceptual summary is:

Cayley–Dickson associator curvature is irreversible rank birth followed by reversible one-bit transport.

The most useful atlas statement is:

Approximate coherence is the problem of covering early rank-birth quantiles.

2 Sources, status labels, and notation

2.1 Source base

This report is built on source artifacts:

- *Associator Field Theory Master Document v2*: the original research notebook through compression and marginal chart laws.
- *Recovered Technical Supplement*: full six- and eight-element multiplication tables, branch-and-bound verifier logic, Cayley–Dickson twist convention, formal atlas lower-bound proof, and precise suffix indexing.
- *finite_magma_verifier_recovered.py*: exact verifier for the finite one-bug examples.

2.2 Status labels

Label	Meaning
Proved in framework	A derivation is given from the fixed Cayley–Dickson convention and finite-geometric arguments. Formal proof-assistant verification may still be pending.
Machine-certified	Checked by exact finite enumeration, branch-and-bound, or automaton dynamic programming.
Theorem candidate	Strongly derived or tested but still needs a polished proof or independent verification.
Conjecture	Supported by computation or structural evidence but not proved.

2.3 Notation

All vector spaces are over $\mathbb{F}_2$. Let

$$V_d = \mathbb{F}_2^d, \quad \text{Gr}(k, V_d) = \{k\text{-planes in } V_d\}.$$

The Gaussian binomial coefficient is written

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \# \text{Gr}(k, \mathbb{F}_2^n).$$

Coordinates are ordered so that e_d is the newest Cayley–Dickson top-bit coordinate. Define suffix gauges

$$\sigma_m = e_{m+1} + e_{m+2} + \cdots + e_d, \quad 3 \leq m \leq d,$$

with $\sigma_d = 0$. The canonical exact atlas is

$$A_d = \{\sigma_3, \sigma_4, \dots, \sigma_d\}.$$

This is the recovered supplement's atlas $\{0, e_d, e_d + e_{d-1}, \dots, e_d + \dots + e_4\}$, reindexed by birth coordinate.

3 Finite one-bug magmas and hidden-state curvature

3.1 Associator field and semigroup distance

For a magma $(A, *)$, define

$$\alpha_*(x, y, z) = e_{(xy)z} - e_{x(yz)}.$$

The index of nonassociativity is

$$ns(A) = \#\{(x, y, z) : (xy)z \neq x(yz)\}.$$

The semigroup distance is the table-entry Hamming distance to the nearest associative operation:

$$sdist(A) = \min\{\# \text{ changed entries needed to make } A \text{ associative}\}.$$

A *one-bug magma* has $ns(A) = 1$.

3.2 Compact fan family

For $m \geq 1$, define

$$F_m = \{0, a, b, c\} \cup \{x_i, u_i : 1 \leq i \leq m\}, \quad |F_m| = 2m + 4.$$

All unspecified products are zero. The nonzero products are

$$\begin{aligned} aa &= b, & ab &= c, \\ ax_i &= u_i, & bx_i &= c, & au_i &= c, & x_i a &= b, & u_i a &= c, & x_i u_j &= c. \end{aligned}$$

Only (a, a, a) fails associativity. A repair can change aa and the au_i 's, using $m + 1$ edits; disjoint witnesses force at least $m + 1$. Thus

$$\boxed{ns(F_m) = 1, \quad sdist(F_m) = m + 1.}$$

For infinite cardinal κ , the same fan construction gives cardinal Hamming distance κ , so $D_\kappa = \kappa$.

3.3 Recovered small hard examples

The recovered supplement gives full tables and exact branch-and-bound logs. The search is complete because every currently failing triple must change at least one of the four dependency positions

$$(i, j), \quad ((ij), k), \quad (j, k), \quad (i, (jk)).$$

Example	Size	Bad triple	Distance	Failed radii	Repairs
Six-element G	6	(a, a, x)	3	0,1,2	21 at radius 3
Eight-element A	8	(x, x, a)	5	0,1,2,3,4	5 at radius 5

The eight-element example has a hidden-state collapse: adding one hidden state h with $ha = c$ and changing $xx := h$ repairs at distance one in the lifted system. Hence internal repair distance and hidden-state completion distance can diverge sharply.

3.4 Projection Associator Equation

Let $(S, \cdot)$ be associative and let $\pi : S \rightarrow A$ be an idempotent projection. Define

$$x \circ y = \pi(x \cdot y).$$

Then the associator of $\circ$ is exactly projection curvature:

$$\alpha_{\circ}(x, y, z) = R_{\pi}(xy, z) - L_{\pi}(x, yz),$$

where

$$R_{\pi}(s, z) = e_{\pi(\pi(s)z)} - e_{\pi(sz)}, \quad L_{\pi}(x, t) = e_{\pi(x\pi(t))} - e_{\pi(xt)}.$$

This equation separates hidden-state curvature from table-entry bugs and cocycle curvature.

4 Cayley–Dickson convention and Boolean curvature

The recovered supplement fixes the pair multiplication convention

$$(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c}).$$

For basis labels $u, v \in \mathbb{F}_2^n$, write

$$e_u e_v = (-1)^{f_n(u, v)} e_{u+v}.$$

With newest top bit last, $u = (u', U)$, $v = (v', V)$, and $\eta(x) = 0$ if $x = 0$, 1 otherwise, the recursive twist is

$$\begin{aligned} f_n((u, 0), (v, 0)) &= f_{n-1}(u, v), \\ f_n((u, 0), (v, 1)) &= f_{n-1}(v, u), \\ f_n((u, 1), (v, 0)) &= f_{n-1}(u, v) + \eta(v), \\ f_n((u, 1), (v, 1)) &= f_{n-1}(v, u) + \eta(v) + 1. \end{aligned}$$

The Boolean associator is

$$\Phi_n(u, v, w) = f_n(u, v) + f_n(u + v, w) + f_n(v, w) + f_n(u, v + w).$$

Then $\Phi_n = 1$ exactly when $(e_u e_v) e_w \neq e_u (e_v e_w)$.

5 Rank-Birth Theorem and 16-state automaton

5.1 Triangular Rank Recurrence

Split newest coordinates:

$$u = (u', U), \quad v = (v', V), \quad w = (w', W).$$

Let $r = \text{rank}(u', v', w')$. Then

$$\Phi_n(u, v, w) = \begin{cases} \Phi_{n-1}(u', v', w') + U + W, & r = 3, \\ \alpha U + \beta V + \gamma W, & r = 2, \quad \alpha u' + \beta v' + \gamma w' = 0, \\ 0, & r \leq 1. \end{cases}$$

In the rank-2 case, $(\alpha, \beta, \gamma) \neq 0$ is the unique relation among u', v', w' .

Proof sketch. Substitute the recovered recursive formula for f_n into $\Phi_n = \delta f_n$ and enumerate the eight top-bit cases. The dependence on u', v', w' collapses to whether they have rank 3, rank 2, or lower. Direct exhaustive checks matched the recurrence through $n = 7$.

5.2 Rank-Birth closed form

Let

$$P = \langle u, v, w \rangle.$$

If $\text{rank}(u, v, w) < 3$, then $\Phi_n(u, v, w) = 0$. If $\text{rank}(u, v, w) = 3$, define

$$\beta(P) = \min\{m : \text{rank}(\pi_m u, \pi_m v, \pi_m w) = 3\}.$$

Then

$$\boxed{\Phi_n(u, v, w) = 1 + \sigma_{\beta(P)}(u) + \sigma_{\beta(P)}(w).}$$

Thus curvature is born once, at the first coordinate where the triple becomes independent, and then transported by suffix parity.

5.3 Minimal 16-state column-span automaton

Let

$$A = \mathbb{F}_2^3, \quad c_i = (u_i, v_i, w_i), \quad \kappa = (1, 0, 1), \quad K_\kappa = \kappa^\perp.$$

The state set is

$$\boxed{\mathcal{S} = \{H : H < A \text{ proper subspace}\} \cup \{\Omega\}.$$

There are 15 proper subspaces of A , plus Ω , so $|\mathcal{S}| = 16$. The initial state is $\{0\}$; the output is 1 only in state Ω . For proper H ,

$$H \xrightarrow{c} \begin{cases} H, & c \in H, \\ H + \langle c \rangle, & c \notin H, \dim(H + \langle c \rangle) < 3, \\ \Omega, & H + \langle c \rangle = A. \end{cases}$$

For Ω ,

$$\Omega \xrightarrow{c} \begin{cases} \Omega, & \kappa \cdot c = 0, \\ K_\kappa, & \kappa \cdot c = 1. \end{cases}$$

The full-rank zero-curvature state is behaviorally identical to the unborn rank-2 state K_κ , giving a collapse from a naive 17 states to 16. Partition refinement confirms the automaton is minimal.

5.4 Automaton state distribution

Let $x = 2^n$. The number of length- n column words ending in each final state type is:

Final state	Number of such states	Words ending in each / total
$\{0\}$	1	1
Line $L \leq A$	7	$x - 1$ each
Plane $H \neq K_\kappa$	6	$(x - 1)(x - 2)$ each
Special plane K_κ	1	$\frac{(x-2)(x^2-10x+30)}{2}$
Curvature-one Ω	1	$\frac{(x-2)(x-4)(x+6)}{2}$

The Ω row is the positive-label failure count:

$$B_n = (2^n - 2)(2^n - 4)(2^{n-1} + 3).$$

For signed basis-unit loops Q_n ,

$$ns(Q_n) = 8B_n.$$

5.5 Birth-time distribution

The probability that a random triple has rank birth exactly at coordinate m is

$$\mathbb{P}(\beta = m) = \frac{7}{2^m} - \frac{42}{4^m} + \frac{56}{8^m} = \frac{168}{8^m} \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2.$$

The mean and variance are

$$\mathbb{E}[\beta] = \frac{94}{21}, \quad \text{Var}(\beta) = \frac{1150}{441}.$$

If $m < n$, the post-birth transport randomizes the final curvature bit:

$$\mathbb{P}(\Phi_n = 1 \mid \beta = m) = \frac{1}{2}.$$

If $m = n$, no transport remains and $\mathbb{P}(\Phi_n = 1 \mid \beta = n) = 1$. Therefore

$$B_{n,m} = \begin{cases} 84 \cdot 8^{n-m} \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2, & 3 \leq m < n, \\ 168 \begin{bmatrix} n-1 \\ 2 \end{bmatrix}_2, & m = n. \end{cases}$$

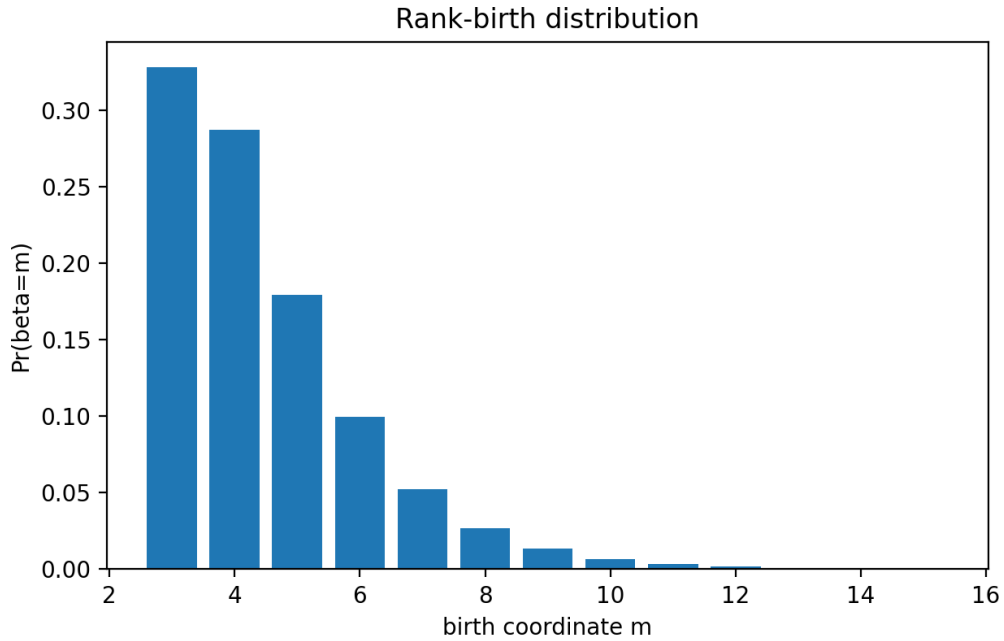


Figure 1: Rank-birth probability mass function. The suffix atlas compression curve is the cumulative distribution of this law.

6 Local potentials and finite sheaf structure

6.1 Three-plane potential law

For every 3-plane $P \leq V_d$, define

$$\theta_P = \sigma_{\beta(P)}|_P.$$

Then

$$\boxed{\Phi_d|_P = \Delta_P(u, v, w)(1 + \theta_P(u) + \theta_P(w)),}$$

where $\Delta_P = 1$ exactly when u, v, w are an ordered basis of P . This makes θ_P the local potential of the associator field.

If $\theta_P = 0$, all 168 ordered bases of P fail; call P octonionic. If $\theta_P \neq 0$, exactly 72 ordered bases fail; call P quasioctonionic.

The plane counts are

$$\boxed{O_d = \begin{bmatrix} d-1 \\ 2 \end{bmatrix}_2 + \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2, \quad Q_d = 7 \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2.}$$

6.2 Vertical and graph laws

If

$$P = R \oplus \langle e_d \rangle, \quad \dim R = 2,$$

then $\beta(P) = d$, hence

$$\boxed{\theta_P = 0.}$$

If $P = \text{graph}(\ell)$ over a lower 3-plane P_0 , then

$$\boxed{\theta_P = \theta_{P_0} + \ell.}$$

These two laws are the local mechanisms behind the exact atlas theorem and all compression laws.

7 Holonomy, Fourier scars, and sensitivity

7.1 One-gauge holonomy spectrum

A global gauge $\lambda \in V_d^*$ covers P if

$$\lambda|_P = \theta_P.$$

Let $N_d(\lambda)$ be the number of covered 3-planes. Scanning $\lambda_d, \lambda_{d-1}, \dots, \lambda_3$, let k be the first zero. Define

$$A_k = \begin{bmatrix} k-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} k-1 \\ 2 \end{bmatrix}_2, \quad B_k = \begin{bmatrix} k-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} k-2 \\ 2 \end{bmatrix}_2.$$

Then

$$N_d(\lambda) = \begin{cases} 8^{d-k} A_k, & \lambda_1 = \dots = \lambda_{k-1} = 0, \\ 8^{d-k} B_k, & (\lambda_1, \dots, \lambda_{k-1}) \neq 0, \\ 0, & \text{if no zero appears among } \lambda_d, \dots, \lambda_3. \end{cases}$$

The holonomy polynomial is

$$H_d(z) = 7 + \sum_{k=3}^d z^{8^{d-k} A_k} + \sum_{k=4}^d (2^{k-1} - 1) z^{8^{d-k} B_k}.$$

The maximum is

$$\max_{\lambda} N_d(\lambda) = 8^{d-3}.$$

For $d \geq 4$, exactly two gauges attain it: σ_3 and σ_4 . There are seven dead gauges with zero coverage.

7.2 Holonomy–Walsh duality

Define

$$\widehat{\Phi}_d(a, b, c) = \sum_{u, v, w \in V_d} (-1)^{\Phi_d(u, v, w) + a(u) + b(v) + c(w)}.$$

Then for every gauge λ ,

$$\widehat{\Phi}_d(\lambda, 0, \lambda) = L_d(\lambda) + 24 \begin{bmatrix} d \\ 3 \end{bmatrix}_2 - 192 N_d(\lambda).$$

Here

$$L_d(0) = 2^{3d} - 168 \begin{bmatrix} d \\ 3 \end{bmatrix}_2,$$

while for $\lambda \neq 0$,

$$L_d(\lambda) = (2^d - 2)(2^d - 4).$$

Thus coherence scars are negative Fourier scars. The dominant spectral coefficient is conjecturally and computationally verified through high dimensions as

$$\max |\widehat{\Phi}_d| = \frac{13}{56} 2^{3d} + 4 \cdot 2^d - \frac{48}{7} \quad (d \geq 4).$$

7.3 Terminal-color ANF law

Let coordinate variables be

$$X_{i,u} = u_i, \quad X_{i,v} = v_i, \quad X_{i,w} = w_i.$$

For a support $S = \{i_1 < \dots < i_k\}$ and a color word $a_1 \dots a_k \in \{u, v, w\}^k$, the monomial

$$\prod_{j=1}^k X_{i_j, a_j}$$

appears in the algebraic normal form of Φ_n if and only if $k \geq 3$ and the final color is determined by the prefix:

$$a_k = \begin{cases} \text{the missing color,} & \text{if the prefix uses exactly two colors,} \\ v, & \text{if the prefix uses all three colors,} \\ \text{none,} & \text{if the prefix is constant.} \end{cases}$$

Therefore

$$\text{CurvSpec}_n(t) = \sum_{k=3}^n (3^{k-1} - 3) \binom{n}{k} t^k.$$

The total ANF monomial count is

$$M_n = \frac{4^n + 8}{3} - 3 \cdot 2^n + 2n.$$

7.4 Influence and noise laws

Exact influence formulas derived from the derivative 16-state automaton are

$$I_u(n) = I_w(n) = n - \frac{141}{49} + \frac{12}{2^n} + \frac{8n - 48}{4^n} + \frac{-\frac{384}{7}n + \frac{4992}{49}}{8^n}$$

and

$$I_v(n) = \frac{188}{147} + \frac{16n - \frac{128}{3}}{4^n} + \frac{-\frac{576}{7}n + \frac{6144}{49}}{8^n}.$$

Thus

$$I_{\text{tot}}(n) = 2n - \frac{94}{21} + \frac{24}{2^n} + \frac{32n - \frac{416}{3}}{4^n} + \frac{-192n + \frac{2304}{7}}{8^n}.$$

The constant 94/21 is the mean birth time. Sensitivity is birth-limited and transport-dominated: u, w are transport channels, while v is mostly a birth channel.

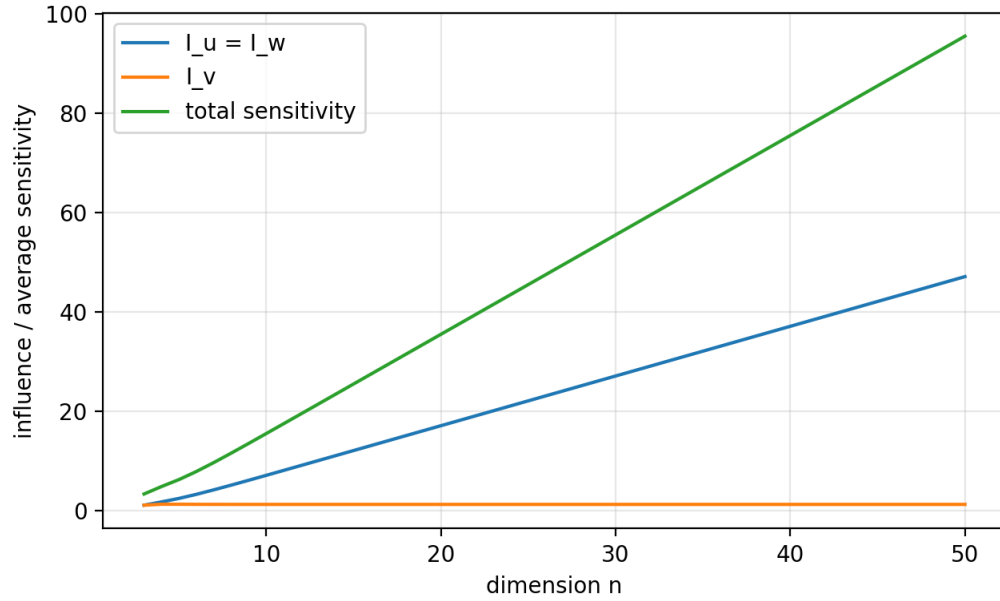


Figure 2: Influence curves from the exact finite-state derivative automaton. The middle variable has bounded total influence, while the outer variables grow linearly.

The paired noise automaton supports the scaling law

$$\text{NS}_n(c/n) \rightarrow \frac{1}{2}(1 - e^{-4c}).$$

8 Atlas theory: exact coherence, compression, and optimality

8.1 Exact atlas theorem

An atlas is a set of gauges such that every local 3-plane potential is covered. The exact atlas number is

$$\tau_d = \min\{|A| : \forall P \in \text{Gr}(3, V_d), \exists \lambda \in A \text{ with } \lambda|_P = \theta_P\}.$$

The exact theorem is

$$\tau_d = d - 2.$$

A canonical atlas is

$$A_d = \{\sigma_3, \sigma_4, \dots, \sigma_d\}.$$

The upper bound follows from vertical/graph recursion. The lower bound uses the top-zero/top-one split: vertical faces require top-zero gauges, top-one gauges saturate graph fibers, and fewer than $d - 2$ gauges cannot cover all lower faces and all eight graph lifts.

8.2 Schubert and suspension laws

The r -suffix atlas is

$$A_{d,r} = \{\sigma_3, \sigma_4, \dots, \sigma_{r+2}\}.$$

Let $\pi_{r+2} : V_d \rightarrow V_{r+2}$. Then

$$P \text{ is covered by } A_{d,r} \iff \dim \pi_{r+2}(P) = 3.$$

Hence suffix coverage is the big Schubert cell of an $(r + 2)$ -dimensional core:

$$\text{Cov}_{d,r} = 8^{d-r-2} \begin{bmatrix} r+2 \\ 3 \end{bmatrix}_2.$$

Equivalently,

$$\text{Cov}_{d,r} = \sum_{m=3}^{r+2} 8^{d-m} \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2.$$

The marginal gain is

$$\text{Cov}_{d,r} - \text{Cov}_{d,r-1} = 8^{d-r-2} \begin{bmatrix} r+1 \\ 2 \end{bmatrix}_2.$$

Asymptotically,

$$F_r = 1 - \left(\frac{7}{2^{r+2}} - \frac{14}{4^{r+2}} + \frac{1}{8^{r+1}} \right).$$

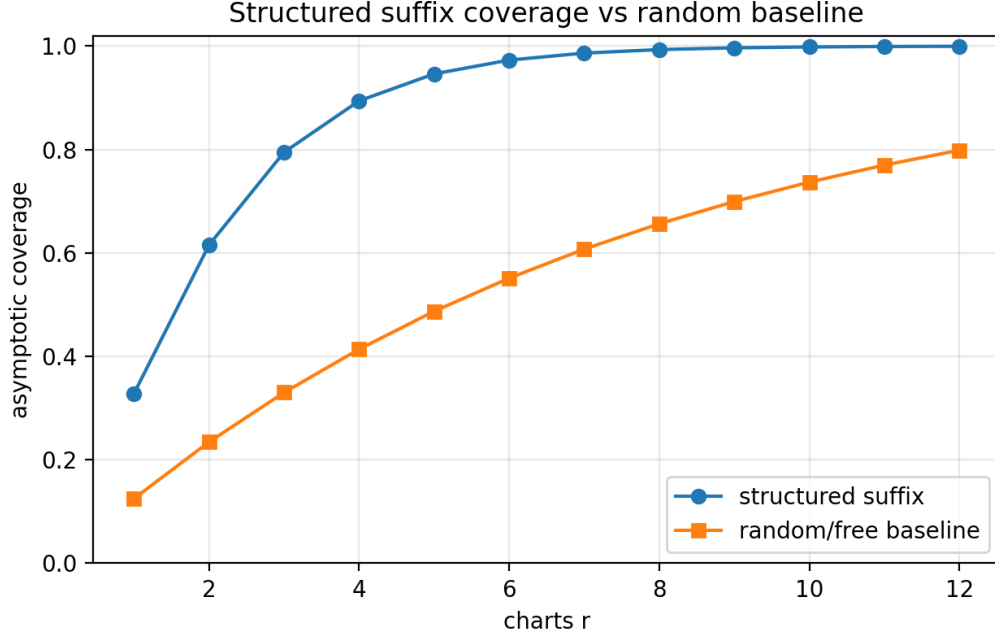


Figure 3: Structured suffix coverage versus random/free chart coverage. Structured charts capture birth layers; random charts sample colors.

8.3 Birth-CDF interpretation

The marginal chart law is the rank-birth mass function:

$$\Delta_r = F_r - F_{r-1} = \mathbb{P}(\beta = r + 2).$$

Therefore

$$F_r = \mathbb{P}(\beta \leq r + 2).$$

An r -chart suffix atlas covers exactly the curvature born by time $r + 2$. This is the clearest causal explanation for logarithmic approximate coherence.

8.4 All-gauge suffix optimality

The central budgeted theorem is:

$$\text{Opt}_{d,r} = \max_{|\Gamma|=r} \text{Cov}_d(\Gamma) = 8^{d-r-2} \begin{bmatrix} r+2 \\ 3 \end{bmatrix}_2.$$

The suffix atlas attains the value.

8.4.1 Affine-rank coverage law

Let

$$a(\Gamma) = \dim \text{Aff}(\Gamma).$$

Then

$$\text{Cov}_d(\Gamma) \leq 8^{d-a(\Gamma)-3} \begin{bmatrix} a(\Gamma) + 3 \\ 3 \end{bmatrix}_2.$$

Since $a(\Gamma) \leq r - 1$, the budgeted theorem follows.

The marginal affine-rank gap is

$$C_d(a) - C_d(a - 1) = 8^{d-a-3} \begin{bmatrix} a + 2 \\ 2 \end{bmatrix}_2.$$

Thus breadth is bought by affine dimension, not by redundant chart count.

8.4.2 Proof chain for the affine-rank law

Let $L = \lambda_0 + M \subset V_d^*$ be an affine a -flat and $K = M^\perp$, $\dim K = d - a$. For a plane P , let $c = \dim(P \cap K)$. The Affine Kernel Criterion says

$$M_L(P) = \begin{cases} 2^{a-3+c}, & (\lambda_0 + \sigma_{\beta(P)})|_{P \cap K} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Stratify by quotient image $S = \pi(P) \leq V_d/K$, $\dim S = 3 - c$. The Relative Kernel Shadow Law bounds each fiber:

$$N_{c,S}(L) \leq 2^{(3-c)^2} \begin{bmatrix} 3 \\ c \end{bmatrix}_2 8^{d-a-3}.$$

The endpoints $c = 0, 3$ are transverse-lift and holonomy statements. The middle entries are the Line Shadow Theorem and Plane Singleton Theorem:

$$N_2^{(1)} \leq 14 \cdot 8^{k-3}, \quad N_1^{(2)} \leq 112 \cdot 8^{k-3}.$$

Summing over quotient subspaces gives

$$\text{Cov}_d(L) \leq 8^{d-a-3} \sum_{c=0}^3 2^{(3-c)^2} \begin{bmatrix} a \\ 3-c \end{bmatrix}_2 \begin{bmatrix} 3 \\ c \end{bmatrix}_2.$$

By q -Vandermonde,

$$\sum_{c=0}^3 2^{(3-c)^2} \begin{bmatrix} a \\ 3-c \end{bmatrix}_2 \begin{bmatrix} 3 \\ c \end{bmatrix}_2 = \begin{bmatrix} a + 3 \\ 3 \end{bmatrix}_2,$$

so

$$\text{Cov}_d(L) \leq 8^{d-a-3} \begin{bmatrix} a + 3 \\ 3 \end{bmatrix}_2.$$

Finally $\text{Cov}_d(\Gamma) \leq \text{Cov}_d(\text{Aff}(\Gamma))$.

9 Redundancy, reliability, and atlas weight enumerators

9.1 Random-walk redundancy law

For the exact canonical atlas A_d , if a 3-plane is born at m , then σ_m covers it deterministically and each later canonical chart independently covers it with probability $1/8$. Thus

$$M_{A_d}(P) \mid \beta(P) = m = 1 + \text{Binomial}(d - m, 1/8).$$

The exact multiplicity count is

$$M_{d,k} = \sum_{m=3}^{d-k+1} \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2 \binom{d-m}{k-1} 7^{d-m-k+1}.$$

The generating polynomial is

$$P_d(y) = y \sum_{m=3}^d \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2 (7+y)^{d-m}.$$

The mean and variance are

$$\mathbb{E}[M_d] = \frac{d}{8} + \frac{37}{84} + O(d^2 2^{-d}),$$

$$\text{Var}(M_d) = \frac{7d}{64} - \frac{3167}{7056} + O(d^2 2^{-d}).$$

A central limit theorem follows from the conditional binomial law.

9.2 Multivariate atlas weight enumerator

Let $C(P)$ be the set of canonical charts covering P . Define

$$\mathcal{W}_d(x_3, \dots, x_d) = \sum_{P \in \text{Gr}(3, V_d)} \prod_{m \in C(P)} x_m.$$

Then

$$\mathcal{W}_d(x_3, \dots, x_d) = \sum_{m=3}^d \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2 x_m \prod_{j=m+1}^d (7+x_j).$$

For nonuniform survival probabilities p_j , the expected number left uncovered is

$$\mathcal{U}_d(\mathbf{p}) = \mathcal{W}_d(1-p_3, \dots, 1-p_d) = \sum_{m=3}^d \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2 (1-p_m) \prod_{j=m+1}^d (8-p_j).$$

9.3 Redundant coverage thresholds

For a canonical subatlas J , let $h_m(J) = \#\{j \in J : j > m\}$. Then

$$\text{Cov}_d^{(\geq s)}(J) = \sum_{m=3}^d 8^{d-m} \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2 \mathbb{P} \left[\mathbf{1}_{m \in J} + \text{Binomial} \left(h_m(J), \frac{1}{8} \right) \geq s \right].$$

Within the canonical atlas, the best r -chart s -redundant design is the first $r-s+1$ birth charts plus any $s-1$ later return charts. The exact optimum is

$$\text{Cov}_{d,r}^{(\geq s)} = 8^{d-r-2} \sum_{h=s-1}^{r-1} \begin{bmatrix} r+1-h \\ 2 \end{bmatrix}_2 \sum_{i=s-1}^h \binom{h}{i} 7^{h-i}.$$

For $s=1$, this is the suffix coverage law. For $s \geq 2$, residuals are return-limited rather than birth-limited.

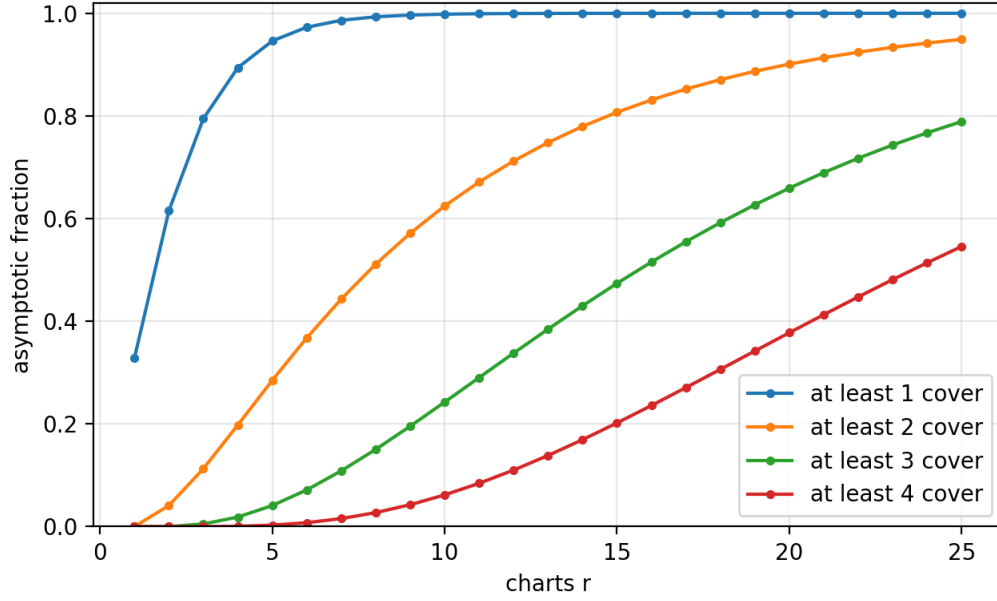


Figure 4: Threshold coverage under optimal canonical redundant designs. First coverage is cheap; robust multi-coverage requires many more charts.

10 Breadth versus depth: affine closures and Pareto structure

10.1 Affine closure multiplicity law

Define the suffix affine closure

$$\Lambda_a = \sigma_{a+3} + \text{span}\{e_4, e_5, \dots, e_{a+3}\}.$$

It has 2^a charts and is the affine closure of the $(a+1)$ -chart suffix block. It adds no new first-covered faces:

$$\text{Cov}_d(\Lambda_a) = 8^{d-a-3} \begin{bmatrix} a+3 \\ 3 \end{bmatrix}_2.$$

It only increases multiplicity. Let

$$b_a(P) = \dim \pi_{\langle e_4, \dots, e_{a+3} \rangle}(P).$$

Then

$$M_{\Lambda_a}(P) = \begin{cases} 2^{a-b_a(P)}, & \beta(P) \leq a+3, \\ 0, & \beta(P) > a+3. \end{cases}$$

The exact multiplicity count is

$$\#\{P : M_{\Lambda_a}(P) = 2^{a-b}\} = 8^{d-a-3} 2^{b^2} \begin{bmatrix} a \\ b \end{bmatrix}_2 \begin{bmatrix} 3 \\ b \end{bmatrix}_2.$$

Thus suffix charts create breadth; affine closure creates depth without adding breadth.

10.2 Consensus and Pareto frontier

For r gauges, full-consensus coverage is optimized by clustering gauges inside the smallest affine cube:

$$\text{Opt}_{d,r}^{(\geq r)} = 8^{d - \lceil \log_2 r \rceil - 3}.$$

For $r = 3$, the exact frontier is

$$\text{Opt}_{d,3}^{(\geq 1)} = 155 \cdot 8^{d-5},$$

$$\text{Opt}_{d,3}^{(\geq 2)} = 3 \cdot 8^{d-4},$$

$$\text{Opt}_{d,3}^{(\geq 3)} = 8^{d-5}.$$

So suffix designs maximize breadth, while affine clusters maximize repeated coverage on a common core.

11 Rigidity and minimal hitting skeletons

11.1 Flag-jump fiber law

Inside Λ_a , define

$$M_a = \text{span}\{e_4, \dots, e_{a+3}\}$$

and suffix path

$$p_t = e_{t+4} + e_{t+5} + \dots + e_{a+3}, \quad 0 \leq t \leq a.$$

For $S \leq M_a$, define prefix rank

$$\rho_S(t) = \dim \pi_t(S)$$

and jump set

$$J(S) = \{0\} \cup \{t \geq 1 : \rho_S(t) > \rho_S(t-1)\}.$$

Solution fibers over S are exactly

$$F_{S,t} = \{\lambda \in \Lambda_a : \lambda|_S = p_t|_S\}, \quad t \in J(S).$$

Thus the suffix basis is a rank-jump hitting skeleton for the coordinate flag.

11.2 Suffix Simplex Rigidity conjectures

Computations through affine rank $a = 5$ support:

Conjecture 11.1 (2D Jump Rigidity). *An affine basis of Λ_a hitting every 2-dimensional jump fiber must be the suffix flag basis.*

Consequently, for $2 \leq r \leq d-2$, the canonical suffix block is conjecturally the unique r -chart all-gauge optimizer. For $r = 1$, the two best gauges σ_3 and σ_4 tie.

The best non-suffix affine basis appears to have core-coverage gap

$$G_a = 2^a(2^{a+1} - 1).$$

This was exact for $a = 2, 3, 4$ and checked on natural candidates for $a = 5$.

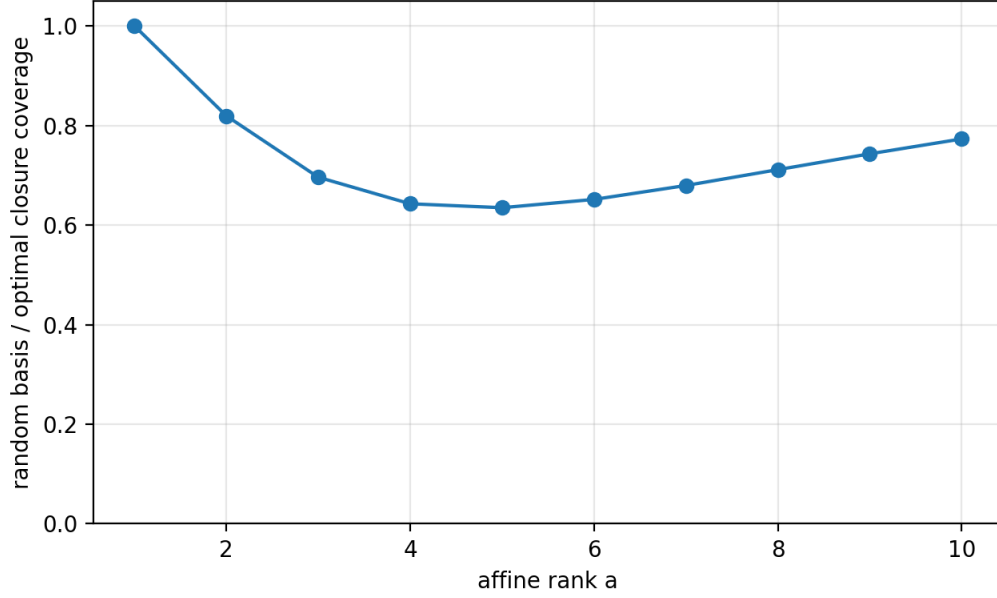


Figure 5: Expected coverage ratio of a random affine basis inside the optimal affine flat. Full affine rank is not enough; the suffix flag basis is special.

12 Algorithms and diagnostics

12.1 Streaming associator evaluation

The 16-state automaton gives an $O(n)$ algorithm:

1. Scan columns $c_i = (u_i, v_i, w_i)$.
2. Before birth, maintain their span in $\mathbb{F}_2^3$.
3. When the span first reaches $\mathbb{F}_2^3$, set curvature state Ω .
4. After birth, update the curvature bit by $u_i + w_i$.

No recursive signs need to be evaluated.

12.2 Spectral chart discovery

The diagonal Walsh formula gives

$$N_d(\lambda) = \frac{L_d(\lambda) + 24 \begin{bmatrix} d \\ 3 \end{bmatrix}_2 - \hat{\Phi}_d(\lambda, 0, \lambda)}{192}.$$

Thus a good one-chart repair can be found by searching for the most negative diagonal Walsh coefficient:

$$\sum_{u,v,w} (-1)^{\Phi(u,v,w) + \lambda(u) + \lambda(w)}.$$

This suggests a black-box algorithm:

sample associator $\rightarrow$ estimate diagonal scars $\rightarrow$ choose chart
 $\rightarrow$ repeat on residual.

12.3 Source-separation pipeline

The overall diagnostic pipeline is:

detect associator field $\rightarrow$ diagnose source $\rightarrow$ fit local potentials
 $\rightarrow$ test gluing $\rightarrow$ find scars $\rightarrow$ build atlas $\rightarrow$ patch residual

A small number of structured charts may repair most coherent curvature, while hidden-state defects require state splitting and table bugs require local edits.

13 Applications

13.1 Finite geometry

The affine-rank theorem is a finite shadow theorem over $\text{Gr}(3, \mathbb{F}_2^d)$: an affine-rank a family of dual gauges cannot cast a shadow larger than a suspended $(a + 3)$ -dimensional core.

13.2 Coding theory

Gauge families are parity-check-like objects. First coverage is controlled by affine rank; redundancy is controlled by affine closure and kernel intersection. The 8-state graph fibers behave like local syndrome alphabets.

13.3 Fault-tolerant algebraic repair

Design rule:

use suffix skeletons for breadth and affine closures for confidence.

A correction atlas can be robust even if individual charts fail, because exact atlases have multiplicity $\sim d/8$ on typical faces.

13.4 Scientific computing

Approximate tensor contractions, reduced-order operators, projection algebras, and floating-point products can fail associativity. The theory suggests separating birth errors from transport errors: cover early birth layers for broad repair, and harden transport channels for robustness.

13.5 AI and tool composition

If tools, memory states, plans, or latent embeddings compose nonassociatively, dense failures may still be generated by a small automaton. Structured correction contexts should target birth states and transport features, not random failing triples.

14 Corroborating data tables

14.1 Cayley–Dickson failure counts

n	Positive bad triples B_n	Signed failures $8B_n$	Density among positive triples
1	0	0	0
2	0	0	0
3	168	1,344	21/64
4	1,848	14,784	231/512
5	15,960	127,680	1995/4096
6	130,200	1,041,600	16275/32768
7	1,046,808	8,374,464	–
8	8,385,048	67,080,384	–

14.2 Suffix coverage versus random coverage

Charts r	Structured suffix coverage F_r	Random/free baseline $1 - (7/8)^r$
1	0.328125	0.125000
2	0.615234	0.234375
3	0.794678	0.330078
4	0.894012	0.413818
5	0.946163	0.487091
6	0.972869	0.551205
7	0.986381	0.607304
8	0.993177	0.656391
9	0.996585	0.699342
10	0.998292	0.736924

14.3 Multiplicity counts in exact atlas

d	Multiplicity counts $M_{d,k}$, $k = 1, 2, \dots$
3	1
4	14, 1
5	133, 21, 1
6	1086, 280, 28, 1
7	8253, 3046, 476, 35, 1
8	60438, 29575, 6378, 721, 42, 1
9	433861, 267463, 74221, 11425, 1015, 49, 1

14.4 Influence check values

n	$I_u = I_w$	I_v	Total sensitivity
8	5.169548	1.280182	11.619278
10	7.134198	1.279023	15.547418
20	17.122460	1.278912	35.523832
30	27.122449	1.278912	55.523810

15 Open problems and next targets

1. **Formal proof assistant route.** Encode the recovered f_n recurrence, verify the top-bit expansion, and derive the 16-state automaton, local potential law, and atlas theorems.
2. **Suffix rigidity.** Prove 2D Jump Rigidity: a minimal basis of the optimal affine flat that hits every 2D jump fiber must be the suffix flag basis.
3. **Second-best basis theorem.** Prove the gap formula $2^a(2^{a+1} - 1)$ for the best non-suffix affine basis.
4. **Cohomological repair distance.** Compute exact $d_{\text{cocyc}}(n)$ for $n \geq 5$.
5. **Hidden-state resistance.** Find one-bug magmas with both large internal semigroup distance and large completion number.
6. **Source-separation algorithms.** Turn the diagnostic pipeline into algorithms for finite operation tables, learned composition systems, and numerical operator products.
7. **Automaton-atlas universality.** Test whether other nonassociative systems generated by small rank-birth automata also have birth-indexed optimal atlases.

16 Final synthesis

The entire continuation can be compressed into the following chain:

Cayley–Dickson twist $\Rightarrow$ rank-birth automaton $\Rightarrow$ local potentials $\Rightarrow$ finite sheaf $\Rightarrow$ optimal suffix atlas.

The key causal picture is:

curvature is born once, then transported by $u + w$.

The key compression law is:

structured atlas coverage is the rank-birth CDF.

The key optimality law is:

first-coverage breadth is an affine-rank invariant.

And the key design law is:

suffix skeleton for breadth; affine closure for depth.

17 Round 2 Corrected Appendix: Identity Spectroscopy and Rank-Detector Laws

Status. Exploratory theorem candidates. The formulas below were derived from the local-potential law in Section 6 of the consolidated report, checked by direct Cayley-Dickson sign enumeration through $n=5$ for the displayed identities, and checked by symbolic associahedron enumeration for all same-word reassociation identities of lengths 4, 5, and 6.

17.1 Correction to the previous Moufang note

The previous informal continuation correctly identified Moufang-type laws as probes of the local potential, but one displayed vector was misstated. For the middle Moufang comparison

$$(xy)(zx) = (x(yz))x$$

the defect on an independent ordered basis x,y,z of a 3-plane P is

$$\begin{aligned} \text{def} &= \text{Phi}(x,y,z) + \text{Phi}(x+y,z,x) \\ &= (1+\text{theta}_P(x)+\text{theta}_P(z)) + (1+\text{theta}_P(x+y)+\text{theta}_P(x)) \\ &= \text{theta}_P(x+y+z). \end{aligned}$$

So this particular Moufang meter reads $\text{theta}_P(x+y+z)$, not $\text{theta}_P(y+z)$. Other quartic identities read $\text{theta}_P(y+z)$, $\text{theta}_P(x)$, $\text{theta}_P(y)$, and every other nonzero coordinate of the local potential. The correction strengthens the result: quartic identities form a complete coordinate system for local transport potential.

17.2 Coordinate-meter laws

Let $\Delta_3(x,y,z)=1$ when x,y,z are independent in $V_n=F_2^n$, and 0 otherwise. On an independent triple spanning P , the local law is $\text{Phi}(x,y,z)=1+\text{theta}_P(x)+\text{theta}_P(z)$. The following identity defects vanish on dependent triples and, on independent triples, read the displayed coordinate of theta_P .

identity comparison	defect on independent triples	name
$((xy)z) = x(yz)$	$1+\text{theta}_P(x)+\text{theta}_P(z)$	associator
$y((xy)z) = (yx)(yz)$	$\text{theta}_P(x)$	X-meter
$x((yx)z) = (xy)(xz)$	$\text{theta}_P(y)$	Y-meter
$(xy)(zy) = ((x(yz))y)$	$\text{theta}_P(z)$	Z-meter
$x(y(xz)) = ((xy)x)z$	$\text{theta}_P(x+y)$	XY-meter
$x(z(xy)) = ((xz)x)y$	$\text{theta}_P(x+z)$	XZ-meter
$((xy)z)y = x(y(zy))$	$\text{theta}_P(y+z)$	YZ-meter
$x((yz)x) = (xy)(zx)$	$\text{theta}_P(x+y+z)$	total meter

Thus a small panel of laws can recover the full local potential. If independence is known, the three coordinate meters X,Y,Z recover theta_P . If independence is not known, add the associator and recover birth as well.

17.3 Observable birth-transport split

Define observable meter bits X,Y,Z by the three coordinate-meter identity defects above. Extending theta_P by zero on non-born triples gives

$$\begin{aligned} X &= \Delta_3 * \text{theta}_P(x), \\ Y &= \Delta_3 * \text{theta}_P(y), \\ Z &= \Delta_3 * \text{theta}_P(z). \end{aligned}$$

The associator separates into a birth bit plus two endpoint transport bits:

$$\begin{aligned}\Phi &= \Delta_3 + X + Z. \\ \text{Equivalently, } \Delta_3 &= \Phi + X + Z.\end{aligned}$$

All addition is mod 2. This is an identity-only rank-birth detector built from four equality tests. It also gives a syndrome code:

$$\begin{aligned}S(x,y,z)=(\Phi,X,Y,Z) \text{ is either } 0000 \\ \text{or } (1+a+c, a, b, c) \text{ for } a,b,c \text{ in } F_2.\end{aligned}$$

The missing patterns are forbidden syndromes. In an observed black-box system, forbidden syndromes indicate that the failures are not explained by the exact Cayley-Dickson local-potential model alone; possible sources include table noise, hidden-state projection, numerical perturbation, or a different cocycle.

17.4 General associator line-integral equation

For any same-word bracketing comparison L in variables x,y,z , move between the two bracketings through the associahedron. Each local rotation $A(BC) \leftrightarrow (AB)C$ contributes $\Phi(A,B,C)$, where A,B,C are the F_2 -sums of the leaves in the three rotated blocks. Therefore

$$\text{Def}_L(T, T_{\text{prime}}) = \sum_{e \in \gamma} \Phi(A_e, B_e, C_e).$$

This is a finite Stokes law for parenthesization: bracketing difference is the line integral of associator curvature. On a 3-plane it collapses further to a two-parameter probe equation

$$\text{Def}_L(x,y,z) = \Delta_3(x,y,z) * (c_L + \theta_P(\ell_L)).$$

Here c_L is a birth parity in F_2 and ℓ_L is a vector in $\text{span}\{x,y,z\}$. This gives four classes of laws: cold $(0,0)$, potential meters $(0,\ell)$, primitive associator probes $(1,\ell)$, and pure birth detectors $(1,0)$.

17.5 Exceptional discovery: a one-test rank detector

The first pure birth detectors appear at word length 5. A compact example is

$$R(x,y,z): x((yx)(zx)) = ((xy)(xz))x.$$

Its defect bit is exactly the rank-3 independence bit:

$$\text{Def}_R(x,y,z) = \Delta_3(x,y,z).$$

So the identity fails precisely when x,y,z are linearly independent, and holds whenever their span has dimension 0, 1, or 2. A proof path contributes

$$\Phi(x, x+y, x+z) + \Phi(y, z, x) + \Phi(x, y, x) + \Phi(x+y, x, z).$$

Dependent triples make all terms zero. On an independent triple, $\Phi(x,y,x)=0$, while the other terms become

$$(1+\theta_P(z)) + (1+\theta_P(x+y)) + (1+\theta_P(x+y+z)) = 1.$$

The transport terms cancel and exactly one birth bit remains. This gives a multiplication-only rank test; no coordinate access or recursive sign evaluation is needed.

17.6 Enumeration evidence

The symbolic enumeration below uses all words over x,y,z , not only words using all three variables. It generates all Catalan bracketings and all unordered bracketing comparisons for each word.

leaves	bracketings	identity comparisons	cold	potential meters	primitive probes	pure birth detectors
4	5	810	576	54	180	0
5	14	22,113	14,337	1,944	5,610	222
6	42	627,669	354,633	82,752	181,740	8,544

For the smaller repeated-variable quartic family with multiset $\{x,x,y,z\}$, the 120 laws split as 42 cold, 18 potential-meter, and 60 primitive-associator laws. No fourth behavior appears at length 4; pure birth detectors begin at length 5.

Global temperatures for the three non-cold channels

$$\begin{aligned} B_n &= (2^{n-2})(2^{n-4})(2^{n-1}+3) \text{ primitive associator,} \\ M_n &= 4(2^{n-1}-1)(2^{n-1}-2)(2^{n-1}-4) \text{ potential meter,} \\ I_n &= (2^{n-1})(2^{n-2})(2^{n-4}) \text{ pure birth detector,} \\ I_n &= B_n + M_n. \end{aligned}$$

The pure birth detector has temperature $I_n/2^{3n}$, which tends to 1. This is signal-hot, not chaotic-hot: almost every high-dimensional triple is independent, and the failure set has a very short description.

17.7 Applications

Black-box finite geometry. The rank-detector identity recovers the rank-3 independence relation from multiplication alone. For nonzero distinct basis units, dependence is collinearity in $PG(n-1,2)$, so

$$\text{Line}(a,b,c) \text{ iff } a,b,c \text{ are distinct nonzero basis units and } \text{Def}_R(a,b,c)=0.$$

Defect classification. The syndrome (Φ, X, Y, Z) distinguishes no-birth triples, born octonionic triples, quasioctonionic transport, and forbidden out-of-model noise. This turns hot/cold testing into source separation.

Error-correcting algebra. Coordinate meters act like parity checks for local potentials, while the suffix atlas acts like a structured repair code for coherent curvature. A repair system should fix coherent heat with gauges and reserve state splitting or table edits for forbidden syndromes.

Parenthesis-safe compilers and numerical kernels. In nonassociative arithmetic, tensor contractions, floating-point-like reductions, reduced-order models, and hypercomplex signal processing, the line-integral equation turns parenthesization choice into a coldest-path problem on the associahedron.

AI/tool-composition diagnostics. Tool calls, memory updates, retrieval states, and latent

ial meters

find stable transport/context gauges; forbidden syndromes point to brittle state collapse or data repair needs.

17.8 Round summary

1. The Moufang vector in the previous informal note is corrected: the middle Moufang law reads $\theta_P(x+y+z)$.
2. Quartic identities give a complete set of coordinate meters for θ_P .
3. The observable split $\Phi = \Delta_3 + X + Z$ makes rank birth measurable from identity tests.
4. The general probe equation $\text{Def}_L = \Delta_3(c_L + \theta_P(\ell_L))$ classifies same-word reassociation laws by what they measure.
5. The exceptional new law $x((yx)(zx)) = ((xy)(xz))x$ fails exactly on rank-3 independent triples.
6. The practical theory becomes: identities are instruments. Some measure birth, some measure transport, some verify coherence, and some expose out-of-model defects.

Round 3 appendix: Associator curl, minimal spectrometers, and syndrome applications

Status: theorem candidates inside the fixed Cayley-Dickson convention; exact symbolic enumeration of reassociation signatures through word length 6; exact Cayley-Dickson sign-recursion tests over all triples through $n = 6$. All additions are mod 2 unless stated otherwise.

18 Round 3: from identity instruments to minimal spectrometers

Round 2 showed that same-word reassociation identities have probe signatures. This round turns that observation into an instrument-design theorem. The main discovery is that rank birth is the cyclic curl of the primitive associator, and that a provably minimal four-test panel recovers the complete local state of a triple.

18.1 Cyclic Curl Birth Law

Let

$$\begin{aligned} A_{xyz} &= \text{Def}(((xy)z) = x(yz)) = \text{Phi}(x, y, z) \\ A_{yzx} &= \text{Def}(((yz)x) = y(zx)) = \text{Phi}(y, z, x) \\ A_{zxy} &= \text{Def}(((zx)y) = z(xy)) = \text{Phi}(z, x, y). \end{aligned}$$

Extend $\Delta_3(x, y, z)$ by $\Delta_3 = 1$ exactly when x, y, z are linearly independent over F_2 , and 0 otherwise. Then the cyclic curl equation is

$$\Delta_3(x, y, z) = A_{xyz} + A_{yzx} + A_{zxy}.$$

Proof sketch. On dependent triples all three associator bits vanish by rank birth. On an independent triple in the 3-plane P , the local potential law gives

$$\begin{aligned} A_{xyz} &= 1 + \theta_P(x) + \theta_P(z), \\ A_{yzx} &= 1 + \theta_P(y) + \theta_P(x), \\ A_{zxy} &= 1 + \theta_P(z) + \theta_P(y). \end{aligned}$$

Adding the three equations cancels every transport term twice and leaves one birth bit. Thus the primitive associator is not merely a hot law: its cyclic curl is an exact rank-birth detector.

The paired curls are transport gradients:

$$\begin{aligned} A_{xyz} + A_{yzx} &= \Delta_3 * \theta_P(y+z), \\ A_{yzx} + A_{zxy} &= \Delta_3 * \theta_P(x+z), \\ A_{zxy} + A_{xyz} &= \Delta_3 * \theta_P(x+y). \end{aligned}$$

This gives a discrete calculus: cyclic curl measures birth, while pairwise differences measure transported potential along edges of the local F_2 triangle.

18.2 The 13-leaf Minimal Complete Spectrometer

Use the three primitive associators above and one quartic coordinate meter

$$X = \text{Def}(y((xy)z) = (yx)(yz)) = \Delta_3 * \theta_P(x).$$

Then define D , Θ_x , Θ_y , Θ_z by

$$\begin{aligned} D &= A_{xyz} + A_{yzx} + A_{zxy}, \\ \Theta_x &= X, \\ \Theta_y &= A_{yzx} + D + X, \\ \Theta_z &= A_{xyz} + D + X. \end{aligned}$$

The output is exactly

$$(D, \Theta_x, \Theta_y, \Theta_z) = (\Delta_3, \Delta_3 * \theta_P(x), \Delta_3 * \theta_P(y), \Delta_3 * \theta_P(z)).$$

Thus four equality tests recover the whole local state: no birth, or birth plus all three coordinates of the local potential. The associated signature matrix, with columns (birth, x , y , z), is

	birth	x	y	z
A_xyz	1	1	0	1
A_yzx	1	1	1	0
A_zxy	1	0	1	1
X-meter	0	1	0	0

This matrix has rank 4 over F_2 . The panel is complete: it separates all eight born potential states and the one unborn state.

18.3 Optimality of the four-test panel

There are nine observable local states: one unborn state and eight born states indexed by θ_P . Three binary tests can distinguish at most eight outcomes, so any complete local spectrometer needs at least four equality tests.

No identity of word length below 3 gives a nonzero associator probe. Word length 3 gives only the three primitive cyclic associator signatures

$(1, x+z), (1, x+y), (1, y+z),$

whose span has dimension 3. A rank-4 complete panel therefore needs at least one word-length-4 probe. Consequently, under the cost model "total variable occurrences across equality tests",

minimum complete spectrometer cost $\geq 3 + 3 + 3 + 4 = 13$.

The panel in Section 18.2 attains this lower bound. Hence it is optimal in this cost model.

Exceptional point. The one-test rank detector from Round 2 is still valuable: it compiles birth into a single length-5 identity. But for a diagnostic panel, three length-3 associators are cheaper and expose more structure. The length-5 law is a compiled curl; the 13-leaf panel is the minimal full spectrometer.

18.4 Probe signatures as a linear code

Every same-word reassociation identity L in variables x,y,z has a signature

$s_L = (c_L, ell_L)$ in $F_2 + F_2^3$

and returns the bit

$b_L = \Delta_3 * (c_L + \theta_P(ell_L))$.

A panel of laws is therefore a binary measurement code. If S is the matrix of row signatures and $q = (1, \theta_x, \theta_y, \theta_z)^T$, then

observed syndrome $b = \Delta_3 * S q$.

A panel is complete exactly when $\text{rank}(S) = 4$. After the inverse transform of Section 18.2, the valid standardized syndromes are

0000, and 1abc for a,b,c in F_2 .

Therefore any observed pattern with $D = 0$ but $(\theta_x, \theta_y, \theta_z)$ nonzero is a forbidden syndrome. Forbidden syndromes are useful because they separate exact Cayley-Dickson curvature from table noise, hidden-state projection, numerical perturbation, or a different cocycle.

18.5 Enumeration of reassociation signatures

A symbolic associahedron enumeration was run for all words over x,y,z , all Catalan bracketings, and all unordered bracketing comparisons. Edge labels used the local rotation signature $(1, A+C)$ when the three rotated block sums A,B,C are independent, and $(0,0)$ otherwise. Path consistency held in all cases tested.

word length	comparisons	cold	potential meters	primitive probes	pure birth	signature conclusion
3	27	21	0	6	0	only three cyclic associator signatures; span rank 3
4	810	576	54	180	0	all non-birth signatures appear; span rank 4
5	22,113	14,337	1,944	5,610	222	all signatures appear; first one-test birth detectors
6	627,669	354,633	82,752	181,740	8,544	many redundant realizations of all signatures

This table refines the Round 2 search. The important distinction is between one-law availability and panel availability. Length 4 still has no single pure-birth detector, but length-4 meters combined with length-3 associators already span the whole signature space.

18.6 Exact Cayley-Dickson verification and state distribution

The spectrometer equations were checked using the recovered recursive Cayley-Dickson sign convention. For every ordered triple in V_n through $n = 6$, the program verified

```
D = rank_3(x,y,z),
X = Delta_3*theta_P(x),
Theta_y = Delta_3*theta_P(y),
Theta_z = Delta_3*theta_P(z).
```

The same computation reveals an exact syndrome distribution. Let $[n \text{ choose } k]_2$ be the Gaussian binomial coefficient. The standardized spectrometer states have counts

```
N_unborn = 2^(3n) - (2^(n-1))(2^(n-2))(2^(n-4)),
N_000     = 168 * ( [n-1 choose 2]_2 + [n-1 choose 3]_2 ),
N_s       = 168 * [n-1 choose 3]_2 for each nonzero s in F_2^3.
```

Here N_{000} means born with theta coordinates 000 in the sampled basis. Each nonzero potential coordinate state has the same count. The theta = 000 state has an extra octonionic-plane contribution.

n	total triples	unborn	born theta=000	each nonzero theta	all checks
3	512	344	168	0	pass
4	4,096	1,576	1,344	168	pass
5	32,768	6,728	8,400	2,520	pass
6	262,144	27,784	52,080	26,040	pass
7	2,097,152	112,904	343,728	234,360	formula
8	16,777,216	455,176	2,432,304	1,984,248	formula

Conditioned on birth, the eight potential coordinate states become asymptotically uniform. Thus high-dimensional curvature is not random noise; it is a nearly uniform three-bit potential signal with a small and exactly described octonionic bias.

18.7 Applications of the spectrometer law

Black-box finite geometry. The cyclic curl gives rank-3 independence from associator tests alone. For nonzero basis units, dependence is projective collinearity over F_2 , so the spectrometer can reconstruct the line relation of $PG(n-1,2)$ without coordinate labels.

Algebraic anomaly detection. Run the four-test panel on sampled triples. Transform raw outputs to $(D, \text{Theta}_x, \text{Theta}_y, \text{Theta}_z)$. The residual

```
rho_forbid = Pr[ D = 0 and (Theta_x, Theta_y, Theta_z) != 000 ]
```

is zero for exact Cayley-Dickson basis curvature. A nonzero value is a direct certificate of out-of-model behavior: table corruption, hidden-state collapse, floating-point perturbation, or a different multiplication cocycle.

Repair-code design. A complete probe panel is a parity-check matrix for local curvature. Choose short probes for fast diagnosis; choose redundant probes for fault tolerance. The existing suffix atlas can then repair coherent transport heat, while forbidden residuals are routed to state splitting or table edits.

Parenthesis-safe computation. In nonassociative numerical kernels, the cyclic curl separates intrinsic birth from gauge-like transport. Parenthesization optimizers should avoid rank-birth triples when possible and use meter data to choose coldest transport gauges when birth cannot be avoided.

AI and tool-composition diagnostics. Treat tools, retrieval calls, memory updates, or latent transforms as

transport, and forbidden hidden-state failures. This suggests a lightweight audit for brittle tool chains.

18.8 Round 3 summary

1. Cyclic curl law: $\Delta_3 = \Phi(x,y,z) + \Phi(y,z,x) + \Phi(z,x,y)$.
2. Pairwise associator differences are transport gradients: they measure θ_P of pair sums.
3. A four-test, 13-leaf panel recovers the full local state $(\Delta_3, \Delta_3 \theta_x, \Delta_3 \theta_y, \Delta_3 \theta_z)$.
4. The 13-leaf panel is optimal among same-word reassociation equality tests under total variable-occurrence cost.
5. Probe signatures form a binary measurement code; completeness is rank 4.
6. Forbidden syndromes give a practical out-of-model detector.
7. Exact verification through $n = 6$ and closed-form counts through all n support the new spectrometer distribution law.

Round 4 Appendix

Defect cohomology, Bianchi closure, and the rank-ladder law

Status note. This appendix continues the Round 3 identity-spectrometer framework. The new results below are proof-level derivations inside the fixed Cayley-Dickson sign convention, supported by exact finite checks and randomized tests. They are exploratory and have not been peer reviewed.

1. Exceptional discovery: associator curvature has a cold Bianchi law

Let $\Phi(a,b,c)$ be the Boolean associator defect for basis labels in $V_n = F_2^n$. Define the group-cohomology coboundary of any 3-cochain F by

$$(\delta F)(a,b,c,d) = F(b,c,d) + F(a+b,c,d) + F(a,b+c,d) + F(a,b,c+d) + F(a,b,c).$$

For the Cayley-Dickson basis twist, $\Phi = \delta f$ for the binary sign cochain f . Therefore $\delta^2 f = 0$ gives the exact pentagon/Bianchi identity

$$\delta \Phi = 0.$$

Interpretation. The associator can be hot, but its second-order boundary is cold. Nonassociativity may be present on many triples, yet every four-variable pentagon closes exactly. This is the first source-separation test: if an observed associator field has nonzero $\delta \Phi$, it is not merely curvature from a binary sign twist over the given label group.

2. Rank detectors as cochains

For $r \geq 1$, define $\Delta_r(x_1, \dots, x_r) = 1$ exactly when the r labels are linearly independent over F_2 , and 0 otherwise. Applying the same coboundary to Δ_r gives an unexpectedly simple parity law:

$$\delta \Delta_r = (r \bmod 2) \Delta_{r+1}.$$

Thus odd-rank detectors give the next rank detector, while even-rank detectors are closed cocycles:

$$\delta \Delta_1 = \Delta_2, \delta \Delta_2 = 0, \delta \Delta_3 = \Delta_4, \delta \Delta_4 = 0.$$

Proof sketch. If $x_1, \dots, x_{r+1}$ are independent, all $r+2$ coboundary terms are independent, so the parity is $r+2$, i.e. $r \bmod 2$. If the span has rank below r , no term is independent. If the span has rank exactly r , let $a_1, \dots, a_{r+1}$ be the unique relation. The drop terms are independent when the endpoint coefficient is 1; the merge term at i is independent when $a_i \neq a_{i+1}$. The parity is $a_1 + \sum_i (a_i + a_{i+1}) + a_{r+1} = 0$.

3. Cyclic Bianchi rank law

Round 3 found that the cyclic curl of the primitive associator detects rank birth:

$$\text{cyc } \Phi(a,b,c) = \Phi(a,b,c) + \Phi(b,c,a) + \Phi(c,a,b) = \Delta_3(a,b,c).$$

Combining this with the rank-ladder law produces a new four-variable detector:

$$\delta(\text{cyc } \Phi) = \delta \Delta_3 = \Delta_4.$$

This is exceptional because a four-dimensional independence test is assembled from local three-plane associator measurements. It is a finite-geometric analogue of measuring higher-dimensional structure by taking the boundary of a lower-dimensional curl.

4. Explicit tetrahedral detector

For four labels a,b,c,d, the rank-4 detector is the five-window boundary of the rank-3 detector:

$$\text{Delta}_4(a,b,c,d) = \text{Delta}_3(b,c,d) + \text{Delta}_3(a+b,c,d) + \text{Delta}_3(a,b+c,d) + \text{Delta}_3(a,b,c+d) + \text{Delta}_3(a,b,c).$$

Substituting $\text{Delta}_3 = \text{cyc } \Phi$ gives 15 primitive associator probes. The five probes in the original orientation sum to $\text{delta } \Phi$, which is zero by the Bianchi law. Therefore the detector compiles to a 10-term tetrahedral curl. Let

$$P = \{(b,c,d), (a+b,c,d), (a,b+c,d), (a,b,c+d), (a,b,c)\}.$$

Then

$$\text{Delta}_4(a,b,c,d) = \sum_{\{(A,B,C) \text{ in } P\}} [\Phi(B,C,A) + \Phi(C,A,B)].$$

Operational form. If $R_3(x,y,z)$ is the Round 2/3 one-test rank-3 identity defect, then

$$\text{Delta}_4(a,b,c,d) = R_3(b,c,d) + R_3(a+b,c,d) + R_3(a,b+c,d) + R_3(a,b,c+d) + R_3(a,b,c).$$

So rank 4 can be detected with five rank-3 identity probes, or with ten primitive associator probes after Bianchi cancellation. This is a practical black-box dimension test whenever the XOR label operation can be formed, for example by multiplying basis units and forgetting sign.

5. The defect cohomology diagnostic

The emerging theory is no longer just a list of identities. It is a cochain complex of defects:

$$\text{binary sign twist } f \rightarrow \text{associator } \Phi = \text{delta } f \rightarrow \text{Bianchi residual } \text{delta } \Phi.$$

For Cayley-Dickson curvature:

$$\text{delta } \Phi = 0, \text{ cyc } \Phi = \text{Delta}_3, \text{ delta}(\text{cyc } \Phi) = \text{Delta}_4.$$

This separates three different failure sources:

- **Coherent sign-twist curvature:** Φ may be hot, but $\text{delta } \Phi$ is cold.
- **Wrong local rank geometry:** $\text{delta } \Phi$ may still vanish, but $\text{cyc } \Phi$ or $\text{delta}(\text{cyc } \Phi)$ disagrees with Delta_3 or Delta_4 .
- **Out-of-model composition:** $\text{delta } \Phi$ is nonzero, indicating hidden state, label corruption, projection curvature, or nonstationarity not representable by a fixed binary sign twist.

6. Experimental checks

The equations above have short proofs, but I also ran direct finite checks using the recovered Cayley-Dickson sign recursion and GF(2) rank enumeration.

6.1 Bianchi closure checks

n	mode	quadruples checked	nonzero delta Phi
1	exhaustive	16	0
2	exhaustive	256	0
3	exhaustive	4,096	0
4	exhaustive	65,536	0
5	random	20,000	0
6	random	20,000	0
7	random	20,000	0
8	random	20,000	0

6.2 Rank-ladder checks

r	ambient n	mode	tests	mismatches
1	5	exhaustive	1,024	0
2	5	exhaustive	32,768	0
3	4	exhaustive	65,536	0
4	6	random	50,000	0
5	7	random	50,000	0

6.3 Tetrahedral detector checks

n	mode	tests	10-term mismatches	15-term mismatches
4	exhaustive	65,536	0	0
5	random	20,000	0	0
6	random	20,000	0	0
7	random	20,000	0	0
8	random	20,000	0	0

The 15-term form is $\delta(\text{cyc } \Phi)$. The 10-term form is the Bianchi-cancelled tetrahedral curl.

7. Rank-detector temperatures

The rank detectors have their own temperature scale. For random ordered labels in V_n :

$$\Pr[\Delta_r=1] = \prod_{i=0}^{r-1} (1 - 2^i / 2^n).$$

Thus rank detectors become almost always hot in high dimension because random small tuples are almost surely independent. This is not a defect of the detector: it means the ambient geometry is high enough that dependence is rare.

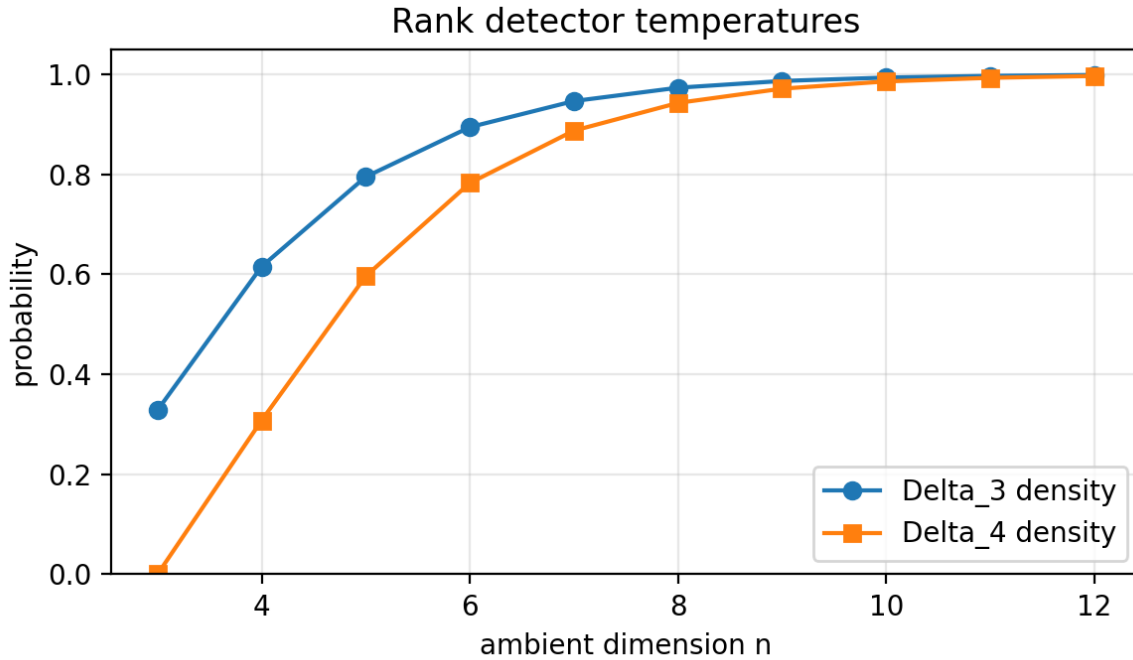


Figure 6. Δ_3 and Δ_4 detector densities by ambient dimension. Δ_3 matches the rank-birth CDF already used by the suffix atlas; Δ_4 is the next-dimensional gluing detector.

n	$\Pr[\Delta_3=1]$	$\Pr[\Delta_4=1]$
3	0.328125	0.000000
4	0.615234	0.307617
5	0.794678	0.596008
6	0.894012	0.782261
7	0.946163	0.887028
8	0.972869	0.942467
10	0.993177	0.985418
12	0.998292	0.996342

8. Applications of the new knowledge

The Bianchi and rank-ladder laws add a second-order layer to the earlier atlas/spectrometer theory.

8.1 Black-box finite geometry tomography

The Round 3 spectrometer recovers rank-3 birth and local potential on a plane. The new formula $\Delta(\text{cyc } \Phi) = \Delta_4$ lets overlapping plane measurements detect whether four unknown directions span a 4-space. Repeating this gives a way to reconstruct a projective/matroid skeleton from multiplication and equality tests rather than coordinates.

8.2 Algebraic anomaly detection

For a coherent sign-twist algebra, $\Delta \Phi$ must vanish. A nonzero Bianchi residual is therefore a sharp alarm: the observed composition is not explainable by any fixed binary sign table over the assumed label group. This can reveal hidden-state projection, changing context, table-index corruption, or noisy nonstationary behavior.

8.3 Error correction and missing associator data

Every pentagon parity equation lets one missing associator bit be inferred from four others under the coherent-cocycle hypothesis. This is analogous to a parity-check code for associator observations. The rank-ladder then tells which cyclic-curl residuals should be hot because the four directions are genuinely independent.

8.4 Parenthesis-safe numerical and compiler systems

For reductions, tensor contractions, approximate operator products, or compiler rewrites, local associativity tests do not identify whether failures compose coherently. Pentagon loops do. If $\Delta \Phi$ is zero, a rewrite system may be correctable by a gauge-like chart. If $\Delta \Phi$ is nonzero, the problem is path-dependent and needs state repair, not just a better parenthesization.

8.5 AI and tool-composition diagnostics

Treat tools, memory updates, retrieval calls, or latent transforms as operations. Triple tests measure local

Bianchi residual says that the result depends on the whole path through tool space, not merely on local pairwise composition.

9. New theory refinement: defect cohomology

The useful new object is not just the temperature of a law, but the coboundary profile of its defect field. A law defect is a cochain; applying delta reveals whether its failures are coherent, whether they have higher-rank boundary, and whether they can be repaired by a lower-dimensional gauge.

Defect cohomology law: $\delta \Phi = 0$, $\text{cyc } \Phi = \Delta_3$, $\delta(\text{cyc } \Phi) = \Delta_4$, $\delta \Delta_r = (r \bmod 2) \Delta_{r+1}$.

In words: Cayley-Dickson associator curvature is closed, but its cyclic curl has a boundary, and that boundary is exactly rank-4 independence.

This turns the earlier identity-spectrometer into a multi-scale instrument:

- Level 1: primitive associator probes measure local parenthesis failure.
- Level 2: cyclic curls detect rank-3 birth.
- Level 3: Bianchi boundaries of cyclic curls detect rank-4 gluing.
- Level 4: even-rank detectors are cold parity checks; odd-rank detectors generate the next rank.

10. Open targets for the next round

- Find or prove the shortest possible primitive-associator expression for Δ_4 . The current natural Bianchi-cancelled expression has 10 terms.
- Compile the five-window Δ_4 parity into a single product identity, analogous to the Round 2 one-test rank-3 identity.
- Generalize identity probes for Δ_5 , Δ_6 , and higher rank, using the rank-ladder law as the target equation.
- Build a statistical Bianchi test for approximate systems where associator bits are noisy probabilities rather than exact Boolean values.
- Apply the cohomology residual to one-bug magmas after imposing or learning a group-like skeleton, to separate table bugs from hidden-state curvature.

Round 4 summary

The main discovery is the Bianchi/rank-ladder layer. The associator field is a closed 3-cochain; its cyclic curl is a rank-3 detector; the coboundary of that cyclic curl is exactly a rank-4 detector. This supplies a new family of cold laws, hot rank tests, and practical diagnostics for coherent versus out-of-model nonassociativity.

Round 5 Appendix: Local Holonomy Quantization

Cold-face spectra, four-plane thermal phases, and applications

Status: theorem candidates with proof sketches and finite verification.

R5.1 Question resumed from Round 4

Round 4 produced the closed-curvature law

$$\delta\Phi = 0,$$

and the rank ladder

$$\text{cyc } \Phi = \Delta_3, \quad \delta(\text{cyc } \Phi) = \Delta_4.$$

The next question is not only whether an individual three-plane is octonionic or quasioctonionic, but how these local phases assemble inside a four-dimensional neighborhood. A four-plane has exactly

$$\begin{bmatrix} 4 \\ 3 \end{bmatrix}_2 = 15$$

three-dimensional faces. If cold and hot behavior were independent face-by-face, many face counts should occur. They do not.

R5.2 Definitions: cold faces and local gauge coverage

Let $W \leq V_d$ be a four-plane. For a three-face $P \leq W$, let θ_P be the local potential from the three-plane potential law,

$$\Phi|_P = \Delta_P(1 + \theta_P(u) + \theta_P(w)).$$

Call P a *cold face* of W when $\theta_P = 0$. Define

$$C(W) = \#\{P \leq W : \dim P = 3, \theta_P = 0\}.$$

Equivalently, $C(W)$ is the number of octonionic three-faces of W .

More generally, each local gauge $\lambda \in W^*$ covers a face P when

$$\lambda|_P = \theta_P.$$

Define the local holonomy coverage number

$$h_W(\lambda) = \#\{P \leq W : \dim P = 3, \lambda|_P = \theta_P\}.$$

Then

$$C(W) = h_W(0).$$

R5.3 Exceptional discovery: universal local holonomy spectrum

Law (Local holonomy quantization). For every four-plane $W \leq V_d$, the multiset of local gauge coverage counts is

$$\text{Spec}_{\lambda \in W^*} h_W(\lambda) = \{8, 8, \underbrace{2, \dots, 2}_{7 \text{ times}}, \underbrace{0, \dots, 0}_{7 \text{ times}}\}.$$

Thus every four-dimensional packet has exactly two best local gauges, seven weak local gauges, and seven dead local gauges.

This is the local version of the one-gauge holonomy spectrum. The global report showed that one gauge can cover at most an 8-face shadow and that the best gauges are special. Round 5 localizes this: every four-plane internally carries the same $8^2 2^7 0^7$ spectrum.

The cold-face count is the special case $\lambda = 0$. Therefore zero can be a best, weak, or dead local gauge, but nothing else:

$$C(W) \in \{0, 2, 8\}.$$

This is the first new result of this round. It says that cold three-plane laws cluster in four-dimensional packets. They are not independent Bernoulli defects.

R5.4 Canonical 16-lift code

The spectrum above appears through a simple graph-lift rule. Let $W_0 \leq V_d$ be a four-plane and lift it to V_{d+1} as a graph

$$W_0^\ell = \{x + \ell(x)e_{d+1} : x \in W_0\}, \quad \ell \in W_0^*.$$

For every face $P \leq W_0$, the graph law gives

$$\theta_{P^\ell} = \theta_P + \ell|_P.$$

Hence a lifted face is cold exactly when the lifting form ℓ equals the old local potential on that face:

$$\theta_{P^\ell} = 0 \iff \ell|_P = \theta_P.$$

Therefore

$$C(W_0^\ell) = h_{W_0}(\ell).$$

The universal spectrum implies the *16-lift transition law*

$$\boxed{\{C(W_0^\ell) : \ell \in W_0^*\} = \{8, 8, \underbrace{2, \dots, 2}_7, \underbrace{0, \dots, 0}_7\}.$$

In words: every lower four-plane has sixteen graph lifts; exactly two lifts have eight cold faces, seven lifts have two cold faces, and seven lifts have none. The transition is memory-erasing: the distribution of lift phases does not depend on the old value of $C(W_0)$.

In an adapted basis this code has the canonical form below. Let ν be the distinguished rank-four birth covector and let $H \leq W^*$ be the seven-point hyperplane complementary to the ν -direction. Then the lift gauge ℓ falls into three classes:

$$\begin{aligned} \ell = 0 \text{ or } \nu &\implies C = 8, \\ 0 \neq \ell \in H &\implies C = 2, \\ \ell \notin H, \ell \neq \nu &\implies C = 0. \end{aligned}$$

This table is useful because it turns local repair into a finite 16-word code.

R5.5 Global cold-face phase distribution

Let

$$N_j(d) = \#\{W \in \text{Gr}(4, V_d) : C(W) = j\}.$$

Then the exact phase counts are

$$\boxed{N_0(d) = N_2(d) = 7 \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2,}$$

$$\boxed{N_8(d) = \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2 + 2 \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2.}$$

These sum to

$$\begin{bmatrix} d \\ 4 \end{bmatrix}_2 = 16 \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2 + \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2,$$

which is the usual terminal-or-graph split for four-planes.

Proof sketch. A four-plane in V_d is either terminal, meaning rank four is born at coordinate d , or it is a graph lift of a four-plane in V_{d-1} . The terminal planes are counted by $\begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2$. In a terminal four-plane, seven faces contain the new kernel line and are born at coordinate d , hence have $\theta = 0$; among the remaining eight graph faces, exactly one matches the inherited local potential. Thus every terminal four-plane has $C = 8$.

Every nonterminal four-plane is one of the sixteen graph lifts of a lower four-plane. The 16-lift code gives seven lifts with $C = 0$, seven with $C = 2$, and two with $C = 8$. Since there are $\begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2$ lower four-planes, the formulas follow.

R5.6 Four-plane heat quantization

Inside a four-plane W , the ordered independent triples are partitioned by the fifteen three-faces. Each octonionic face contributes 168 associator failures and 0 Moufang failures. Each quasioctonionic face contributes 72 associator failures and 96 Moufang failures. Hence, with $C = C(W)$,

$$A(W) = 168C + 72(15 - C) = 1080 + 96C,$$

$$M(W) = 96(15 - C) = 1440 - 96C.$$

Since a four-plane has

$$(16 - 1)(16 - 2)(16 - 4) = 2520$$

ordered independent triples, the local associator and Moufang temperatures are

$$\tau_A(W) = \frac{A(W)}{2520} = \frac{45 + 4C(W)}{105},$$

$$\tau_M(W) = \frac{M(W)}{2520} = \frac{60 - 4C(W)}{105}.$$

Thus only three local thermal phases are possible:

$C(W)$	cold-face phase	$\tau_A(W)$	$\tau_M(W)$
0	dead-zero phase	3/7	4/7
2	dipole phase	53/105	52/105
8	octet phase	11/15	4/15

Associator and Moufang heat remain complementary:

$$\tau_A(W) + \tau_M(W) = 1.$$

As $d \rightarrow \infty$, the phase probabilities converge to

$$\Pr(C = 0), \Pr(C = 2), \Pr(C = 8) \rightarrow \frac{7}{16}, \frac{7}{16}, \frac{1}{8}.$$

The mean number of cold faces tends to

$$\mathbb{E}[C] \rightarrow \frac{15}{8},$$

but the variance tends to

$$\text{Var}(C) \rightarrow \frac{399}{64},$$

far larger than the binomial baseline $15(1/8)(7/8) = 105/64$. Cold faces therefore appear in clusters, not as independent noise.

R5.7 Experimental checks

The following table gives exact exhaustive counts for all four-planes through $d = 7$, together with the closed formulas.

d	$\# \text{Gr}(4, V_d)$	N_0	N_2	N_8
4	1	0	0	1
5	31	7	7	17
6	651	217	217	217
7	11811	4557	4557	2697

Formula extension gives:

d	$\# \text{Gr}(4, V_d)$	N_0	N_2	N_8
8	200787	82677	82677	35433
9	3309747	1405509	1405509	498729
10	53743987	23168229	23168229	7407529

Additional verification:

- The local holonomy spectrum $8^2 2^7 0^7$ was exhaustively checked for all four-planes in V_4, V_5, V_6 .
- The 16-lift transition law was exhaustively checked for every four-plane in V_4, V_5, V_6 lifted to V_5, V_6, V_7 .
- Random spot checks in dimensions 8, 10, 12 found no cold-face count outside $\{0, 2, 8\}$.

R5.8 Applications

1. Local anomaly detection. In a Cayley-Dickson-like black-box system, sample a four-dimensional packet and classify its fifteen three-faces as octonionic or quasioctonionic. A coherent rank-birth sign-twist model permits only

$$C \in \{0, 2, 8\}.$$

Any observed count such as 1, 3, 4, 5, 6, 7, 9, $\dots$, 15 is a sharp out-of-model signal: table corruption, hidden-state projection, nonstationary composition, or a non-Cayley-Dickson cocycle.

2. Error-correcting face code. The fifteen face labels form a local syndrome word. The valid syndrome words are not arbitrary; locally they arise from the 16-word holonomy code. This gives parity-check-like constraints for repairing measured associator data. If one face is misclassified, the weight usually jumps outside $\{0, 2, 8\}$, making the error detectable. The two best gauges provide a short candidate list for correction.

3. Black-box geometry reconstruction. Round 3 showed how identities detect rank-three birth, and Round 4 showed how cyclic boundaries detect rank-four birth. Round 5 adds a four-plane phase fingerprint. This helps reconstruct projective neighborhoods: a true four-plane must carry the local spectrum $8^2 2^7 0^7$, so fake or merged neighborhoods can be rejected.

4. Parenthesis-safe computation. A four-operation computation can be routed through its three-face packets. In an octet phase ($C = 8$), many subcontexts have cold potential and are good parenthesization anchors. In a dead-zero phase ($C = 0$), no zero-gauge face is safe, so a nonzero local gauge or state refinement is required.

5. AI and tool-composition diagnostics. If retrieval calls, tools, memory updates, or latent transformations compose nonassociatively, four-step packets can be tested by their fifteen triple subpackets. Independent random failures should not produce the 0, 2, 8 spectrum with the asymptotic $7/16, 7/16, 1/8$ phase frequencies. Seeing this spectrum would indicate coherent transport-like nonassociativity; violating it would point to hidden context leakage or unstable state.

R5.9 New theory direction: thermal sheaf phases

Round 5 upgrades the local-potential theory from individual three-planes to four-plane neighborhoods. The emerging object is a *thermal sheaf phase*: a local packet of faces, potentials, and gauge-coverage numbers.

The most useful new equations are

$$\text{Spec}_{\lambda \in W^*} h_W(\lambda) = 8^2 2^7 0^7,$$

$$C(W) = h_W(0) \in \{0, 2, 8\},$$

$$\tau_A(W) = \frac{45 + 4C(W)}{105}, \quad \tau_M(W) = \frac{60 - 4C(W)}{105}.$$

The guiding interpretation is:

three-planes carry potentials; four-planes carry phase spectra.

This suggests a general theory of hot/cold laws where local neighborhoods have quantized thermal fingerprints rather than independent law failures.

R5.10 Next targets

1. Prove the local holonomy spectrum in a fully coordinate-free way from the three-plane potential law and the Bianchi identity.
2. Classify the actual fifteen-bit cold-face syndrome words, not just their weights.
3. Extend from four-plane neighborhoods to five-plane neighborhoods and determine whether higher thermal phase spectra are also quantized.
4. Build a practical decoder: given noisy face classifications on a four-plane, return the nearest valid holonomy syndrome and the two best local gauges.
5. Test whether one-bug finite magmas with hidden-state curvature exhibit analogous local phase spectra around their unique defect.

Round 6 Appendix: Universal Packet Spectra

Status. The main result below is a theorem candidate with a short induction proof from the graph law. It was also checked by exact enumeration for all five-planes in dimensions 5, 6, 7, all six-planes in dimension 7, and by random checks in higher dimensions. It extends the Round 5 four-plane phase law from a special $k = 4$ packet to arbitrary packet dimension $k \geq 3$.

R6.1 Local holonomy is homogeneous

Let $U \leq V_d$ be a k -plane, $k \geq 3$. For each local gauge $\lambda \in U^*$ define

$$h_U(\lambda) = \#\{P \in \text{Gr}(3, U) : \lambda|_P = \theta_P\}.$$

Thus $h_U(\lambda)$ counts how many three-faces of the packet U are repaired by the gauge λ .

Universal Local Holonomy Spectrum. For every k -plane $U \leq V_d$,

$$\boxed{\text{Spec}_{\lambda \in U^*} h_U(\lambda) = \mathcal{H}_k,}$$

where the multiset $\mathcal{H}_k$ is exactly the one-gauge holonomy spectrum of the full model in dimension k :

$$\boxed{\mathcal{H}_k(z) = 7 + \sum_{m=3}^k z^{8^{k-m} A_m} + \sum_{m=4}^k (2^{m-1} - 1) z^{8^{k-m} B_m}.$$

Here

$$A_m = \begin{bmatrix} m-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} m-1 \\ 2 \end{bmatrix}_2, \quad B_m = \begin{bmatrix} m-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} m-2 \\ 2 \end{bmatrix}_2.$$

The 7 term means seven gauges have coverage zero.

Interpretation. The Round 5 law

$$8^2 2^7 0^7$$

was the $k = 4$ instance. Round 6 shows this was not an isolated four-plane accident. Every k -dimensional packet carries the same internal gauge spectrum as the full Cayley–Dickson space of dimension k , regardless of where the packet sits in V_d .

R6.2 First spectra

The first packet spectra are:

k	$\# \text{Gr}(3, V_k)$	$\mathcal{H}_k$ as a multiset
3	1	$1^1 0^7$
4	15	$8^2 2^7 0^7$
5	155	$64^2 50^1 22^{15} 16^7 0^7$
6	1395	$512^2 400^1 310^1 190^{31} 176^{15} 128^7 0^7$
7	11811	$4096^2 3200^1 2480^1 2046^1 1550^{63} 1520^{31} 1408^{15} 1024^7 0^7$

A notation such as 64^2 means two gauges cover 64 three-faces.

Exceptional point. The spectra are not binomial and not random. They are sparse, quantized, and recursive. For a five-packet, the zero gauge can make exactly

$$0, 16, 22, 50, 64$$

three-faces cold, and no other number. For a six-packet the possible cold counts are

$$0, 128, 176, 190, 310, 400, 512.$$

This gives a packet-level fingerprint of coherent Cayley–Dickson curvature.

R6.3 Proof mechanism: the gauge wheel

Let $U_0 \leq V_{d-1}$ be a k -plane and let $\ell \in U_0^*$ define a graph lift

$$U_\ell = \{u + \ell(u)e_d : u \in U_0\} \leq V_d.$$

The graph law for local potentials says that every three-face potential is translated by the same local functional:

$$\theta_P^{U_\ell} = \theta_{P_0}^{U_0} + \ell|_{P_0}.$$

Consequently the zero-gauge cold count of the lift is

$$\boxed{C_k(U_\ell) = h_{U_0}(\ell),}$$

where

$$C_k(U) = h_U(0) = \#\{P \in \text{Gr}(3, U) : \theta_P = 0\}.$$

As the 2^k graph lifts vary, the zero gauge walks through the entire local holonomy spectrum once. This is the **gauge-wheel transition**. Terminal k -planes add one fixed phase,

$$\boxed{A_k = \begin{bmatrix} k-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} k-1 \\ 2 \end{bmatrix}_2.}$$

Thus a phase in dimension d is either inherited by a graph-wheel rotation from dimension $d-1$, or born terminally at phase A_k .

R6.4 Global phase distribution for arbitrary packet dimension

Let

$$N_s^{(k)}(d) = \#\{U \in \text{Gr}(k, V_d) : C_k(U) = s\},$$

and let

$$m_k(s) = \#\{\lambda \in V_k^* : h_{V_k}(\lambda) = s\}$$

be the multiplicity of s in the universal spectrum $\mathcal{H}_k$. Then

$$\boxed{N_s^{(k)}(d) = m_k(s) \begin{bmatrix} d-1 \\ k \end{bmatrix}_2 + \mathbf{1}_{s=A_k} \begin{bmatrix} d-1 \\ k-1 \end{bmatrix}_2.}$$

This formula specializes exactly to the Round 5 four-plane phase law. For $k = 4$, $A_4 = 8$ and $m_4(0), m_4(2), m_4(8) = (7, 7, 2)$, so

$$N_0 = N_2 = 7 \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2, \quad N_8 = 2 \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2 + \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2.$$

Five-packet distribution. For $k = 5$, $A_5 = 50$ and

$$\mathcal{H}_5 = 64^2 50^1 22^{15} 16^7 0^7.$$

Therefore

$$N_0^{(5)}(d) = N_{16}^{(5)}(d) = 7 \begin{bmatrix} d-1 \\ 5 \end{bmatrix}_2,$$

$$N_{22}^{(5)}(d) = 15 \begin{bmatrix} d-1 \\ 5 \end{bmatrix}_2, \quad N_{64}^{(5)}(d) = 2 \begin{bmatrix} d-1 \\ 5 \end{bmatrix}_2,$$

$$N_{50}^{(5)}(d) = \begin{bmatrix} d-1 \\ 4 \end{bmatrix}_2 + \begin{bmatrix} d-1 \\ 5 \end{bmatrix}_2.$$

For $d = 5, 6, 7, 8$ this gives:

d	$\# \text{Gr}(5, V_d)$	N_0	N_{16}	N_{22}	N_{50}	N_{64}
5	1	0	0	0	1	0
6	63	7	7	15	32	2
7	2667	441	441	945	714	126
8	97155	18669	18669	40005	14478	5334

As $d \rightarrow \infty$, the terminal term disappears and the asymptotic probabilities are the normalized spectrum multiplicities:

$$\Pr(C_5 = 0, 16, 22, 50, 64) \rightarrow \left(\frac{7}{32}, \frac{7}{32}, \frac{15}{32}, \frac{1}{32}, \frac{2}{32} \right).$$

R6.5 General packet heat quantization

Let

$$T_k = \# \text{Gr}(3, V_k) = \begin{bmatrix} k \\ 3 \end{bmatrix}_2.$$

A k -packet with $C = C_k(U)$ cold three-faces has associator and Moufang failure counts

$$A_{\text{fail}}(U) = 72T_k + 96C, \quad M_{\text{fail}}(U) = 96(T_k - C).$$

Since $168T_k$ is the number of ordered independent triples in U ,

$$\tau_A(U) = \frac{3}{7} + \frac{4C}{7T_k}, \quad \tau_M(U) = \frac{4}{7} \left(1 - \frac{C}{T_k} \right).$$

Again,

$$\tau_A(U) + \tau_M(U) = 1.$$

For five-packets, $T_5 = 155$ and the allowed heat phases are:

$C_5(U)$	$\tau_A(U)$	$\tau_M(U)$
0	3/7	4/7
16	529/1085	556/1085
22	79/155	76/155
50	19/31	12/31
64	103/155	52/155

R6.6 Experimental checks

Exact enumeration found one universal local spectrum for every tested packet:

k	d	#packets tested	result
5	5	1	$\mathcal{H}_5$ only
5	6	63	$\mathcal{H}_5$ only
5	7	2667	$\mathcal{H}_5$ only
6	7	127	$\mathcal{H}_6$ only

Random checks also matched the universal spectrum:

k	d	#random packets	result
6	8	20	$\mathcal{H}_6$ only
6	9	20	$\mathcal{H}_6$ only
7	8	5	$\mathcal{H}_7$ only

The exhaustive five-packet cold-count distributions were:

d	# $\text{Gr}(5, V_d)$	distribution of C_5
5	1	50 : 1
6	63	0 : 7, 16 : 7, 22 : 15, 50 : 32, 64 : 2
7	2667	0 : 441, 16 : 441, 22 : 945, 50 : 714, 64 : 126

R6.7 Applications

1. Packet fingerprints for black-box systems. In a Cayley–Dickson-like black-box algebra, choose a k -dimensional packet and test all of its three-faces. The number of cold faces must lie in $\text{supp}(\mathcal{H}_k)$. For $k = 5$ the only legal values are 0, 16, 22, 50, 64. For $k = 6$ the only legal values are 0, 128, 176, 190, 310, 400, 512. Any other value is a certificate of out-of-model behavior.

2. Coherence-preserving feature extensions. Adding one new bit or feature corresponds locally to a graph lift. The gauge-wheel transition says that the 2^k possible lifts cycle through the whole spectrum. Thus, instead of adding a feature randomly, one can choose a lift that puts the packet in a high-coherence phase. For a five-packet, the best lifts repair 64 of 155 three-faces while seven lifts repair none.

3. Error-correcting algebra. The spectrum is a parity-check-like code for local curvature. A single mistaken face classification will often move the packet count outside the legal support. Larger k gives a stronger checksum because the legal support remains sparse inside the full interval $[0, T_k]$.

4. Parenthesis-safe computation. The heat formula shows exactly how much local associator heat remains after choosing a gauge. A compiler or numerical kernel can select local gauges/parenthesizations that place computation in a lower-temperature packet phase.

5. AI and tool-composition diagnostics. If tools, memory updates, retrieval calls, or latent transforms compose nonassociatively, then three-way tests define a curvature field. Packet spectra give a multi-step diagnostic: coherent transport-like nonassociativity should produce legal packet counts, while context leakage, hidden-state collapse, or nonstationarity should violate the spectrum.

R6.8 Summary

The Round 5 principle

three-planes carry potentials; four-planes carry phase spectra

now extends to

$$\boxed{\text{every } k\text{-plane carries the universal spectrum } \mathcal{H}_k.}$$

The most useful equations from this round are

$$\boxed{\text{Spec}_{\lambda \in U^*} h_U(\lambda) = \mathcal{H}_k,}$$

$$\boxed{C_k(U_\ell) = h_{U_0}(\ell),}$$

$$\boxed{N_s^{(k)}(d) = m_k(s) \begin{bmatrix} d-1 \\ k \end{bmatrix}_2 + \mathbf{1}_{s=A_k} \begin{bmatrix} d-1 \\ k-1 \end{bmatrix}_2,}$$

$$\boxed{\tau_A = \frac{3}{7} + \frac{4C}{7 \begin{bmatrix} k \\ 3 \end{bmatrix}_2}, \quad \tau_M = \frac{4}{7} \left(1 - \frac{C}{\begin{bmatrix} k \\ 3 \end{bmatrix}_2} \right).}$$

Conceptually, this turns identity spectroscopy into packet spectroscopy: local defects are not only measured on triples and glued on four-planes; they now form a universal hierarchy of quantized k -packet phases.

Round 7 Appendix: Overlap Holonomy and Thermal Equipartition

Pair gluing, frustrated pencils, and moment laws for packet spectroscopy

Status note. This round continues the Round 6 theorem-candidate that every k -packet carries a universal local holonomy spectrum H_k . The new layer asks not only how many faces a gauge covers, but why local repairs fail to glue. The computations below were checked by exact GF(2) enumeration for the displayed dimensions. The formulas have proof sketches, but still deserve independent verification.

R7.1 Question resumed from Round 6

Round 6 converted three-face potentials into a packet spectrum:

$$h_W(\lambda) = \#\{P \in \text{Gr}(3, W) : \lambda|_P = \theta_P\}.$$

The universal spectrum H_k tells how many three-faces each local gauge repairs. The next question is more local:

When two or three local repairs agree pairwise, do they necessarily glue to one gauge?

The answer is no. The smallest obstruction is already visible inside a four-plane: three three-faces can be pairwise compatible on their overlaps and still fail to have a common gauge. This is the first genuine Čech-type holonomy obstruction in the packet theory.

R7.2 Pair-overlap compatibility law

For two three-faces $P, Q \leq W$, define them to be compatible when their local potentials agree on the intersection:

$$\theta_P|_{P \cap Q} = \theta_Q|_{P \cap Q}.$$

Let $s = \dim(P \cap Q)$. The new empirical/theorem-candidate law is that the compatibility fraction depends only on s , not on k or on the ambient position:

$$\boxed{p_0 = 1, \quad p_1 = \frac{101}{155}, \quad p_2 = \frac{3}{5}, \quad p_3 = 1.} \quad (\text{R7.1})$$

Here p_s is the fraction of ordered pairs (P, Q) with $\dim(P \cap Q) = s$ that are compatible.

The ordered number of such pairs in a k -packet is

$$\boxed{T_s(k) = \begin{bmatrix} k \\ 3 \end{bmatrix}_2 \begin{bmatrix} 3 \\ s \end{bmatrix}_2 2^{(3-s)^2} \begin{bmatrix} k-3 \\ 3-s \end{bmatrix}_2.} \quad (\text{R7.2})$$

Thus the number of compatible ordered pairs is $p_s T_s(k)$.

Interpretation. Disjoint faces always glue because they impose no common boundary condition. Identical faces always glue. The nontrivial sheaf friction occurs only for line and plane overlaps. Plane overlaps are compatible 3/5 of the time. Line overlaps are compatible 101/155 of the time.

R7.3 Exceptional discovery: frustrated three-face pencils

Let $L \leq U$ with $\dim L = 2$ and $\dim U = 4$. The three hyperplanes of U that contain L are three three-faces:

$$P_a = L + \langle a \rangle, \quad P_b = L + \langle b \rangle, \quad P_{a+b} = L + \langle a + b \rangle,$$

where a, b are independent modulo L . This triple is called a **three-face pencil**.

If the three face potentials agree on the common core L , then the pencil is pairwise compatible. Pairwise compatibility is not enough. The residual holonomy bit is

$$\boxed{\mathcal{H}_L(a, b) = \theta_{P_a}(a) + \theta_{P_b}(b) + \theta_{P_{a+b}}(a + b).} \quad (\text{R7.3})$$

This bit is independent of the chosen representatives a, b once the restrictions to L agree. A common gauge exists on U if and only if

$$\boxed{\mathcal{H}_L(a, b) = 0.} \quad (\text{R7.4})$$

So the smallest true gluing failure is a triangle: each pair can pass its overlap test, while the three together fail around the quotient projective line $U/L \cong \mathbb{F}_2^2$.

R7.4 Four-plane pencil quantization

Every four-plane contains

$$\left[\begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right]_2 = 35$$

three-face pencils. The new quantized law is

$$\boxed{21 \text{ open pencils,} \quad 7 \text{ flat pairwise-compatible pencils,} \quad 7 \text{ frustrated pairwise-compatible pencils.}} \quad (\text{R7.5})$$

Here:

- **open** means at least one pair already disagrees on the common two-plane;
- **flat** means all three pairwise overlaps agree and $\mathcal{H} = 0$;
- **frustrated** means all three pairwise overlaps agree but $\mathcal{H} = 1$.

Consequently, every k -packet contains

$$\boxed{N_{\text{open}} = 21 \left[\begin{smallmatrix} k \\ 4 \end{smallmatrix} \right]_2, \quad N_{\text{flat}} = 7 \left[\begin{smallmatrix} k \\ 4 \end{smallmatrix} \right]_2, \quad N_{\text{hol}} = 7 \left[\begin{smallmatrix} k \\ 4 \end{smallmatrix} \right]_2.} \quad (\text{R7.6})$$

This is because a pencil is equivalently a pair (L, U) with $L \in \text{Gr}(2, U)$ and $U \in \text{Gr}(4, W)$, and each four-plane contributes the same $21 : 7 : 7$ split.

Exceptional point. Pairwise law-consistency is not a certificate of global repair. In the coherent Cayley-Dickson curvature model, exactly half of the pairwise-compatible pencils are still globally frustrated.

R7.5 Thermal equipartition over local gauges

Let W be any k -packet and choose a local gauge $\lambda \in W^*$ uniformly. Let

$$H = h_W(\lambda), \quad G_k = \left[\begin{smallmatrix} k \\ 3 \end{smallmatrix} \right]_2.$$

The Round 6 universal spectrum implies the exact mean law

$$\boxed{\mathbb{E}_\lambda H = \frac{G_k}{8}.} \quad (\text{R7.7})$$

Thus a random local gauge repairs exactly one eighth of the three-faces on average.

More surprisingly, the second moment has a closed form. With $x = 2^k$,

$$\boxed{\text{Var}_\lambda(H) = \frac{(x-1)(x-2)(x-4)(47x^2 + 174x - 680)}{5713920}.} \quad (\text{R7.8})$$

Equivalently,

$$\boxed{\text{Var} \left(\frac{H}{G_k} \right) = \frac{49(47x^2 + 174x - 680)}{9920(x-1)(x-2)(x-4)} = \frac{2303}{9920} 2^{-k} + O(4^{-k}).} \quad (\text{R7.9})$$

Since packet temperatures are

$$\tau_A = \frac{3}{7} + \frac{4H}{7G_k}, \quad \tau_M = \frac{4}{7} \left(1 - \frac{H}{G_k} \right),$$

we get the exact local-gauge equipartition law

$$\boxed{\mathbb{E}_\lambda \tau_A = \mathbb{E}_\lambda \tau_M = \frac{1}{2}.} \quad (\text{R7.10})$$

The common variance is

$$\boxed{\text{Var}(\tau_A) = \text{Var}(\tau_M) = \frac{47x^2 + 174x - 680}{620(x-1)(x-2)(x-4)} = \frac{47}{620}2^{-k} + O(4^{-k}).} \quad (\text{R7.11})$$

Interpretation. As packet dimension grows, almost every local gauge looks thermally balanced: associator heat and Moufang heat both approach $1/2$. The exceptional best gauges remain as rare scars: exactly two gauges cover 8^{k-3} faces, which is asymptotically a fraction $21/64$ of all three-faces.

R7.6 Proof sketch for the moment law

Write

$$H = \sum_{P \in \text{Gr}(3, W)} I_P(\lambda),$$

where $I_P = 1$ when $\lambda|_P = \theta_P$. For one face,

$$\mathbb{P}(I_P = 1) = 2^{-3} = \frac{1}{8}.$$

For two faces P, Q with $s = \dim(P \cap Q)$, the two restrictions are jointly satisfiable exactly when the pair is compatible. If compatible, they impose $\dim(P + Q) = 6 - s$ independent equations, so

$$\mathbb{P}(I_P = I_Q = 1) = 2^{-(6-s)}.$$

Using the pair counts $T_s(k)$ and compatibility fractions p_s from (R7.1) gives

$$\mathbb{E}H^2 = \sum_{s=0}^3 p_s T_s(k) 2^{-(6-s)},$$

which simplifies to

$$\boxed{\mathbb{E}H^2 = \frac{(x-1)(x-2)(x-4)(155x^3 + 1218x^2 + 10696x - 34560)}{279982080}.} \quad (\text{R7.12})$$

Subtracting $(G_k/8)^2$ gives (R7.8).

R7.7 Experimental checks

The following exact checks used independent GF(2) row-reduced enumeration, suffix-potential computation, compatibility tests on intersections, and linear-system consistency tests.

k	G_k	mean H	variance H	moment formula	comments
3	1	0.125	0.109	yes	full H_3
4	15	1.875	6.234	yes	full H_4
5	155	19.375	241.609	yes	full H_5
6	1395	174.375	8324.859	yes	full H_6
7	11811	1476.375	274909.359	yes	spectrum formula
8	97155	12144.375	8923990.359	yes	spectrum formula
9	788035	98504.375	287517032.359	yes	spectrum formula
10	6347715	793464.375	9231061516.359	yes	spectrum formula

k	overlap s	ordered pairs	compatible pairs	compatibility fraction
3	3	1	1	1
4	2	210	126	3/5
4	3	15	15	1
5	1	17360	11312	101/155
5	2	6510	3906	3/5
5	3	155	155	1
6	0	714240	714240	1
6	1	1093680	712656	101/155
6	2	136710	82026	3/5
6	3	1395	1395	1

k	open pencils	flat pencils	frustrated pencils	predicted split
4	21	7	7	21 : 7 : 7
5	651	217	217	$21 \begin{bmatrix} 5 \\ 4 \end{bmatrix}_2 : 7 \begin{bmatrix} 5 \\ 4 \end{bmatrix}_2 : 7 \begin{bmatrix} 5 \\ 4 \end{bmatrix}_2$
6	13671	4557	4557	$21 \begin{bmatrix} 6 \\ 4 \end{bmatrix}_2 : 7 \begin{bmatrix} 6 \\ 4 \end{bmatrix}_2 : 7 \begin{bmatrix} 6 \\ 4 \end{bmatrix}_2$

R7.8 Applications

Pairwise tests are insufficient. A system can pass every pairwise overlap test in a local triangle and still fail to glue. This is a direct warning for algebraic repair, distributed consistency, and AI tool-composition diagnostics: pairwise agreement does not certify global coherence.

Minimal holonomy test. The pencil bit $\mathcal{H}_L(a, b)$ is a three-face test for hidden obstruction. In a black-box operation system, testing the three faces around a two-dimensional core can reveal path dependence that no pair test sees.

Packet checksums. A coherent Cayley-Dickson-type four-packet must have the 21 : 7 : 7 pencil split. Deviations indicate table corruption, noisy equality, state projection, nonstationary context, or a different cocycle.

Repair planning. Pair-incompatible pencils require boundary repair first. Pairwise-compatible but frustrated pencils require a holonomy-changing repair, not merely local smoothing. Flat pencils can be safely patched by a common gauge.

Statistical baselines for hot/cold laws. The exact mean and variance of H give a null model for local repair coverage. A random local gauge covers 1/8 of faces on average; best gauges cover $\sim 21/64$. Observed systems outside this mean-variance envelope are not coherent rank-birth packets.

AI and tool-composition diagnostics. Treat tools, memory writes, retrieval calls, and latent transforms as composable operations. Pairwise agreement between two routes can hide a triangular loop failure. The pencil holonomy bit is a compact test for context leakage and hidden state drift.

R7.9 Round 7 summary

1. Pair compatibility is quantized by overlap dimension:

$$p_0 = 1, \quad p_1 = 101/155, \quad p_2 = 3/5, \quad p_3 = 1.$$

2. The smallest genuine gluing obstruction is a three-face pencil around a two-plane core.
3. Every four-plane has exactly 21 open pencils, 7 flat pairwise-compatible pencils, and 7 frustrated pairwise-compatible pencils.
4. Every k -packet therefore has $21 \begin{bmatrix} k \\ 4 \end{bmatrix}_2$ open pencils, $7 \begin{bmatrix} k \\ 4 \end{bmatrix}_2$ flat pencils, and $7 \begin{bmatrix} k \\ 4 \end{bmatrix}_2$ frustrated pencils.
5. Over local gauges, packet heat obeys exact equipartition:

$$\mathbb{E}\tau_A = \mathbb{E}\tau_M = 1/2.$$

6. The heat variance decays as

$$\text{Var}(\tau_A) = \text{Var}(\tau_M) = \frac{47}{620} 2^{-k} + O(4^{-k}).$$

The theory direction is now:

$$\boxed{\text{identity spectroscopy} \rightarrow \text{packet spectra} \rightarrow \text{overlap holonomy}.}$$

Three-planes carry potentials. k -planes carry spectra. Four-plane pencils reveal the first nontrivial gluing cycles.

R7.10 Addendum: support-code derivative law

The pencil obstruction above is local gluing. There is also a global support-code view. For a packet U , define the repair support of a gauge

$$S_U(\lambda) = \{P \in \text{Gr}(3, U) : \lambda|_P = \theta_P\}.$$

Let $M \leq U^*$ have dimension r and let $K = M^\perp$. Then

$$\boxed{\bigcap_{\mu \in M} S_U(\lambda_0 + \mu) = S_K(\lambda_0|_K).} \quad (\text{R7.13})$$

A common repaired face for an affine r -flat of gauges is therefore exactly a repaired face in the lower-dimensional packet K of dimension $k - r$. Thus the intersection spectrum is a dimension derivative:

$$\boxed{\mathcal{I}_{k,r} = \begin{bmatrix} k \\ r \end{bmatrix}_2 H_{k-r}.} \quad (\text{R7.14})$$

For $r = 1$, every nonzero difference δ has the same pair-overlap spectrum:

$$\boxed{\text{Spec}_{\lambda/\langle \delta \rangle} |S(\lambda) \cap S(\lambda + \delta)| = H_{k-1}.} \quad (\text{R7.15})$$

This gives a second checksum family: coherent packets must have the legal support weights H_k , the legal pair overlaps $(2^k - 1)H_{k-1}$, and more generally the legal affine-intersection spectra $\begin{bmatrix} k \\ r \end{bmatrix}_2 H_{k-r}$.

The support family is a nonlinear binary code of length $G_k = \begin{bmatrix} k \\ 3 \end{bmatrix}_2$. Seven dead gauges collapse to the all-zero support. Exact checks for $k = 3, 4, 5, 6$ found $2^k - 6$ distinct support words and minimum positive distances

$$1, \text{hspace0.5em} 2, \text{hspace0.5em} 16, \text{hspace0.5em} 128,$$

matching the candidate law

$$\boxed{d_{\min}^+ = 2 \cdot 8^{k-4} \quad (k \geq 4).} \quad (\text{R7.16})$$

So the positive support code corrects up to $8^{k-4} - 1$ face-bit errors, while the seven dead gauges remain intentionally indistinguishable unless extra probes are added.

Application. The support-code layer strengthens packet diagnostics. A noisy or corrupted system may pass the scalar cold-count test but fail pair-overlap or affine-intersection checks. In AI/tool-composition language, individual correction contexts may look plausible, while their overlap code reveals context collapse, state leakage, or nonstationary behavior.

Round 8 Appendix - Law Spectrum Beyond Associativity

This round extends identity spectroscopy from associativity to other algebraic rules: commutativity, distributivity, and the Lie/Jacobi law of the commutator bracket. The main point is that nonassociativity was not a special case; it was the first member of a broader law-spectrum method. Different identities act as different instruments. Some detect rank-2 order-sensitivity, some detect rank-3 grouping-sensitivity, and some separate true algebraic failure from a phase-skeleton artifact.

Status. The commutator and twist-distributivity formulas below are theorem candidates with direct proof sketches from the fixed Cayley-Dickson sign recurrence. Counts were machine-checked by exact enumeration for $n \leq 8$. The Jacobi support law has a short structural proof and was also checked through $n = 6$.

1. Commutativity is a pure rank-2 detector

For basis labels u, v in $V_n = F_2^n$, write $e_u e_v = (-1)^{f_n(u,v)} e_{u+v}$. Define the Boolean commutativity defect

$$K_n(u, v) = f_n(u, v) + f_n(v, u).$$

Then the exact law is

$$K_n(u, v) = \Delta_2(u, v),$$

where $\Delta_2(u, v) = 1$ exactly when u and v are linearly independent over F_2 . Equivalently, since rank-2 independence for two vectors means nonzero and unequal,

$$K_n(u, v) = \eta(u) + \eta(v) + \eta(u+v),$$

$$\eta(x) = 0 \text{ if } x = 0, \eta(x) = 1 \text{ otherwise.}$$

Thus the basis-unit Cayley-Dickson commutativity law is maximally hot but not chaotic. It fails on exactly the projectively distinct pairs. The phase-corrected anticommutativity law is cold on those same rank-2 pairs.

$$e_u e_v = -e_v e_u \text{ for } u, v \text{ independent.}$$

The ordered-pair count and temperature are

$$N_{\text{comm}}(n) = (2^n - 1)(2^n - 2),$$

$$\tau_{\text{comm}}(n) = 1 - 3/2^n + 2/4^n.$$

n	comm defects	tau_comm
2	6	0.375000
3	42	0.656250
4	210	0.820312
5	930	0.908203
6	3906	0.953613
7	16002	0.976685
8	64770	0.988312

2. Rank detectors are finite derivatives of pointedness

The commutator formula is the $r=2$ case of a general finite-difference identity. For r vectors, define $\Delta_r = 1$ iff the r vectors are independent. Then

$$\Delta_r(x_1, \dots, x_r) = \sum_{\emptyset \neq I \subseteq \{1, \dots, r\}} \eta(\sum_{i \in I} x_i) \pmod{2}.$$

If the vectors are independent, all $2^r - 1$ nonempty subset sums are nonzero, so the parity is 1. If their span has lower rank, each nonzero vector in the span has an even number of preimages among nonempty subset sums, so the parity is 0.

This gives a clean hierarchy: commutativity exposes Δ_2 ; the cyclic associator curl exposes Δ_3 ; the Round 4 boundary law exposes Δ_4 . In this view, rank itself is measurable through algebraic law-failure instruments.

```

commutator defect      -> Delta_2
cyclic associator curl -> Delta_3
tetrahedral curl       -> Delta_4

```

3. Distributivity: cold in the algebra, hot in the sign skeleton

There are two different meanings of distributivity here. The full Cayley-Dickson algebra is bilinear, so ordinary distributivity over vector addition is cold:

$$a(b+c)=ab+ac, \quad (a+b)c=ac+bc.$$

But the sign skeleton f_n is not bilinear. Its non-additivity splits the associator into two measurable phase-distributivity defects:

$$\begin{aligned} D_R(u;v,w) &= f(u,v+w)+f(u,v)+f(u,w), \\ D_L(u,v;w) &= f(u+v,w)+f(u,w)+f(v,w). \end{aligned}$$

Their sum is exactly the associator curvature:

$$\Phi(u,v,w) = D_R(u;v,w) + D_L(u,v;w).$$

On an independent triple spanning a three-plane P with local potential θ_P , the local laws are

$$\begin{aligned} D_R(u;v,w) &= 1 + \theta_P(u), \\ D_L(u,v;w) &= \theta_P(w), \\ \Phi(u,v,w) &= 1 + \theta_P(u) + \theta_P(w). \end{aligned}$$

So twist non-distributivity separates the associator into two components: a birth-plus-left-end channel D_R and a pure right-end transport channel D_L . This is a sharper instrument than the associator alone.

4. Exact counts for phase-distributivity

Exact enumeration matched the closed forms below. D_R has the same count as the Cayley-Dickson associator. D_L has the same count as the Moufang/transport defect found earlier.

$$\begin{aligned} N_R(n) &= B_n = (2^{n-2})(2^{n-4})(2^{n-1}+3), \\ N_L(n) &= M_n = 4(2^{n-1}-1)(2^{n-1}-2)(2^{n-1}-4). \end{aligned}$$

n	D_R count	D_L count	Phi count
3	168	0	168
4	1,848	672	1,848
5	15,960	10,080	15,960
6	130,200	104,160	130,200
7	1,046,808	937,440	1,046,808
8	8,385,048	7,936,992	8,385,048

Interpretation: true distributivity is cold in the full algebra, but the basis-sign phase has a measurable nonlinearity. That phase nonlinearity is not noise; it is precisely the source decomposition of associator curvature.

5. Noncommutativity creates a rank-3 non-Lie detector

The commutator bracket of basis units is supported exactly on Δ_2 :

$$\begin{aligned} [e_u, e_v] &= e_u e_v - e_v e_u = 0 & \text{if } \Delta_2(u,v)=0, \\ [e_u, e_v] &= \pm 2 e_{\{u+v\}} & \text{if } \Delta_2(u,v)=1. \end{aligned}$$

Now form the Jacobiator

$$J(u,v,w) = [e_u, [e_v, e_w]] + [e_v, [e_w, e_u]] + [e_w, [e_u, e_v]].$$

Then the support law is

$$J(u,v,w) \neq 0 \quad \text{iff} \quad \Delta_3(u,v,w)=1.$$

Proof sketch: if $\text{rank}(u,v,w) < 3$, at least one required bracket collapses or terms repeat into zero. If $\text{rank}=3$, all three nested-bracket terms land on $e_{\{u+v+w\}}$; each has coefficient ± 4 , and a sum of three signs cannot be zero. Therefore the Jacobi identity fails exactly on rank-3 triples. The coefficient is always one of ± 4 or ± 12 .

$$\begin{aligned} N_J(n) &= (2^{n-1})(2^{n-2})(2^{n-4}), \\ \tau_J(n) &= 1 - 7/2^n + 14/4^n - 8/8^n. \end{aligned}$$

This is a new bridge between noncommutativity and nonassociativity: the commutator bracket is born at rank 2, while its failure to be Lie is born at rank 3.

6. New synthesis: Law Spectrum Theory

The broader object is no longer a single hot/cold law. It is a vector of law-defect fields, each with its own birth rank, temperature, boundary, repair pressure, and null-model-normalized heat.

$$\text{Spec}(A) = \{ \text{Def}_L : L \text{ is an identity in the chosen algebraic theory} \}.$$

For the Cayley-Dickson basis-sign skeleton, the first law-spectrum ladder is:

unit law	cold
commutativity defect K	Δ_2
commutator Jacobi support	Δ_3
associator cyclic curl	Δ_3
phase-right-distributivity D_R	$\Delta_3 \cdot (1 + \theta_P(u))$
phase-left-distributivity D_L	$\Delta_3 \cdot \theta_P(w)$
associator Φ	$D_R + D_L$

This suggests a useful classification principle: laws should be chosen as probes, not merely as pass/fail axioms. A good probe isolates a component: order-sensitivity, grouping-sensitivity, phase nonlinearity, hidden-state curvature, or gluing obstruction.

7. Applications

Black-box algebra classification. Test commutativity, associativity, distributivity, and Jacobi defects as a law-spectrum vector. The pattern distinguishes groups, loops, rings, nonassociative algebras, projection algebras, and noisy learned operators more sharply than any one identity.

Scientific computing. Full distributivity may be cold in exact algebra but hot in approximate numerical kernels. D_R and D_L style split-tests can identify whether error comes from left input aggregation, right input aggregation, or grouping.

Compiler and tensor optimizer safety. Reordering, regrouping, and distributing expressions are different transformations. Law spectroscopy gives separate tests for order-safety, parenthesis-safety, and expansion-safety.

AI/tool-composition diagnostics. Tool pipelines have analogues of noncommutativity (A then B differs from B

tasks changes output). A law-spectrum panel can diagnose whether failures come from ordering, grouping, splitting, retrieval context, or hidden state.

Error-correcting algebra. Δ_2 , Δ_3 , and Δ_4 detectors form a rank-ladder of parity checks. They can be used to identify impossible local syndromes and separate coherent curvature from table corruption.

8. Next targets

The natural continuation is a general law-spectrum atlas: choose a family of identities across associativity, commutativity, distributivity, identity/inverse laws, Moufang/Malcev laws, and Jordan laws; compute their defect fields. A focused Law Spectrum Theory study would make sense if the goal is to expand beyond the current program.

Round 8 summary

$$\begin{aligned} K_n(u,v) &= \Delta_2(u,v). \\ \Delta_r(x_1, \dots, x_r) &= \sum_{\emptyset \neq I \subseteq \{1, \dots, r\}} \epsilon(I) \sum_{i \in I} x_i \pmod 2. \\ \Phi &= D_R + D_L. \end{aligned}$$

$$D_R = \Delta_3(1+\theta_P(u)), \quad D_L = \Delta_3\theta_P(w).$$

$$\text{Jacobi-support}([\ , \]) = \Delta_3.$$

The main conceptual upgrade is: hot/cold laws should become a multi-law spectrum. Noncommutativity, nonassociativity, and phase non-distributivity are not separate curiosities; they are consecutive instruments in a rank-birth diagnostic ladder.

Appendix: Round 9 - Lie-Malcev Spectrum and Amplitude Spectroscopy

Preserved continuation between the Round 8 law spectrum and Round 10 polarized alternativity

April 30, 2026

1. Purpose

Round 8 showed that commutativity is a rank-two detector and that distributivity of the sign skeleton splits associator curvature. Round 9 studies what the commutator bracket sees:

$$[e_u, e_v] = e_u e_v - e_v e_u.$$

The main discovery is that the Malcev identity is a local-potential meter. It detects a coordinate of θ_P even when the original product is hidden and only the bracket is observable.

2. Lie-Malcev ladder

For Cayley-Dickson basis units, the commutator support is exactly rank two:

$$[e_x, e_y] \neq 0 \iff \Delta_2(x, y) = 1.$$

Define the Jacobiator

$$J(x, y, z) = [x, [y, z]] + [y, [z, x]] + [z, [x, y]].$$

Then the support law is

$$J(e_x, e_y, e_z) \neq 0 \iff \Delta_3(x, y, z) = 1.$$

Thus the bracket is born at rank two, and its failure to be Lie is born at rank three.

3. Malcev identity as a potential meter

Define the Malcev defect

$$\text{Mal}(x, y, z) = J(x, y, [x, z]) - [J(x, y, z), x].$$

For an independent triple x, y, z spanning a three-plane P , exact expansion gives

$$\text{Mal}(e_x, e_y, e_z) \neq 0 \iff \theta_P(y) = 1.$$

More precisely,

$$\text{Mal}(e_x, e_y, e_z) \in \{0, \pm 16e_{y+z}\}.$$

So the Malcev identity measures the middle coordinate of the local potential:

$$\mu(x, y, z) = \Delta_3(x, y, z)\theta_P(y),$$

where $\mu = 1$ means that the Malcev identity fails.

4. Bracket-only spectrometer

Permuting the inputs gives a bracket-only way to recover the full local state:

$$(J_{\text{supp}}(x, y, z), \mu(y, x, z), \mu(x, y, z), \mu(x, z, y)) = (\Delta_3, \Delta_3\theta_P(x), \Delta_3\theta_P(y), \Delta_3\theta_P(z)).$$

This mirrors the earlier associator spectrometer but uses only the commutator bracket. It distinguishes:

- no rank-three birth;
- octonion-like Malcev behavior, where $\theta_P = 0$;
- higher Cayley-Dickson behavior, where $\theta_P \neq 0$.

5. Amplitude spectroscopy

For independent x, y, z ,

$$J(e_x, e_y, e_z) = ce_{x+y+z},$$

with

$$c \in \{\pm 4, \pm 12\}.$$

The magnitude rule is

$$|c| = 12 \iff \theta_P(x) = \theta_P(y) = \theta_P(z).$$

Otherwise,

$$|c| = 4.$$

Thus support alone is not the whole law spectrum. The amplitude of failure carries hidden local-potential information.

6. Corroborating counts

The Jacobi support count is

$$N_J(n) = (2^n - 1)(2^n - 2)(2^n - 4).$$

The Malcev support count is

$$N_{\text{Mal}}(n) = 4(2^{n-1} - 1)(2^{n-1} - 2)(2^{n-1} - 4).$$

n	Jacobi failures	Malcev failures
3	168	0
4	2,520	672
5	26,040	10,080
6	234,360	104,160
7	1,984,248	937,440
8	16,322,040	7,936,992

The $n = 3$ line is conceptually important: the octonion commutator fails Jacobi but satisfies Malcev. Starting at $n = 4$, Malcev failure appears exactly in the quasioctonionic local-potential channel.

7. Applications

Black-box bracket diagnostics. If only a bracket operation is available, the bracket-only spectrometer can still recover rank-three birth and local potential coordinates.

Lie versus Malcev classification. Jacobi support detects non-Lie behavior; Malcev support separates octonion-like Malcev brackets from higher Cayley-Dickson brackets.

Amplitude-aware anomaly detection. Two systems may have the same Boolean failure support but different coefficient amplitudes. Amplitude therefore becomes an additional syndrome channel.

AI/tool composition. Noncommutativity, Jacobi failure, and Malcev failure correspond to order-dependence, three-way composition failure, and higher coherence failure. These should be tested separately.

8. Round 9 summary

The useful equations are

$$[e_x, e_y] \neq 0 \iff \Delta_2(x, y) = 1,$$

$$J(e_x, e_y, e_z) \neq 0 \iff \Delta_3(x, y, z) = 1,$$

$$\text{Mal}(e_x, e_y, e_z) \neq 0 \iff \Delta_3(x, y, z)\theta_P(y) = 1.$$

The conceptual upgrade is

laws fail with structured support, structured amplitude, and structured diagnostic meaning.

Appendix: Round 10 - Polarized Alternativity, Leibniz Defects, and Jordan Collision Spectra

Continued development of Associator Field Theory / Law Spectrum Theory

April 30, 2026

1. Purpose of this round

The previous rounds expanded from associator heat into commutativity, phase-distributivity, Jacobi support, Malcev failure, and amplitude spectroscopy. This round asks what happens to the classical “nearby” rules:

$$(xx)y = x(xy), \quad (yx)x = y(xx), \quad [x, yz] = [x, y]z + y[x, z],$$

and to the symmetric or Jordan product

$$\{x, y\} = xy + yx.$$

The exceptional point is that ordinary basis alternativity is cold, but its polarized version is hot and exactly measures gradients of the local potential. This creates a new distinction:

absolute probes measure θ ; polarized probes measure $d\theta$.
--

2. Setup and notation

Use the fixed Cayley-Dickson basis convention

$$e_u e_v = s(u, v) e_{u+v}, \quad s(u, v) = (-1)^{f(u, v)}, \quad u, v \in V_n = \mathbb{F}_2^n.$$

The Boolean associator defect is

$$\Phi(u, v, w) = f(u, v) + f(u + v, w) + f(v, w) + f(u, v + w).$$

For a rank-three plane $P = \langle x, y, z \rangle$, the local potential law is

$$\Phi(x, y, z) = 1 + \theta_P(x) + \theta_P(z).$$

Here $\theta_P = \sigma_{\beta(P)}|_P$ is the local transported potential born at the first coordinate where the triple becomes independent.

3. Polarized left alternativity is a potential-gradient meter

Basis-unit left alternativity is cold:

$$(e_x e_x) e_z = e_x (e_x e_z).$$

This coldness is deceptive. Polarize the repeated variable by setting

$$A = e_x + e_y$$

and define the left polarized alternator

$$\mathcal{A}_L(x, y; z) = ((AA)e_z) - A(Ae_z).$$

Expanding and cancelling the repeated cold terms gives

$$\mathcal{A}_L(x, y; z) = ((e_x e_y) e_z + (e_y e_x) e_z - e_x (e_y e_z) - e_y (e_x e_z)).$$

All four terms have label $x + y + z$, so the result is a scalar multiple of e_{x+y+z} . Exact enumeration and sign expansion give the support law

$$\mathcal{A}_L(x, y; z) \neq 0 \iff \Delta_3(x, y, z)(\theta_P(x) + \theta_P(y)) = 1.$$

When it is nonzero, the coefficient has magnitude 2:

$$\mathcal{A}_L(x, y; z) \in \{0, \pm 2e_{x+y+z}\}.$$

Thus left polarized alternativity does not measure an absolute coordinate of θ_P . It measures the edge-gradient

$$d\theta_P(x, y) = \theta_P(x) + \theta_P(y).$$

4. Polarized right alternativity measures the next edge

Similarly, set

$$B = e_y + e_z$$

and define the right polarized alternator

$$\mathcal{A}_R(x; y, z) = (e_x B)B - e_x(BB).$$

Then

$$\mathcal{A}_R(x; y, z) \neq 0 \iff \Delta_3(x, y, z)(\theta_P(y) + \theta_P(z)) = 1,$$

and again

$$\mathcal{A}_R(x; y, z) \in \{0, \pm 2e_{x+y+z}\}.$$

So the two polarized alternativity probes recover two local gradients:

$$g_{xy} = \theta_P(x) + \theta_P(y), \quad g_{yz} = \theta_P(y) + \theta_P(z).$$

Together they recover the relative potential pattern up to adding a constant. A Malcev, Moufang, or associator coordinate probe supplies the missing absolute offset.

5. Polarized flexibility stays cold

For comparison, polarize flexibility by putting the repeated variable on both sides:

$$C = e_x + e_z,$$

$$\mathcal{F}(x; y; z) = ((Ce_y)C) - C(e_yC).$$

Exact sign expansion and enumeration give

$$\mathcal{F}(x; y; z) = 0 \quad \text{for all basis labels } x, y, z.$$

This separates Cayley-Dickson behavior into three classes:

$$\text{basis alternativity cold; polarized alternativity hot; polarized flexibility cold.}$$

That is a useful law-spectrum fingerprint.

6. Count law for polarized alternativity

A quasioctonionic three-plane has a nonzero linear potential θ_P . On its 168 ordered bases, the condition $\theta_P(x) + \theta_P(y) = 1$ occurs exactly 96 times. Octonionic planes contribute zero polarized-alternativity failures. Since

$$Q_n = 7 \binom{n-1}{3}_2,$$

the global polarized left/right alternativity count is

$$N_{\text{pAlt}}(n) = 96Q_n = 672 \binom{n-1}{3}_2.$$

Equivalently,

$$N_{\text{pAlt}}(n) = 4(2^{n-1} - 1)(2^{n-1} - 2)(2^{n-1} - 4).$$

This is the same support count previously found for Moufang/Malcev transport probes, but the meaning is different: polarized alternativity measures a gradient, not an absolute coordinate.

7. Leibniz/derivation defect of the commutator

In an associative algebra, the adjoint map $\text{ad}_x(y) = [x, y]$ is a derivation:

$$[x, yz] = [x, y]z + y[x, z].$$

Define the Leibniz defect

$$\mathcal{L}(x; y, z) = [e_x, e_y e_z] - [e_x, e_y] e_z - e_y [e_x, e_z].$$

For Cayley-Dickson basis labels,

$$\mathcal{L}(x; y, z) \neq 0 \iff \Delta_3(x, y, z) = 1.$$

So the commutator fails to be a derivation exactly at rank-three birth.

The amplitude splits by the equality pattern of the local potential:

$$|\mathcal{L}(x; y, z)| = 6 \iff \theta_P(x) = \theta_P(y) = \theta_P(z),$$

$$|\mathcal{L}(x; y, z)| = 2 \quad \text{otherwise.}$$

This is a half-amplitude version of the Jacobi amplitude scar from the previous round. It gives a practical rule:

$$\text{Leibniz support detects birth; Leibniz amplitude detects potential collapse/equality.}$$

8. Symmetric/Jordan product behaves differently

Define the symmetric product

$$\{e_u, e_v\} = e_u e_v + e_v e_u.$$

Since the commutativity defect is Δ_2 , the symmetric product cancels all independent pairs:

$$\{e_u, e_v\} = 0 \quad \text{when } u, v \text{ are nonzero and distinct.}$$

More explicitly,

$$\begin{aligned}\{e_0, e_v\} &= 2e_v, \\ \{e_u, e_u\} &= -2e_0 \quad (u \neq 0), \\ \{e_u, e_v\} &= 0 \quad (\text{rank}(u, v) = 2).\end{aligned}$$

Thus the Jordan skeleton is not a rank-three curvature probe. It is a collision probe.

Define the symmetric associator

$$J_+(x, y, z) = \{\{e_x, e_y\}, e_z\} - \{e_x, \{e_y, e_z\}\}.$$

Then

$$\boxed{J_+(x, y, z) \neq 0}$$

exactly in the two adjacent-collision cases

$$\boxed{x = y \neq 0, \quad z \notin \{0, x\}}$$

or

$$\boxed{y = z \neq 0, \quad x \notin \{0, y\}}.$$

The coefficient is $-4e_z$ in the first case and $+4e_x$ in the second, with labels interpreted through the displayed symmetric product expansion. The total number of nonzero basis symmetric-associator cases is

$$\boxed{N_{J_+}(n) = 2(2^n - 1)(2^n - 2)}.$$

So symmetric associativity has low-rank collision heat, not rank-birth heat.

Random sparse integer-vector tests through $n = 5$ found no failure of the Jordan identity

$$\{\{a, b\}, \{a, a\}\} = \{a, \{b, \{a, a\}\}\},$$

but this round treats that only as computational evidence, not as a proved theorem.

9. Corroborating data

The following exact enumeration checks used the recovered Cayley-Dickson sign recursion. Counts are over positive basis labels $0, \dots, 2^n - 1$.

n	rank-3 triples	pAlt failures	Leibniz nonzero	Leibniz high amp	Jordan assoc. nonzero
3	168	0	168	168	84
4	2,520	672	2,520	1,512	420
5	26,040	10,080	26,040	10,920	1,860
6	234,360	104,160	234,360	78,120	7,812

The high-amplitude Leibniz count is

$$\boxed{168O_n + 24Q_n},$$

where

$$O_n = \binom{n-1}{2}_2 + \binom{n-1}{3}_2, \quad Q_n = 7 \binom{n-1}{3}_2.$$

The low-amplitude Leibniz count is

$$\boxed{144Q_n}.$$

The total is

$$168O_n + 168Q_n = (2^n - 1)(2^n - 2)(2^n - 4),$$

as expected from the rank-three support law.

10. New synthesis: absolute, gradient, amplitude, and collision probes

The law spectrum now separates into diagnostic layers:

commutativity	$\longrightarrow$	Δ_2 ,
associator / Jacobi support	$\longrightarrow$	Δ_3 ,
Moufang / Malcev	$\longrightarrow$	θ_P coordinate meters,
polarized alternativity	$\longrightarrow$	$d\theta_P$ gradient meters,
Leibniz defect	$\longrightarrow$	Δ_3 support plus equality-amplitude scar,
symmetric/Jordan associator	$\longrightarrow$	rank-one/rank-two collision heat.

The new useful equation is therefore

$$\mathcal{A}_L(x, y; z) \neq 0 \iff \Delta_3(x, y, z)(\theta_P(x) + \theta_P(y)) = 1.$$

Together with

$$\mathcal{A}_R(x; y, z) \neq 0 \iff \Delta_3(x, y, z)(\theta_P(y) + \theta_P(z)) = 1,$$

it upgrades identity spectroscopy into a discrete differential calculus:

$$\text{identities can measure } \theta, d\theta, \Delta_3, \text{ and collision structure separately.}$$

11. Applications

1. Black-box algebra classification. Instead of asking only whether associativity fails, test a panel: commutativity, associativity, polarized alternativity, Leibniz derivation, and symmetric associator. Their support pattern distinguishes rank-birth curvature, transport potential, gradient obstruction, and collision behavior.

2. Error correction and repair. Absolute probes locate a potential value; gradient probes locate disagreements between adjacent local repairs. This is useful when absolute gauge labels are unreliable but relative consistency can still be measured.

3. Scientific computing. Many reduced models and projected operators preserve some laws while breaking others. Polarized alternativity tests whether duplicating an input before applying a product introduces hidden transport error. Leibniz tests whether an update direction acts locally like a derivation.

4. Compiler and tensor optimizer safety. Optimizers often reorder, regroup, duplicate, or distribute expressions. This round suggests separate safety tests for each move: order tests, grouping tests, duplication tests, derivation tests, and symmetrization tests.

5. AI and tool-composition diagnostics. In an AI pipeline, a tool transform may be treated like an operation. Commutativity tests order-dependence; associativity tests grouping-dependence; polarized alternativity tests duplication/copy sensitivity; Leibniz tests whether a local edit distributes across a composed task; symmetric associator tests whether averaging or consensus collapses distinct states.

12. Round 10 summary

The main discovery is:

$$\text{basis alternativity is cold, but polarized alternativity is a hot gradient meter.}$$

The most useful new laws are:

$$\mathcal{A}_L \neq 0 \iff \Delta_3(\theta_x + \theta_y) = 1, \quad \mathcal{A}_R \neq 0 \iff \Delta_3(\theta_y + \theta_z) = 1.$$

The most useful new diagnostic distinction is:

absolute probes recover potential values; polarized probes recover potential gradients.

The broader theory has now become a multi-instrument law spectrum: different algebraic rules fail for different reasons, and those reasons can be separated by choosing the right identity probes.

Round 11 Appendix: Bol-Moufang Gradient Code and Medial Heat

Continuing Associator Field Theory into pure loop laws and entropic/interchange laws

Status: theorem candidates with exhaustive finite verification in small dimensions and exact closed-form counts for the medial law.

11.1 Motivation

Round 10 separated algebraic probes into absolute probes, gradient probes, amplitude probes, and collision probes. The polarized alternativity law was especially useful because it measured a difference of the local potential, but it required a linear combination such as $e_x + e_y$. This round asks whether the same information can be recovered using only basis-unit multiplication and equality tests.

The answer appears to be yes. Classical loop identities, especially Bol and Moufang identities, form a pure basis-level gradient code. In parallel, the four-variable medial/entropic identity supplies a first exact law-temperature formula outside the associator/Moufang family.

Throughout this appendix, $x, y, z, w \in V_n = \mathbb{F}_2^n$, $e_u e_v = (-1)^{f(u,v)} e_{u+v}$, and $P = \langle x, y, z \rangle$. If $\text{rank}(x, y, z) = 3$, let θ_P be the local potential from the three-plane potential law. Write

$$\theta_x = \theta_P(x), \quad \theta_y = \theta_P(y), \quad \theta_z = \theta_P(z).$$

11.2 Exceptional finding: Bol identities are pure gradient meters

Define the left Bol/Moufang-type defect bit

$$B_{xy}(x, y, z) = \text{Def}(x(y(xz))) = ((xy)x)z).$$

This is the Boolean sign difference between the two sides. Exhaustive computation and direct sign expansion give

$$B_{xy}(x, y, z) = \Delta_3(x, y, z)(\theta_x + \theta_y).$$

By permuting variables,

$$B_{xz}(x, y, z) = \Delta_3(x, y, z)(\theta_x + \theta_z),$$

where $B_{xz} = B_{xy}(x, z, y)$.

Now define the middle Moufang defect

$$M(x, y, z) = \text{Def}((xy)(zx) = (x(yz))x).$$

The corrected Round 8–10 formula is

$$M(x, y, z) = \Delta_3(x, y, z)(\theta_x + \theta_y + \theta_z).$$

Thus the three pure loop-law tests

$$(B_{xy}, B_{xz}, M)$$

are an invertible linear encoding of the local potential coordinates. When $\Delta_3 = 1$,

$$\theta_x = B_{xy} + B_{xz} + M, \quad \theta_y = B_{xz} + M, \quad \theta_z = B_{xy} + M.$$

This is the main conceptual advance of the round:

$$\text{Bol/Moufang loop laws recover } d\theta \text{ and } \theta \text{ without linear combinations or brackets.}$$

A pure multiplication/equality oracle can therefore recover the same local potential data that earlier required associators, commutators, or polarized sums.

11.3 The Bol-Moufang gradient code

For independent triples, the map

$$(\theta_x, \theta_y, \theta_z) \mapsto (B_{xy}, B_{xz}, M)$$

is the invertible linear code

$$\begin{pmatrix} B_{xy} \\ B_{xz} \\ M \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \theta_x \\ \theta_y \\ \theta_z \end{pmatrix}.$$

The resulting table is:

$(\theta_x, \theta_y, \theta_z)$	(B_{xy}, B_{xz}, M)
000	000
001	011
010	101
011	110
100	111
101	100
110	010
111	001

If $\Delta_3 = 0$, all three defects vanish. Therefore a complete pure-loop spectrometer consists of one rank-birth detector plus these three loop-law bits. For example, Round 2 provided a basis-only rank detector

$$R(x, y, z) = \text{Def}(x((yx)(zx)) = ((xy)(xz))x = \Delta_3(x, y, z).$$

Then

$$\boxed{(R, B_{xy}, B_{xz}, M)}$$

recovers either “no rank-three birth” or the full local potential θ_P .

11.4 Verification of the loop-law spectrometer

The following checks used the recovered Cayley-Dickson sign recursion. Exhaustive enumeration was run for $n = 3, 4, 5, 6$, and random checks of 200,000 triples were run for $n = 7, 8$.

n	triples tested	formula mismatches	states observed
3	512	0	1
4	4,096	0	8
5	32,768	0	8
6	262,144	0	8
7	200,000 random	0	8
8	200,000 random	0	8

For $n = 6$, the observed independent-triple counts by potential state were

θ	# ordered triples
000	52,080
001, 010, 011, 100, 101, 110, 111	26,040 each.

This matches the earlier packet-spectrum principle: the zero-potential/octonionic state has different mass, while nonzero potentials are equipopulated.

11.5 Medial/interchange law: exact heat outside associativity

The four-variable medial, entropic, or interchange identity is

$$(xy)(zw) = (xz)(yw).$$

For basis units define

$$E(x, y, z, w) = \text{Def}((xy)(zw) = (xz)(yw)).$$

This law is not a pure local-potential meter. It mixes commutativity, associativity, and four-variable interchange. However, its total heat has an unexpectedly compact closed form.

Let $X = 2^n$. The exact number of medial failures among all ordered quadruples is

$$N_E(n) = \frac{1}{2}X^4 + 2X^2 - 24X + 32 = \frac{(X-2)^2(X^2 + 4X + 16)}{2}.$$

Therefore the medial temperature is

$$\tau_E(n) = \frac{1}{2} + \frac{2}{4^n} - \frac{24}{8^n} + \frac{32}{16^n}.$$

So medial heat tends to $1/2$, unlike random magma associativity heat, which tends to 1. The medial law is balanced at high rank but carries small low-rank scars.

11.6 Rank-stratified medial law

The exact failure count decomposes cleanly by rank:

$$N_E(n) = 96 \binom{n}{2}_2 + \left(1260 \binom{n}{3}_2 + 84 \binom{n-1}{2}_2 \right) + 10080 \binom{n}{4}_2.$$

The three summands are the rank-2, rank-3, and rank-4 contributions. Equivalently:

quadruple rank	medial failures
0, 1	0
2	$96 \binom{n}{2}_2$
3	$1260 \binom{n}{3}_2 + 84 \binom{n-1}{2}_2$
4	$10080 \binom{n}{4}_2.$

The rank-4 layer is exactly half of all rank-4 ordered quadruples, because $|GL(4, 2)| = 20160$ and $10080 = 20160/2$. The rank-2 layer has failure fraction $96/210 = 16/35$. The rank-3 layer is almost balanced but has a positive scar

$$\frac{1}{2} \cdot 2520 \binom{n}{3}_2 + 84 \binom{n-1}{2}_2.$$

This suggests a new class of laws:

$$\boxed{\text{interchange laws are high-rank balanced with lower-rank scars.}}$$

11.7 Medial verification table

n	$X = 2^n$	$N_E(n)$	$\tau_E(n)$	rank 2	rank 3	rank 4
2	4	96	0.37500000	96	0	0
3	8	2,016	0.49218750	672	1,344	0
4	16	32,928	0.50244141	3,360	19,488	10,080
5	32	525,600	0.50125122	14,880	198,240	312,480
6	64	8,395,296	0.50039864	62,496	1,770,720	6,562,080
7	128	134,247,456	0.50011075	256,032	14,936,544	119,054,880
8	256	2,147,608,608	0.50002909	1,036,320	122,639,328	2,023,932,960

Exhaustive enumeration was performed for $n = 2, 3, 4, 5$. The $n \geq 6$ rows are formula extensions from the closed form.

11.8 Applications

1. Pure black-box loop spectroscopy. Earlier spectrometers used associators, commutators, or linear combinations. The Bol-Moufang gradient code works with basis multiplication and equality alone. This is important for black-box finite loops, quasigroups, and table systems where addition or scalar access is unavailable.

2. Retry/reinforcement diagnostics in AI tool pipelines. Bol identities repeat one operation, as in $x(y(xz))$. That resembles retry, reinforcement, wrapper, or context-reapplication patterns. The fact that Bol laws measure $\theta_x + \theta_y$ suggests a diagnostic for whether repeated context application introduces a coherent gradient-like error.

3. Parallelization and batching safety. The medial law is the algebraic shape of interchange:

$$(xy)(zw) \stackrel{?}{=} (xz)(yw).$$

A compiler, tensor optimizer, distributed workflow, or tool scheduler implicitly assumes interchange when it batches or swaps independent subcomputations. The exact heat law says that in Cayley-Dickson-type systems, interchange failure is asymptotically balanced and cannot be ignored, but its low-rank scars are structured and therefore diagnosable.

4. Error-correcting algebra. The loop spectrometer gives three parity checks for θ plus a rank bit. The medial rank-stratification gives a separate four-variable checksum. Combining them yields a two-layer code: local-potential recovery on triples and interchange-anomaly detection on quadruples.

5. Law-spectrum design principle. Associator, Moufang, Bol, polarized alternativity, commutator, Malcev, and medial identities now fall into different instrument classes: absolute meters, gradient meters, amplitude meters, collision meters, and high-rank balanced interchange meters. This supports the broader Law Spectrum Theory program.

11.9 Round 11 summary

The most useful new equations are

$$B_{xy} = \Delta_3(\theta_x + \theta_y), \quad B_{xz} = \Delta_3(\theta_x + \theta_z), \quad M = \Delta_3(\theta_x + \theta_y + \theta_z),$$

with inverse

$$\theta_x = B_{xy} + B_{xz} + M, \quad \theta_y = B_{xz} + M, \quad \theta_z = B_{xy} + M.$$

The most useful four-variable law is

$$\tau_E(n) = \frac{1}{2} + \frac{2}{4^n} - \frac{24}{8^n} + \frac{32}{16^n}.$$

The theory direction is now

identity spectroscopy $\rightarrow$ law-spectrum spectroscopy $\rightarrow$ loop and interchange spectroscopy.

In short: Bol-Moufang laws recover local potential using only repeated basis multiplication; the medial law reveals that interchange failure is asymptotically balanced but low-rank scarred.

24 Round 12: Label-Phase Separation and Balanced-Law Filtering

24.1 Purpose of this continuation

Earlier rounds treated associativity, commutativity, Moufang/Bol laws, Malcev failure, polarized alternativity, and the medial/interchange law as separate instruments. The next step is to ask which algebraic laws are worth using as curvature probes. The answer is that a basis-unit law has two layers: a label layer and a phase layer. A law that is unbalanced at the label layer usually becomes almost automatically false in large dimension and is therefore a poor curvature instrument. Balanced laws survive the label filter and expose sign curvature, rank birth, local potential, or higher interchange heat.

This round introduces a general label-phase separation theorem and a balanced-law filter. It also adds a pair-product interchange code for the 24 oriented expressions of the form

$$((ab)(cd)), \quad \{a, b, c, d\} = \{x, y, z, w\}.$$

24.2 Label-phase separation theorem

For any binary-tree monomial word T built from variables $x_1, \dots, x_m$, evaluation on Cayley-Dickson basis units has the form

$$T(e_{x_1}, \dots, e_{x_m}) = (-1)^{\phi_T(x_1, \dots, x_m)} e_{\ell_T(x_1, \dots, x_m)},$$

where the label is the parity occurrence vector

$$\ell_T(x_1, \dots, x_m) = \sum_{i=1}^m \nu_i(T) x_i \quad \text{in } \mathbb{F}_2^n,$$

with $\nu_i(T)$ equal to the number of appearances of x_i in T modulo 2. The phase ϕ_T depends on the Cayley-Dickson twist f and on the bracketing, but the label depends only on occurrence parity.

For an identity $L = R$,

$$\text{Def}_{L=R} = 1$$

if and only if either the labels disagree,

$$\ell_L \neq \ell_R,$$

or the labels agree and the phases disagree,

$$\ell_L = \ell_R \quad \text{and} \quad \phi_L + \phi_R = 1.$$

Thus every law splits into:

label obstruction + phase obstruction.

24.3 Balanced-law filter

Let

$$d_i = \nu_i(L) + \nu_i(R) \in \mathbb{F}_2$$

be the parity mismatch vector of a law. If the d_i span a q -dimensional linear condition over the input labels, then the label equality set has density

$$2^{-qn}.$$

Consequently an unbalanced law has label-temperature

$\tau_{\text{label}} = 1 - 2^{-qn}.$

It becomes asymptotically hot for a trivial reason: the two sides usually land on different basis labels. Such a law may still be useful, but it is not a clean sign-curvature probe.

Balanced laws, where $q = 0$, are the main objects of law-spectrum spectroscopy. They include commutativity, associativity, Moufang/Bol laws, Malcev laws, and medial/interchange laws. These can reveal rank birth, local potential, potential gradients, amplitude scars, and high-rank balanced heat.

24.4 Examples: unbalanced laws are label-hot

Let $X = 2^n$.

law	label condition for possible truth	failure count / temperature
idempotence $xx = x$	$x = 0$	$X - 1, 1 - X^{-1}$
left self-distribution $x(yz) = (xy)(xz)$	$x = 0$	$(X - 1)X^2, 1 - X^{-1}$
right self-distribution $(xy)z = (xz)(yz)$	$z = 0$	$(X - 1)X^2, 1 - X^{-1}$
commutativity $xy = yx$	always label-balanced	phase law $\Delta_2(x, y)$
associativity $(xy)z = x(yz)$	always label-balanced	phase law $\Phi(x, y, z)$
medial $(xy)(zw) = (xz)(yw)$	always label-balanced	high-rank phase law

This explains why self-distributivity should not be grouped with commutativity or associativity in this basis-unit setting. It is almost always false because it changes the output label, not because it reveals a subtle Cayley-Dickson curvature.

24.5 Pair-product phase code

For four labels define the oriented pair-product phase

$$Q_{ab|cd} = f(a, b) + f(c, d) + f(a + b, c + d),$$

so that

$$(e_a e_b)(e_c e_d) = (-1)^{Q_{ab|cd}} e_{a+b+c+d}.$$

The 24 expressions $((ab)(cd))$ are controlled by orientation changes and three basic pairings.

Let

$$K(p, q) = f(p, q) + f(q, p) = \Delta_2(p, q)$$

be the commutativity defect. Then the orientation moves obey

$$Q_{ba|cd} = Q_{ab|cd} + K(a, b),$$

$$Q_{ab|dc} = Q_{ab|cd} + K(c, d),$$

$$Q_{cd|ab} = Q_{ab|cd} + K(a + b, c + d).$$

Thus many four-variable laws that look distinct are generated by rank-2 commutativity scars plus the genuinely cross-paired interchange scar.

24.6 Interchange triangle parity law

There are three unordered pairings:

$$P_0 = (xy|zw), \quad P_1 = (xz|yw), \quad P_2 = (xw|yz).$$

Let

$$E_{ij} = Q_{P_i} + Q_{P_j}.$$

Then

$$E_{01} + E_{02} + E_{12} = 0.$$

Therefore, for every quadruple x, y, z, w , the three basic interchange comparisons have either zero failures or exactly two failures. There is no isolated one-bug interchange triangle.

This is the four-variable analogue of a parity-check code: an observed packet with exactly one failed pair-comparison cannot come from a coherent Cayley-Dickson pair-product phase. It signals measurement error, table corruption, hidden state, or an out-of-model law.

24.7 Closed counts for key balanced four-variable laws

Again let $X = 2^n$.

A single orientation reversal, such as

$$(xy)(zw) = (yx)(zw),$$

or a pair swap,

$$(xy)(zw) = (zw)(xy),$$

fails exactly when the relevant two labels are nonzero and distinct. Therefore

$$N_{\text{single reversal}} = X^2(X-1)(X-2).$$

The full reversal

$$(xy)(zw) = (wz)(yx)$$

has defect

$$K(x, y) + K(z, w) + K(x + y, z + w),$$

and exact count

$$N_{\text{full reversal}} = (X-1)(X-2)(X^2 - 2X + 12).$$

The pure medial/interchange law from Round 11 has

$$N_{\text{medial}} = \frac{1}{2}X^4 + 2X^2 - 24X + 32.$$

So not all balanced four-variable laws are equally hot. Some are rank-2 reversal scars; some are high-rank interchange heat; some are composites.

24.8 Exploratory classification of the 24 oriented pair-product laws

Using the canonical expression $(xy)(zw)$ as reference, there are 23 nontrivial oriented pair-product identities. Exhaustive enumeration for $n = 2, 3, 4, 5$ grouped them into seven heat classes. The table lists one representative per class and the number of laws in that class.

representative right side	class size	failures at $n = 2$	failures at $n = 3$	failures at $n = 4$
$(zw)(yx)$	2	96	1344	13440
$(xz)(yw)$	9	96	2016	32928
$(wy)(zx)$	1	96	2688	37632
$(xy)(wz)$	3	96	2688	53760
$(yx)(wz)$	1	120	1848	19320
$(yw)(xz)$	6	120	1848	31416
$(wz)(yx)$	1	120	2520	49560

The class split shows that four-variable interchange laws are not a single phenomenon. They form a small spectrum built from rank-2 reversals, cross-pairing heat, and composite orientation scars. A full closed-form table for all seven classes is a natural next target.

24.9 Machine checks

The following checks were performed with the same Cayley-Dickson sign recursion used in the main report.

- Label-phase separation was checked by evaluating arbitrary binary trees and confirming that labels are occurrence-parity sums.
- Idempotence and self-distribution counts matched $1 - X^{-1}$ temperature through exhaustive enumeration for $n \leq 6$.
- The orientation equations for $Q_{ab|cd}$ were exhaustively checked for all quadruples through $n = 5$ and randomly checked for $n = 6, 7, 8$.
- The interchange triangle parity $E_{01} + E_{02} + E_{12} = 0$ had zero mismatches in exhaustive tests through $n = 5$.
- The 23 pair-product identities split into the seven heat classes shown above in exhaustive tests through $n = 5$.

24.10 Applications

1. Law selection for black-box algebra. Before using a law as a diagnostic, first pass it through the balanced-law filter. Unbalanced laws mostly measure output-label mismatch. Balanced laws measure phase curvature, local potentials, gradients, or interchange heat.

2. Compiler and optimizer safety. A compiler rewrite can fail because the two sides compute different labels/types/shapes, or because they compute the same label with a different phase/order/path. Label-phase separation gives a clean diagnostic split: first check type/shape parity, then check phase/ordering.

3. AI/tool composition. Tool pipelines often fail for two reasons: the output object changes type, or the same apparent object carries a different hidden state. The label layer models object identity/type. The phase layer models hidden-state or context-dependent order effects.

4. Error-correcting law packets. The interchange triangle parity law means pair-product tests have a built-in parity check. Exactly one failed edge in the triangle is impossible under the coherent model and should trigger retesting or state repair.

5. Discovery-machine design. A law-discovery engine should not merely enumerate identities and count failures. It should classify candidate laws by label balance, phase spectrum, rank support, amplitude, and packet parity checks. This prevents wasting search effort on laws that are hot for trivial label reasons.

24.11 Round 12 synthesis

The central new equations are

$$T(e_{x_1}, \dots, e_{x_m}) = (-1)^{\phi^T} e_{\sum_i \nu_i(T)x_i},$$

$$\tau_{\text{label}} = 1 - 2^{-q^n},$$

$$Q_{ba|cd} = Q_{ab|cd} + K(a, b), \quad Q_{ab|dc} = Q_{ab|cd} + K(c, d), \quad Q_{cd|ab} = Q_{ab|cd} + K(a + b, c + d),$$

$$E_{01} + E_{02} + E_{12} = 0.$$

The new theory direction is

law-spectrum spectroscopy $\rightarrow$ label-phase spectroscopy $\rightarrow$ law-code design.

The practical rule is simple: first ask whether a law is balanced. If not, it is mostly measuring label mismatch. If yes, it can become a true instrument for curvature, potential, gradient, amplitude, or higher interchange structure.

25 Round 13: Rewrite Holonomy and the Balanced-Law Compiler

25.1 Purpose of this continuation

Round 12 separated every basis-unit law into a label layer and a phase layer. This round asks what happens after a law passes the balanced-label filter. The new answer is that balanced monomial laws have a rewrite-holonomy calculus: their defects can be computed as a path integral of two elementary instruments, commutativity defects and associator defects.

This is useful because it turns law discovery into a compiler problem. A proposed identity can be parsed into a syntax tree, checked for label balance, decomposed into elementary swaps and rotations, and assigned a predicted defect field before any large finite table is enumerated.

25.2 Elementary rewrite graph

A monomial expression has two visible parts:

1. a binary bracketing tree;
2. a leaf order, with possibly repeated variables.

Let A, B, C denote subexpressions, and let $a, b, c \in \mathbb{F}_2^n$ be their current output labels. Two elementary rewrite moves generate all balanced monomial transformations.

Sibling swap.

$$AB \longleftrightarrow BA.$$

The phase cost is

$$K(a, b) = f(a, b) + f(b, a).$$

For Cayley-Dickson basis units,

$$K(a, b) = \Delta_2(a, b).$$

Thus a swap measures rank-two birth.

Rotation.

$$(AB)C \longleftrightarrow A(BC).$$

The phase cost is

$$\Phi(a, b, c) = f(a, b) + f(a + b, c) + f(b, c) + f(a, b + c).$$

This is the associator field. It measures rank-three birth plus local-potential transport.

25.3 Balanced-law path-integral theorem

Let T and S be two binary monomial expressions with the same label function. Choose any path γ in the rewrite graph from T to S , using sibling swaps and rotations. Let a_e, b_e, c_e be the current subexpression labels at edge e . Then

$$\text{Def}_{T=S} = \sum_{e \in \gamma \cap \text{swap}} K(a_e, b_e) + \sum_{e \in \gamma \cap \text{rotate}} \Phi(a_e, b_e, c_e) \pmod{2}.$$

The right side is independent of the chosen path.

Proof sketch. For every term T , recursively define a phase ϕ_T by

$$\phi_x = 0, \quad \phi_{AB} = \phi_A + \phi_B + f(\ell_A, \ell_B),$$

where ℓ_A is the output label of A . A sibling swap changes ϕ by $K(\ell_A, \ell_B)$, and a rotation changes ϕ by $\Phi(\ell_A, \ell_B, \ell_C)$. Summing changes along any path gives $\phi_T + \phi_S$, which is exactly the phase defect of the balanced identity.

25.4 Exceptional new coherence equation: the mixed hexagon law

Path independence forces identities among K and Φ . The most important one links noncommutativity and nonassociativity.

Compare the direct swap

$$x(yz) \longrightarrow (yz)x$$

with the longer route that moves x past y and then past z through reassociations:

$$x(yz) \rightarrow (xy)z \rightarrow (yx)z \rightarrow y(xz) \rightarrow y(zx) \rightarrow (yz)x.$$

The result is the mixed hexagon law

$$\boxed{K(x, y + z) = \Phi(x, y, z) + K(x, y) + \Phi(y, x, z) + K(x, z) + \Phi(y, z, x).}$$

Equivalently,

$$\boxed{K(x, y + z) + K(x, y) + K(x, z) + \Phi(x, y, z) + \Phi(y, x, z) + \Phi(y, z, x) = 0.}$$

This is the clean bridge between Round 8's commutativity defect and the associator field. Noncommutativity is not separate from associator curvature: they satisfy a braided coherence constraint.

25.5 Pentagon, hexagon, and the beginning of law cohomology

The associator field already satisfied the cold Bianchi law

$$\boxed{\delta\Phi = 0,}$$

which is the finite pentagon coherence law. The new mixed hexagon law says that K and Φ also fit together under permutation.

Thus the elementary law calculus has the form

$$\boxed{\text{balanced identity defect} = \int_{\gamma} (K, \Phi),}$$

with coherence constraints

$$\boxed{\text{pentagon: } \delta\Phi = 0, \quad \text{hexagon: } dK + \text{braid}(\Phi) = 0.}$$

This suggests a broader “law cohomology”: law failures are not just scalars, but curvature forms on a syntax-rewrite complex.

25.6 Four-leaf balanced-law census

To test the rewrite-calculus viewpoint, I enumerated all 120 binary-tree monomial forms using the four distinct variables x, y, z, w exactly once. There are 5 bracketings and 24 leaf orders. Hence there are

$$\binom{120}{2} = 7140$$

balanced identities between two such forms.

For each identity I computed the rank-stratified failure signature

$$(c_0, c_1, c_2, c_3, c_4),$$

where c_r is the number of failing quadruples of rank r in $\mathbb{F}_2^n$. Exact enumeration at $n = 4$ and $n = 5$ produced exactly 23 distinct signatures.

ambient dimension	quadruples tested per law	laws tested	distinct rank spectra
$n = 4$	$16^4 = 65,536$	7,140	23
$n = 5$	$32^4 = 1,048,576$	7,140	23

Every four-leaf balanced identity was cold on ranks 0 and 1. This is forced: with fewer than two independent labels, all nontrivial swaps and rotations collapse.

At rank 2, the census found only three local heat levels per two-plane:

$$\boxed{0, 96, 120.}$$

Thus rank-two behavior is controlled by commutativity-scar arithmetic.

At rank 4, the spectra were highly quantized. For $n = 4$ the possible rank-four failure fractions were

$$\boxed{0, \frac{4}{15}, \frac{13}{30}, \frac{1}{2}, \frac{17}{30}, \frac{11}{15}, 1.}$$

For $n = 5$ they were

$$\boxed{0, \frac{12}{31}, \frac{29}{62}, \frac{1}{2}, \frac{33}{62}, \frac{19}{31}, 1.}$$

The values pair symmetrically around $1/2$, with two exceptional cold/hot extremes 0 and 1. The non-extreme classes drift toward $1/2$ as dimension increases, consistent with high-rank sign balancing plus low-rank boundary scars.

25.7 Representative rank spectra at $n = 4$

The following examples show how different balanced laws occupy different spectral positions. The entries are counts by rank $(0, 1, 2, 3, 4)$.

representative law	rank-stratified failure signature at $n = 4$
$x(y(zw)) = x((wz)y)$	(0, 0, 3360, 10080, 0)
$x(y(zw)) = x((yw)z)$	(0, 0, 3360, 15456, 5376)
$x(y(zw)) = ((xz)(yw))$	(0, 0, 3360, 19488, 8736)
$x(y(zw)) = ((xw)(yz))$	(0, 0, 3360, 13440, 11424)
$x(y(zw)) = x((yz)w)$	(0, 0, 0, 14784, 14784)
$x(y(zw)) = x(y(wz))$	(0, 0, 3360, 30240, 20160)

This is evidence that four-variable balanced laws form a small but nontrivial code. They are not random identities. They are composite instruments assembled from K and Φ .

25.8 Search result: no single four-leaf identity isolates Δ_4

A natural target was a single balanced four-variable identity whose defect is exactly the rank-four detector

$$\Delta_4(x, y, z, w).$$

The exhaustive four-leaf search found no such identity at $n = 4$, and the same negative result persisted at $n = 5$ for the 7140 tested identities.

This is informative. Round 4 built a rank-four detector from a tetrahedral curl, i.e. a packet of associator measurements. The present search suggests that rank-four independence is not exposed by one ordinary four-leaf monomial identity. It appears to require a loop/packet measurement rather than a single equation.

25.9 Applications

1. Balanced-law compiler. Given a candidate law, a program can now perform the following pipeline:

parse $\rightarrow$ label-balance check $\rightarrow$ rewrite decomposition $\rightarrow$ defect prediction $\rightarrow$ targeted tests.

This avoids brute-force enumeration over all inputs whenever a symbolic rewrite path is available.

2. Compiler and optimizer verification. Program transformations often combine reordering and regrouping. The law calculus says to instrument these separately: swaps produce K -type defects, reassociations produce Φ -type defects, and larger rewrite loops must satisfy pentagon/hexagon checks.

3. AI/tool-chain diagnostics. In tool-using AI systems, a swap corresponds to changing the order of two operations; a rotation corresponds to changing grouping or nesting. The mixed hexagon law gives a concrete test of whether order effects and grouping effects are coherent with one another.

4. Discovery-machine design. A discovery engine should search the rewrite complex, not just the space of finished equations. The elementary costs K and Φ are reusable features. New laws can be generated by composing paths and then filtering for nontrivial spectra, amplitude signatures, or packet checks.

5. Rank-four diagnostics. The negative result for single four-leaf Δ_4 detectors suggests a design principle: higher-rank independence may require closed-loop packet instruments, not individual identities. That points back to the tetrahedral curl as the right rank-four tool.

25.10 Machine checks

The following checks were performed using the recovered Cayley-Dickson sign recursion.

- The mixed hexagon law was exhaustively checked for all triples through $n = 5$ and randomly checked for $n = 6, 7, 8$.
- The path-integral formula was tested on all 7140 four-leaf balanced identities by direct phase evaluation at $n = 4$ and $n = 5$.
- The four-leaf rank-spectrum census found 23 signatures at both $n = 4$ and $n = 5$.
- No tested four-leaf identity had defect exactly equal to Δ_4 at $n = 4$ or $n = 5$.

25.11 Round 13 synthesis

The most useful new equations are

$$\text{Def}_{T=S} = \sum_{e \in \gamma \cap \text{swap}} K(a_e, b_e) + \sum_{e \in \gamma \cap \text{rotate}} \Phi(a_e, b_e, c_e),$$

$$K(x, y + z) = \Phi(x, y, z) + K(x, y) + \Phi(y, x, z) + K(x, z) + \Phi(y, z, x).$$

The theory direction is now

$$\text{law-spectrum spectroscopy} \rightarrow \text{label-phase spectroscopy} \rightarrow \text{rewrite-holonomy calculus}.$$

The key conceptual gain is that balanced algebraic laws are not isolated objects. They are path integrals over a syntax geometry whose elementary curvatures are noncommutativity and nonassociativity.

26 Round 14: Elemental Law Thermography and Slot-Resolved Heat

26.1 Purpose of this continuation

Rounds 8–13 moved from associator-only study to a broader law spectrum: commutativity, distributivity skeletons, Lie/Malcev failures, polarized alternativity, Bol-Moufang probes, medial/interchange laws, label-phase separation, and rewrite holonomy. This round asks a different diagnostic question:

Which individual basis labels are hot, and in which syntactic slots?

The answer is a new layer of law spectroscopy: *elemental thermography*. A law can be globally half-hot while still having highly structured element-by-element and slot-by-slot heat. This matters for black-box diagnosis, because a scalar, a normal basis unit, and the newest Cayley-Dickson generator do not leave the same thermal fingerprint.

Throughout this section let $X = 2^n$, and let e_n denote the newest top-bit basis label, i.e. the vector 2^{n-1} in integer encoding. The scalar basis label is 0.

26.2 Fixed-label defect counts

For a law L of arity m , define the fixed-slot heat of a label a in slot i by

$$H_{L,i}^{(n)}(a) = \#\{(x_1, \dots, x_{m-1}) : L \text{ fails when } a \text{ is inserted in slot } i\}.$$

This is a refinement of ordinary temperature. Ordinary temperature sums over all possible labels in every slot; fixed-slot heat asks which labels are responsible.

26.3 Commutativity thermography

The commutativity defect is

$$K_n(x, y) = f_n(x, y) + f_n(y, x) = \Delta_2(x, y).$$

Therefore, for either slot,

$$H_K^{(n)}(a) = \begin{cases} 0, & a = 0, \\ X - 2, & a \neq 0. \end{cases}$$

Commutativity has a simple thermogram: the scalar is cold and every nonzero basis label is equally hot.

26.4 Associator thermography: the newest generator is an apex scar

Let

$$\Phi_n(x, y, z) = \text{Def}((xy)z = x(yz)).$$

For any of the three slots, the fixed-label associator heat is

$$H_{\Phi}^{(n)}(a) = \begin{cases} 0, & a = 0, \\ \frac{X^2}{2} - 8, & a \neq 0, a \neq e_n, \\ (X - 2)(X - 4), & a = e_n. \end{cases}$$

Thus the basis nucleus is visible immediately:

$$\{a : H_{\Phi}^{(n)}(a) = 0\} = \{0\} \quad (n \geq 3).$$

The exceptional finding is that the newest Cayley-Dickson generator is not merely another nonzero basis label. Its fixed-slot associator heat is asymptotically twice the heat of every other nonzero basis label:

$$\boxed{\frac{H_{\Phi}^{(n)}(e_n)}{H_{\Phi}^{(n)}(a)} = \frac{2(X-2)}{X+4} \longrightarrow 2 \quad (a \neq 0, e_n).}$$

So global associator heat hides a one-point thermal scar. The newest generator is an *apex scar*: it is singled out by how much nonassociativity appears when it is held fixed.

Consistency check with the global count. The positive-label associator failure count from the main report is

$$B_n = (X-2)(X-4)(X/2+3).$$

The fixed-label law sums to it exactly:

$$0 + (X-2) \left(\frac{X^2}{2} - 8 \right) + (X-2)(X-4) = B_n.$$

26.5 Medial thermography: slot asymmetry appears

Now consider the medial/interchange law

$$E(x, y, z, w) = \text{Def}((xy)(zw) = (xz)(yw)).$$

Round 11 found the global count

$$N_E(n) = \frac{1}{2}X^4 + 2X^2 - 24X + 32.$$

Fixed-slot heat reveals a sharper fact: medial heat is not slot-symmetric. Slots 1 and 4 are outer slots; slots 2 and 3 are inner slots.

For an outer slot,

$$\boxed{H_{E,\text{outer}}^{(n)}(a) = \begin{cases} \frac{X^3}{2} + 4X^2 - 26X + 32, & a = 0, \\ \frac{X^3}{2} - 2X, & a \neq 0. \end{cases}}$$

For an inner slot,

$$\boxed{H_{E,\text{inner}}^{(n)}(a) = \begin{cases} B_n = (X-2)(X-4)(X/2+3), & a = 0, \\ \frac{X^3}{2} + 2X - 8, & a \neq 0. \end{cases}}$$

These formulas also sum to the global medial count:

$$H_{E,\text{outer}}(0) + (X-1)H_{E,\text{outer}}(a \neq 0) = N_E(n),$$

$$H_{E,\text{inner}}(0) + (X-1)H_{E,\text{inner}}(a \neq 0) = N_E(n).$$

26.6 Exceptional interpretation: identity can be cold in one law and hot in another

The scalar label 0 is cold for commutativity and associativity. But in the outer slots of the medial law, it is hotter than any nonzero label:

$$H_{E,\text{outer}}(0) - H_{E,\text{outer}}(a \neq 0) = 4X^2 - 24X + 32.$$

In inner slots, the scalar is cooler than a nonzero label:

$$H_{E,\text{inner}}(a \neq 0) - H_{E,\text{inner}}(0) = 16X - 32.$$

This gives a new warning for law discovery:

An element is not absolutely hot or cold; it is hot/cold relative to a law and a slot.

The identity element can neutralize sequential grouping tests while intensifying certain interchange tests. This is easy to miss if one only measures global law temperature.

26.7 Thermographic fingerprint vector

Define a compact thermographic vector for a basis label a :

$$\Theta_n(a) = (H_K(a), H_\Phi(a), H_{E,\text{outer}}(a), H_{E,\text{inner}}(a)).$$

For $n \geq 4$, the three principal element types have distinct fingerprints:

$$\begin{aligned} \Theta_n(0) &= \left(0, 0, \frac{X^3}{2} + 4X^2 - 26X + 32, B_n\right), \\ \Theta_n(a) &= \left(X - 2, \frac{X^2}{2} - 8, \frac{X^3}{2} - 2X, \frac{X^3}{2} + 2X - 8\right) \quad (a \neq 0, e_n), \\ \Theta_n(e_n) &= \left(X - 2, (X - 2)(X - 4), \frac{X^3}{2} - 2X, \frac{X^3}{2} + 2X - 8\right). \end{aligned}$$

Thus a small set of law tests can identify:

1. the scalar label;
2. the newest Cayley-Dickson generator;
3. the outer versus inner positions of the medial/interchange law;
4. corruption or non-Cayley-Dickson behavior if the fingerprint class is illegal.

26.8 Machine checks

The formulas were checked by direct finite enumeration from the recovered Cayley-Dickson sign recursion.

n	regular assoc.	apex assoc.	outer $a = 0$	outer $a \neq 0$	inner $a = 0$	inner $a \neq 0$
3	24	24	336	240	168	264
4	120	168	2688	2016	1848	2072
5	504	840	19680	16320	15960	16440
6	2040	3720	145824	130944	130200	131192

For $n = 3, 4, 5, 6$, all listed fixed-label formulas matched exhaustive enumeration. The commutativity thermogram was exhaustively checked through $n = 6$ as well. The medial tests for $n = 6$ used selected full fixed-label sweeps for the formula classes, each sweep covering $64^3 = 262,144$ triples per fixed label and slot.

26.9 Applications

1. Black-box coordinate calibration. A black-box multiplication table may not reveal which label is the scalar, which label is the newest generator, or which variables occupy outer versus inner slots in an implemented law. Thermographic fingerprints recover these roles by statistics alone.

2. Algebraic anomaly detection. If a finite basis-unit system claims to use the Cayley-Dickson sign convention but its labels do not fall into the three legal thermographic classes, then something is wrong: table corruption, changed convention, hidden state, projection collapse, or measurement error.

3. Compiler and optimizer safety. The identity/no-op element is safe for some rewrites and dangerous for others. In an interchange or batching transformation, placing an identity-like operation in an outer slot can increase law heat. Optimizers should therefore track not only which operations occur, but which syntactic slot each operation occupies.

4. AI/tool-composition diagnostics. In AI tool pipelines, a no-op, identity context, or reset step may be harmless inside a sequential composition but disruptive when placed at the boundary of a parallel/interchange pattern. Thermography suggests testing operations by fixed-slot insertion, not only by random global trials.

5. Discovery-machine feature selection. A discovery engine should not only rank laws by global temperature. It should compute thermograms. Laws with similar global heat may have very different elemental fingerprints, and those fingerprints expose hidden coordinates, privileged generators, and slot asymmetries.

26.10 Round 14 synthesis

The most useful new equations are

$$H_{\Phi}^{(n)}(a) = \begin{cases} 0, & a = 0, \\ X^2/2 - 8, & a \neq 0, e_n, \\ (X - 2)(X - 4), & a = e_n, \end{cases}$$

$$H_{E,\text{outer}}^{(n)}(0) = X^3/2 + 4X^2 - 26X + 32, \quad H_{E,\text{outer}}^{(n)}(a \neq 0) = X^3/2 - 2X,$$

$$H_{E,\text{inner}}^{(n)}(0) = B_n, \quad H_{E,\text{inner}}^{(n)}(a \neq 0) = X^3/2 + 2X - 8.$$

The theory direction is now

$$\text{law-spectrum spectroscopy} \rightarrow \text{rewrite holonomy} \rightarrow \text{elemental thermography}.$$

The key conceptual gain is that law heat is not only a property of an identity. It is also a property of an element, a slot, and a syntax pattern.

Round 15 Appendix: Thermographic Feature Selection and Outer-Pair Barcodes

Associator Field Theory continuation

April 30, 2026

Purpose of this round

Round 14 introduced elemental thermography: instead of ranking a law only by global temperature, one fixes an element in a slot and counts how much heat remains. The observation suggests a sharper discovery-machine principle: rank candidate laws by their *thermographic fingerprints*, not merely by their global failure rate.

This round develops that idea into a feature-selection calculus. The main new result is that one-point thermography is often too coarse, while *outer-pair associator thermography* explodes into a high-resolution barcode with 2^{n-3} possible levels.

Thermographic features

Let L be a law of arity m over a finite operation system A , and let $S \subseteq \{1, \dots, m\}$ be a subset of slots. Define the thermographic feature

$$\Theta_{L,S}(a_S) = \#\{b_{\bar{S}} \in A^{m-|S|} : L(a_S, b_{\bar{S}}) \text{ fails}\}.$$

Thus global temperature is the case $S = \emptyset$, elemental thermography is $|S| = 1$, and pair/packet thermography is $|S| \geq 2$.

A discovery-machine scoring rule should reward a feature for creating many stable distinguishable classes. One simple score is the thermographic entropy

$$\mathcal{I}(L, S) = H(\Theta_{L,S}),$$

where the entropy is taken over uniformly sampled fixed slot assignments a_S . A cost-normalized version is

$$\text{TIG}(L, S) = \frac{H(\Theta_{L,S})}{2^{n(m-|S|)}}.$$

The denominator is the number of oracle tests needed to exhaustively evaluate one fixed assignment in a 2^n -element basis skeleton.

Discovery 1: marginal thermography is a blind spot

In the Round 14 data, one-slot associator thermography separates only three element classes:

$$0, \quad e_n, \quad \text{all other nonzero labels.}$$

A new enumeration of all 7140 balanced four-leaf monomial identities using each of x, y, z, w exactly once found the same limitation at $n = 4$: no single four-leaf law separated the 16 labels into more than these three marginal classes. The best laws had entropy about 0.669 bits, because they singled out 0 and the newest generator e_n and collapsed the remaining 14 elements.

This is a negative but useful discovery:

one-slot thermography detects apex scars, but it is mostly blind to lower internal structure.

Discovery 2: outer-pair associator thermography gives a barcode

Define the outer-pair associator heat

$$H_n(x, z) = \#\{y \in V_n : \Phi_n(x, y, z) = 1\}.$$

Here x and z are the two outer variables of the associator

$$(e_x e_y) e_z = e_x (e_y e_z).$$

For dependent pairs, $H_n(x, z) = 0$. For independent pairs, exhaustive computation through $n = 8$ and a recurrence proof from the rank-birth law give:

$$H_n(x, z) = 4 + 8q_n(x, z), \quad 0 \leq q_n(x, z) < 2^{n-3}.$$

Every level occurs. Thus the outer pair carries a barcode

$$q_n(x, z) = \frac{H_n(x, z) - 4}{8} \in \{0, 1, \dots, 2^{n-3} - 1\}.$$

This is a high-resolution thermographic coordinate. It carries almost $n - 3$ bits of information.

The barcode recurrence

Let $x = (x', X)$ and $z = (z', Z)$ with newest top bits $X, Z \in \mathbb{F}_2$. For independent x, z :

$$H_n(x, z) = \begin{cases} 2H_{n-1}(x', z') + 4, & x', z' \text{ independent and } X = Z, \\ 2^n - 4 - 2H_{n-1}(x', z'), & x', z' \text{ independent and } X \neq Z, \\ 2^n - 4, & x', z' \text{ dependent.} \end{cases}$$

Equivalently, if $q_n = (H_n - 4)/8$ and $M_n = 2^{n-3}$, then

$$q_n = \begin{cases} 2q_{n-1} + 1, & x', z' \text{ independent and } X = Z, \\ M_n - 2 - 2q_{n-1}, & x', z' \text{ independent and } X \neq Z, \\ M_n - 1, & x', z' \text{ dependent.} \end{cases}$$

The initial value is $H_3 = 4$, hence $q_3 = 0$ for every independent pair in V_3 .

Distribution recurrence

Let $C_n(q)$ be the number of ordered independent pairs (x, z) with barcode $q_n(x, z) = q$. Then $C_3(0) = 42$, and for $n \geq 4$:

$$\begin{aligned} C_n(2q + 1) &+= 2C_{n-1}(q), \\ C_n(M_n - 2 - 2q) &+= 2C_{n-1}(q), \\ C_n(M_n - 1) &+= 6(2^{n-1} - 1). \end{aligned}$$

The final term is the number of pairs whose lower projections are dependent but whose top bits make them independent.

Proof sketch

Start from the triangular rank recurrence for the Cayley-Dickson associator. For fixed outer pair (x, z) , sum the recurrence over the middle variable $y = (y', Y)$. If x', z' are independent, then lower triples with y' outside the lower line contribute the lower associator count, but the newest top-bit parity either preserves or complements it depending on $X + Z$. The four lower-line choices supply the constant correction. This yields

$$\begin{aligned} H_n &= 2H_{n-1} + 4 \quad (X = Z), \\ H_n &= 2^n - 4 - 2H_{n-1} \quad (X \neq Z). \end{aligned}$$

If x', z' are dependent, the pair is born only at the newest bit and all admissible middle choices except the four line elements fail, giving $H_n = 2^n - 4$. The quantization and distribution recurrence follow by induction.

Experimental data

n	one-slot classes	one-slot entropy	adjacent-pair classes	adjacent entropy	outer-pair classes	outer entropy
3	2	0.544	1	0.000	1	0.000
4	3	0.669	2	0.722	2	0.971
5	3	0.400	2	0.459	4	1.933
6	3	0.232	2	0.276	8	2.903
7	3	0.132	2	0.161	16	3.882
8	3	0.074	2	0.092	32	4.870

The outer-pair entropy is nearly maximal: it approaches $\log_2(2^{n-3}) = n - 3$ bits. By contrast, one-slot and adjacent-pair thermography become less informative as n grows.

n	ordered independent outer-pair heat distribution
3	4 : 42
4	4 : 84, 12 : 126
5	4 : 252, 12 : 168, 20 : 168, 28 : 342
6	4 : 684, 12 : 504, 20 : 336, 28 : 336, 36 : 336, 44 : 336, 52 : 504, 60 : 870
7	16 levels 4, 12, ..., 124; entropy 3.882 bits
8	32 levels 4, 12, ..., 252; entropy 4.870 bits

Feature-selection principle

The main design law is:

Do not rank laws only by global temperature; rank law-slot packets by thermographic information.

More concretely:

A weak global law can be a strong feature if its conditional heat has high entropy.

The associator illustrates this perfectly. Its one-element marginal heat is low-information at large n , but its outer-pair heat has exponentially many quantized levels. A discovery engine that only scores global temperature or one-slot heat would miss the barcode.

Applications

1. Discovery-machine feature selection

For automated law discovery, each candidate should be evaluated as a feature family

$$(L, S, \Theta_{L,S}, \mathcal{I}, \text{TIG}),$$

not as a single true/false law. The engine should prioritize laws that create stable, high-entropy, low-cost partitions of the input space.

2. Black-box algebra tomography

Outer-pair associator heat gives a coordinate-like barcode using only multiplication, equality, and counting. It can fingerprint a Cayley-Dickson-like basis skeleton even when the actual sign convention or coordinate labels are unknown.

3. Error detection

The legal outer-pair heat levels are exactly

$$4, 12, 20, \dots, 2^n - 4.$$

A measured independent outer pair with heat outside this arithmetic progression cannot come from the coherent Cayley-Dickson rank-birth model. It indicates table corruption, hidden state, projection collapse, nonstationarity, or a different twist.

4. AI and tool composition

In a tool pipeline, one-slot tests correspond to inserting a fixed operation in one position. Outer-pair tests correspond to fixing two boundary operations and sweeping a middle operation. This can reveal path-sensitive transport errors invisible to single-operation thermography.

5. Compiler and optimizer safety

Some rewrite failures appear only when two boundary operands are held fixed and a middle operand varies. This matters for reassociation, batching, and pipeline fusion: a transformation may look safe in marginal tests but produce a high-resolution outer-pair barcode of failure.

Round 15 summary

The new theory layer is

law-spectrum spectroscopy $\rightarrow$ thermographic feature selection $\rightarrow$ barcode tomography.

The best new equations are

$$\Theta_{L,S}(a_S) = \#\{b_{\bar{S}} : L(a_S, b_{\bar{S}}) \text{ fails}\},$$

$$H_n(x, z) = 4 + 8q_n(x, z), \quad 0 \leq q_n < 2^{n-3},$$

$$q_n = \begin{cases} 2q_{n-1} + 1, & X = Z, \\ 2^{n-3} - 2 - 2q_{n-1}, & X \neq Z, \\ 2^{n-3} - 1, & \text{lower projections dependent.} \end{cases}$$

The exceptional discovery is that associator heat has a hidden high-resolution barcode. Global temperature and one-slot thermography miss it; outer-pair thermography reveals it.

Round 16: Barcode Feature Selection and Thermographic Information Gain

This appendix continues the thermographic feature-selection program. Round 15 showed that global temperature and one-slot heat can miss structure, while fixed outer-pair associator heat creates a high-resolution barcode. Round 16 turns that observation into an explicit feature-selection calculus for discovery machines.

Status. The recurrence laws below were derived from the rank-birth/local-potential model and checked by exact recurrence enumeration. The corruption results are synthetic bit-noise tests on the associator field, included only as a diagnostic stress test rather than a theorem about table-entry noise.

16.1 Thermographic information gain

For a candidate law L with slots $1, \dots, m$ and a fixed slot subset S , define the thermographic feature

$$\text{Theta}_{\{L,S\}}(a_S) = \#\{b_{\{\text{not } S\}}: L(a_S, b_{\{\text{not } S\}}) \text{ fails}\}.$$

Global temperature is only the mean of this random variable. Thermographic feature selection uses the whole distribution. The proposed score is:

$$\text{TIG}(L,S) = H(\text{Theta}_{\{L,S\}}) / \text{Cost}(L,S), \text{ with optional penalties for label imbalance, high query cost, or low anomaly sensitivity.}$$

A law-feature pair is valuable when it has high conditional entropy, a legal-value spectrum that is hard to fake accidentally, and an interpretable inverse map. This distinguishes weak global laws from strong diagnostic instruments.

16.2 Exceptional finding: the outer-pair heat is a reversible barcode automaton

Fix an independent outer pair (x,z) and count associator failures while sweeping the middle slot:

$$H_n(x,z) = \#\{y \text{ in } V_n : \Phi_n(x,y,z)=1\}, \quad q_n(x,z)=(H_n(x,z)-4)/8.$$

Round 15 found that q_n takes every value from 0 to $2^{n-3}-1$. The stronger Round 16 observation is that q_n is generated by an almost invertible barcode automaton. Write $x=(x',X)$ and $z=(z',Z)$. If the lower projections are independent, then

$$q_n = 2 q_{n-1} + 1 \text{ if } X=Z; \quad q_n = 2^{n-3} - 2q_{n-1} \text{ if } X \neq Z.$$

If the lower projections are dependent but the full pair is independent, then

$$q_n = 2^{n-3} - 1.$$

Thus the newest output parity of the barcode reveals the newest top-bit relation: even q means $X \neq Z$; odd q means $X=Z$ unless the barcode is in the saturated all-ones birth state. Recursing backward decodes a suffix of the pair-difference word. Conditional on the pair being born at the earliest possible layer, the map from suffix difference words to q is a bijection.

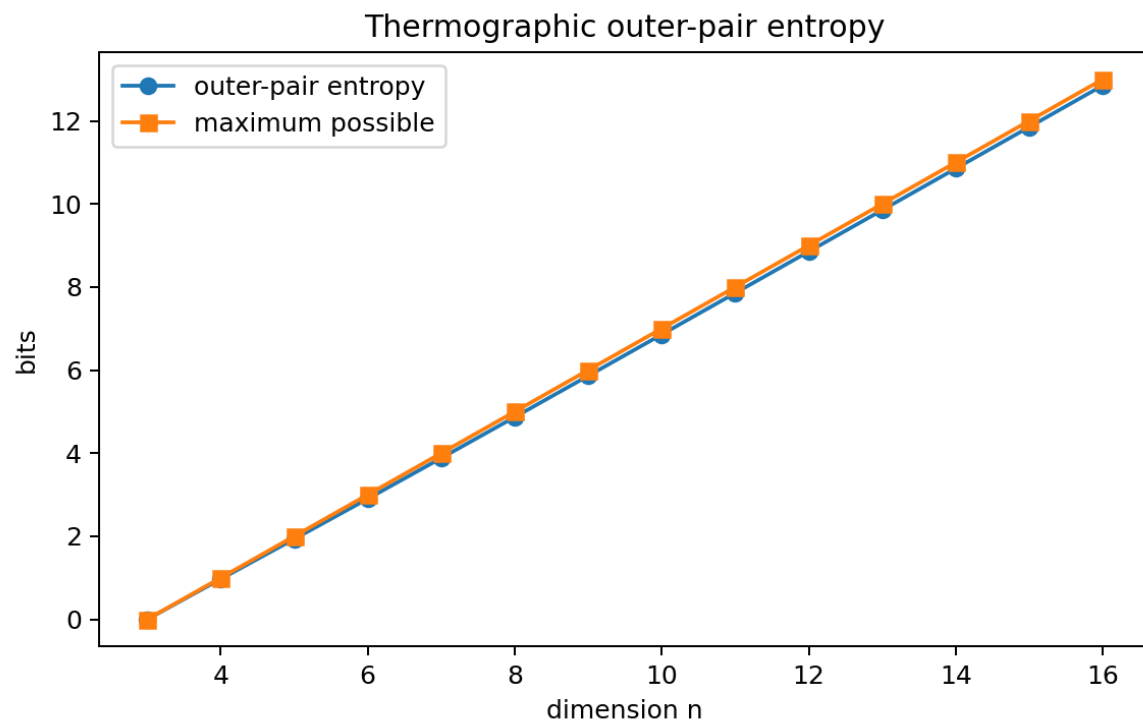


Figure 16.1. The outer-pair entropy is almost maximal. The limiting entropy deficit appears to approach about 0.1471 bits, meaning the barcode extracts nearly $n-3$ bits from a single conditional heat profile.

16.3 Exact multiplicity recurrence

Let $M_n(q)$ be the number of independent ordered outer pairs with barcode q in dimension n . The recurrence is:

$$M_{n+1}(2q+1) = 2 M_n(q), M_{n+1}(2^{n-2}-2-2q) = 2 M_n(q), M_{n+1}(2^{n-2}-1) = 6(2^{n-1}).$$

The final injection term counts pairs whose lower projections are dependent but whose new top-bit lifts become independent. This is why the barcode distribution is almost uniform but carries a persistent boundary scar.

n	classes	entropy bits	max bits	gap bits
3	1	-0.000000	0.0	0.000000
4	2	0.970951	1.0	0.029049
5	4	1.933131	2.0	0.066869
6	8	2.902899	3.0	0.097101
7	16	3.882456	4.0	0.117544
8	32	4.869761	5.0	0.130239
9	64	5.862277	6.0	0.137723
10	128	6.858016	7.0	0.141984
11	256	7.855648	8.0	0.144352
12	512	8.854356	9.0	0.145644

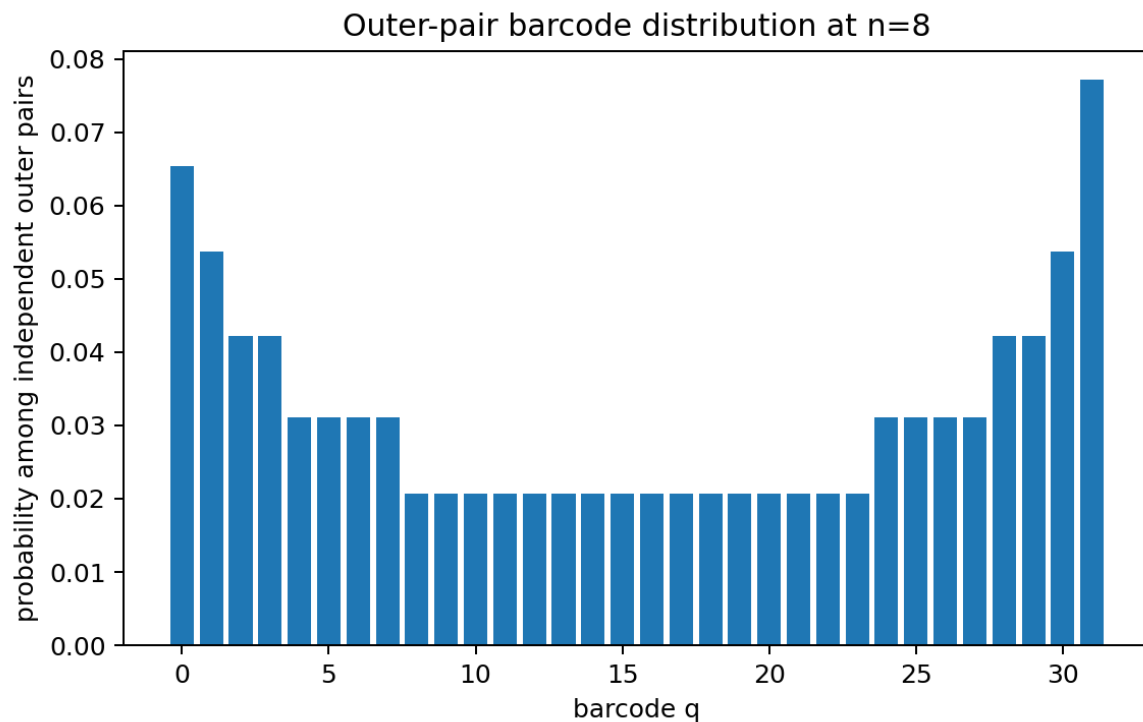


Figure 16.2. Barcode multiplicity at $n=8$. The support is full, but the boundary levels are scarred by late pair birth. This is exactly the kind of structure a scalar temperature would erase.

16.4 Legal-value anomaly detector

For independent outer pairs, the legal heat levels are not arbitrary. They are exactly:

$H_n(x,z) \in \{4, 12, 20, \dots, 2^{n-4}\}.$

This gives a checksum. If an observed independent outer-pair heat is not 4 mod 8, or lies outside the legal interval, the measured system is not a clean Cayley-Dickson rank-birth sign field. Possible causes include table corruption, hidden-state projection, measurement error, nonstationarity, or a different cocycle.

Synthetic stress test. In dimension $n=6$, random independent-triple associator-bit flips with probability epsilon cause the following fraction of independent outer-pair barcodes to become illegal:

bit flip rate epsilon	illegal barcode fraction
0.0005	0.0302
0.0010	0.0607
0.0020	0.1190
0.0050	0.2660
0.0100	0.4443

The illegal-barcode fraction is much larger than the raw bit-flip rate because each fixed outer-pair barcode aggregates an entire middle-slot sweep. Aggregation converts tiny local noise into visible thermographic checksum failure.

16.5 Discovery-machine feature selector

A practical discovery machine should not rank candidate laws by global temperature alone. It should generate candidate law-feature pairs (L,S), reject unbalanced label-hot laws unless label mismatch is the desired signal, estimate thermographic entropy, and keep features with high entropy, legal spectra, and interpretable inverses.

Feature card = (law L, fixed slots S, global temperature, thermographic entropy, legal support, inverse map, anomaly sensitivity, test cost).

For the Cayley-Dickson basis skeleton, the outer-pair associator feature receives a high score: it has full support size 2^{n-3} , near-maximal entropy, an explicit recurrence, a partial inverse, and a simple legality test. One-slot associator heat receives a low score in high dimension despite being mathematically meaningful, because its entropy collapses.

16.6 Applications

Automated law discovery. Instead of asking which identities are hot, the system asks which law-conditioned heat profiles produce high-information fingerprints. This avoids wasting search on laws whose only signal is label imbalance or scalar noise.

Black-box algebra tomography. The outer-pair barcode can infer coordinate-order information from multiplication/equality tests alone. It behaves like a partial decoder for the post-birth difference suffix of two basis labels.

Error-correcting algebra. Legal heat levels form a checksum code. A single observed level outside 4 mod 8 is enough to flag an out-of-model packet.

AI/tool-composition diagnostics. Fix two boundary operations and sweep a middle operation. If the resulting failure-count barcode has high entropy, the workflow has structured boundary-conditioned path sensitivity. If the barcode violates a previously learned legal spectrum, hidden state or context leakage is likely.

Compiler and optimizer safety. Some rewrite rules look safe on average but fail for particular boundary operands. Thermographic pair scans identify dangerous boundary contexts before global optimization rewrites are applied.

16.7 Round 16 summary

$TIG(L,S)=H(\Theta_{\{L,S\}})/Cost(L,S).$

$H_n(x,z)=4+8q_n(x,z),$ with q_n in $\{0,...,2^{n-3}-1\}.$

$q_n=2q_{n-1}+1$ for equal newest bits; $q_n=2^{n-3}-2-2q_{n-1}$ for unequal newest bits; $q_n=2^{n-3}-1$ at late pair birth.

Legal independent outer-pair heat: H_n in $\{4,12,20,...,2^{n-4}\}.$

The conceptual advance is: thermographic fingerprints are not just descriptive statistics. Some are decodable barcodes. A discovery machine should search for laws whose conditional heat profiles are high-entropy, legality-constrained, and invertible enough to reveal hidden structure.

Round 17 Appendix: Line-Barcode Coherence and Thermographic Checksums

Continuation from Round 16. Status: theorem candidates with exact recurrence proofs and finite exhaustive checks through the dimensions stated below. This appendix extends outer-pair barcode tomography from individual pointed pairs to whole projective lines.

1. Motivation

Round 16 showed that the fixed outer-pair associator heat $H_n(x,z) = \#\{y: \Phi_n(x,y,z)=1\}$ is a high-entropy barcode. This round asks the next natural question: if x and z span a two-plane L , what relations hold among the three barcodes attached to the three nonzero points of L ?

2. Pointed-line barcode

For an independent outer pair (x,z) , set $t=x+z$ and $L=\langle x,z \rangle$. The heat depends only on the pointed line (L,t) , not on the order of x and z . Define

$$H_n(L,t) = \#\{y \text{ in } V_n : \Phi_n(x,y,z)=1\}, \text{ where } x+z=t,$$

$$q_n(L,t) = (H_n(L,t)-4)/8, \quad 0 \leq q_n(L,t) < 2^{n-3}.$$

Equivalently, $H_n(L,t) = 4 + 8q_n(L,t)$. The integer q_n is the barcode; its binary expansion has length $n-3$.

3. Exceptional discovery: the line-barcode checksum

Let L be a two-plane and let a,b,c be its three nonzero points, so $a+b+c=0$. Let $\Omega_n = 2^{n-3}-1$, the all-ones barcode of length $n-3$. Then the three pointed-line barcodes obey the exact checksum

$$q_n(L,a) \text{ XOR } q_n(L,b) \text{ XOR } q_n(L,c) = \Omega_n.$$

This is a new coherence law for thermographic fingerprints. Individual outer-pair barcodes may look like arbitrary high-entropy integers, but every projective line carries a parity checksum. A single corrupted pointed-pair heat usually breaks the checksum immediately.

4. Bitwise form and decoding

Write the barcode bits as $b_i(L,t)$, $i=4,\dots,n$. The recurrence from Round 16 implies the suffix-parity rule

$$b_i(L,t) = 1 + t_i + t_{i+1} + \dots + t_n \text{ after the pair has reached rank 2, with the pre-birth prefix constant.}$$

Consequently, along a fixed line L , the linear t_i terms cancel when summed over the three nonzero points, leaving one constant bit in each coordinate. That gives the checksum bit by bit.

The inverse derivative is also useful. If b_i are the barcode bits, then for post-birth coordinates

$$t_i = b_i + b_{i+1} \text{ for } i < n, \text{ and } t_n = 1 + b_n.$$

Thus the barcode is not merely descriptive. It is a compressed suffix-parity transform of the omitted point $t=x+z$.

5. Universal line-type law

Let $L=n-3$ be the barcode length. The unordered triple of barcode values on any projective line is a multiset $\{a,b,c\}$ satisfying

$$a \text{ XOR } b \text{ XOR } c = 2^{L-1}.$$

Conversely, every such multiset occurs. Therefore the number of possible line-barcode types is

$$T_L = 2^L + ((2^{L-1})(2^{L-2}))/6.$$

The terms correspond to: one all-equal type $\{\Omega, \Omega, \Omega\}$; 2^{L-1} two-equal types $\{a, a, \Omega\}$; and $((2^{L-1})(2^{L-2}))/6$ all-distinct types.

6. Exact flag enumerator recurrence

Let $C_n(q)$ be the number of ordered independent outer pairs with barcode q . Then

$$C_3(0)=42,$$

$$C_n(2q+1) += 2C_{n-1}(q),$$

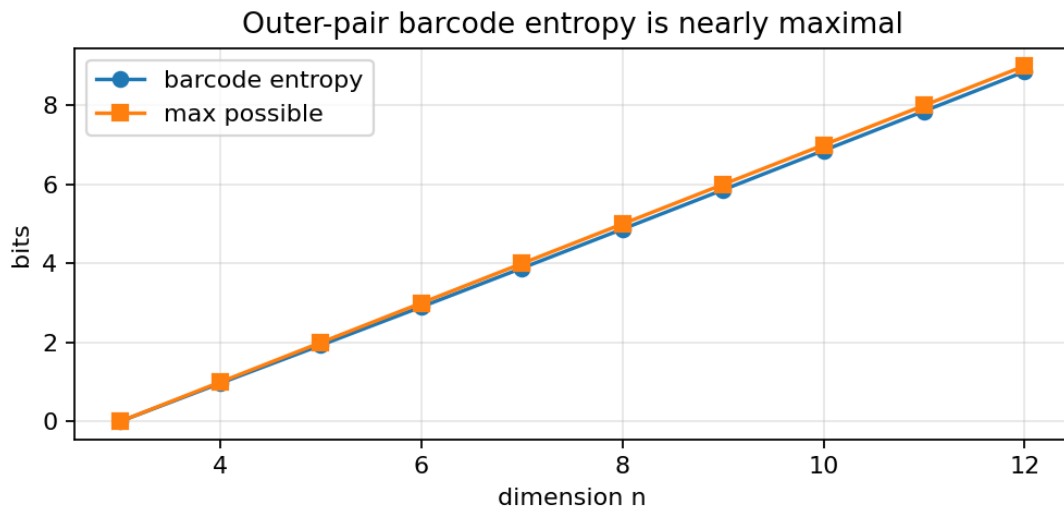
$$C_n(2^{n-3}-2-2q) += 2C_{n-1}(q),$$

$$C_n(2^{n-3}-1) += 6(2^{n-1}-1).$$

Dividing by 2 gives pointed-line flag counts, because two ordered outer pairs represent the same pointed line (L, t) . This recurrence is a compact exact distributional law for barcode thermography.

7. Evidence table

n	barcode length	classes	entropy bits	gap to max	line types
3	0	1	-0.000000	0.000000	1
4	1	2	0.970951	0.029049	2
5	2	4	1.933131	0.066869	5
6	3	8	2.902899	0.097101	15
7	4	16	3.882456	0.117544	51
8	5	32	4.869761	0.130239	187
9	6	64	5.862277	0.137723	715
10	7	128	6.858016	0.141984	2795
11	8	256	7.855648	0.144352	11051
12	9	512	8.854356	0.145644	43947



The entropy remains within about 0.146 bits of the maximum by $n=12$, while the line-type count grows cubically in the number of barcode levels. This makes line-barcode thermography a high-information feature family.

8. Finite checks

The checksum $q(a) \text{ XOR } q(b) \text{ XOR } q(c) = \Omega$ was exhaustively checked by direct Cayley-Dickson associator enumeration through $n=7$ and by recurrence enumeration through $n=12$. The line-type count matched T_L for all tested n . No illegal barcode level appeared; independent pointed-line heat always lay in $\{4, 12, 20, \dots, 2^{n-4}\}$.

9. Applications

Discovery-machine feature selection. A candidate law-feature should be ranked by global temperature, conditional entropy, legality constraints, inverse decodability, and packet checksums. The outer-pair barcode scores high on all five. The line checksum adds a cheap consistency test.

Error-correcting algebra. Given a line L , measure the three pointed-pair heats. Convert them to q -values. If their XOR is not Ω , then at least one measurement, table entry, or model assumption is wrong. This is a parity check for thermographic data.

Black-box coordinate tomography. The inverse derivative $t_i = b_i + b_{(i+1)}$ recovers suffix information about the omitted point $t = x + z$. Repeating this over lines can reconstruct partial coordinate order without direct access to coordinates.

AI/tool-composition diagnostics. Treat x and z as boundary operations and sweep y as the middle operation. The heat barcode detects boundary-conditioned path sensitivity. The three-point line checksum detects inconsistent or nonstationary behavior across equivalent boundary decompositions.

Compiler and optimizer safety. In tensor or rewrite systems, boundary operands can hide high-entropy regrouping risk. Pair thermography and line checksums can identify which boundary pairs are unsafe for reassociation or batching.

10. Round 17 summary

The main new equations are:

$$H_n(L, t) = 4 + 8q_n(L, t),$$

$$q_n(L, a) \text{ XOR } q_n(L, b) \text{ XOR } q_n(L, c) = 2^{(n-3)} - 1,$$

$$T_L = 2^L + ((2^{L-1})(2^{L-2}))/6.$$

The new theory direction is: thermographic feature selection -> barcode tomography -> checksum-coded discovery. The conceptual gain is that high-entropy thermographic fingerprints are not free noise; they obey projective-line parity laws that can be used for decoding and error correction.

Round 18 Appendix: Birth-Truncated Barcode Calculus and Fano-Plane Thermography

Continuation from Round 17. Status: theorem candidates with exact recurrence proofs and finite exhaustive checks through dimension 7, plus random packet checks in dimensions 8, 10, and 12. This round turns the outer-pair barcode from a high-entropy statistic into an explicit transform with line and plane-level checksums.

1. Motivation

Round 17 showed that the outer-pair barcode $q_n(L,t)$ obeys a projective-line checksum. This round asks what the barcode actually encodes. The answer is stronger than expected: each bit is a birth-truncated suffix parity of the omitted point t . So the barcode is not only high entropy; it is partially invertible.

2. Exceptional discovery: the birth-truncated suffix transform

Let L be a projective line in V_n , let t be one of its three nonzero points, and let $\beta_2(L)$ be the first coordinate b for which the projection of L to V_b has rank 2. Write the barcode

$$q_n(L,t) = \sum_{m=4}^n b_m(L,t) 2^{n-m}.$$

Then the exact bit law is

$$b_m(L,t) = 1 + \sigma_{\max(m, \beta_2(L)+1)}(t), \quad 4 \leq m \leq n,$$

where $\sigma_r(t) = t_r + t_{r+1} + \dots + t_n$ over F_2 , and $\sigma_{n+1} = 0$. In words: before the line is born, the barcode has a transported constant prefix; after birth, it records suffix parities of the omitted point.

3. Inverse derivative law

For post-birth coordinates, the barcode can be differentiated to recover the omitted point. If $m > \beta_2(L)$, then

$$t_m = b_m + b_{m+1} \text{ for } m < n,$$

$$t_n = 1 + b_n.$$

Thus a heat profile $H_n(L,t) = 4 + 8q_n(L,t)$ can reveal suffix coordinates of t without direct coordinate access. This is the strongest reason so far to treat thermographic fingerprints as discovery-machine features rather than descriptive curiosities.

4. Line checksum as a corollary

If a, b, c are the three nonzero points on L , then $a+b+c=0$. For every bit m , the three birth-truncated suffix parities cancel and three constants remain. Therefore

$$b_m(L,a) + b_m(L,b) + b_m(L,c) = 1,$$

and hence

$$q_n(L,a) \text{ XOR } q_n(L,b) \text{ XOR } q_n(L,c) = 2^{n-3} - 1.$$

The Round 17 line checksum is therefore not accidental. It is the bitwise shadow of a suffix-parity transform.

5. New Fano-plane checksum

Let U be a 3-plane. It has seven projective lines. XOR the three pointed barcodes on every line and then XOR over all seven lines. Since seven is odd, every 3-plane obeys

$$\text{XOR}_{\{L \leq U, \dim L=2\}} \text{XOR}_{\{t \in L \setminus \{0\}\}} q_n(L,t) = 2^{n-3}-1.$$

Equivalently, the 21 pointed-line barcodes of a Fano plane have a global parity checksum. This gives a second packet-level error check above the single-line checksum.

6. Bit-slice active-line quantization

Fix a barcode coordinate m. A line L inside a 3-plane U is active at m if $\beta_2(L) < m$. The number of active lines is determined only by the rank of the projection of U to $V_{\{m-1\}}$:

rank 0 or 1 -> 0 active lines; rank 2 -> 4 active lines; rank 3 -> 7 active lines.

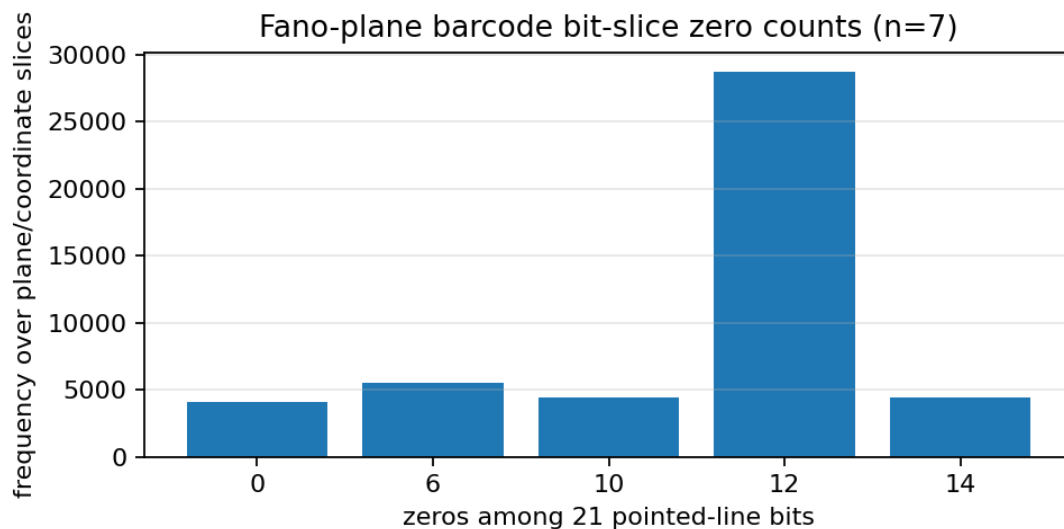
Thus each barcode bit-slice of a Fano plane has a quantized active-line skeleton. The actual zero count among the 21 pointed-line bits is then controlled by how the relevant suffix functional cuts those active lines. In exhaustive tests the possible zero counts were

0, 6, 10, 12, 14.

These five zero-count phases are a new thermographic fingerprint for 3-packets.

7. Exhaustive checks

n	3-planes	bad line checks	bad plane checks	bad bit laws	active-line counts	zero counts
4	15	0	0	0	{4: 7, 7: 8}	{0: 1, 6: 7, 12: 7}
5	155	0	0	0	{0: 7, 4: 119, 7: 184}	{0: 23, 6: 84, 10: 21, 12: 161, 14: 21}
6	1395	0	0	0	{0: 114, 4: 1359, 7: 2717, 10: 339, 6: 717, 10: 378, 12: 2373, 14: 378}	
7	11811	0	0	0	{0: 1237, 4: 13151, 7: 12831, 10: 107, 6: 5550, 10: 4419, 12: 28749, 14: 4419}	



The table shows zero failures for the line checksum, plane checksum, and bit law in exhaustive checks through n=7. The active-line count is always in {0,4,7}; the zero count is always in {0,6,10,12,14}.

8. Random higher-dimensional checks

n	random 3-planes	bad line checks	bad plane checks	bad bit laws	active counts seen	zero counts seen
8	250	0	0	0	[0, 4, 7]	[0, 6, 10, 12, 14]
10	250	0	0	0	[0, 4, 7]	[0, 6, 10, 12, 14]
12	250	0	0	0	[0, 4, 7]	[0, 6, 10, 12, 14]

Random packet checks in dimensions 8, 10, and 12 found no violations. The same five zero-count phases and three active-line phases appeared.

9. Applications

Discovery-machine feature selection. This round strengthens the feature-selection rule. Prefer law-features whose conditional heat is high-entropy, legal-valued, checksum-constrained, and at least partially invertible. The outer-pair barcode satisfies all four requirements.

Black-box coordinate tomography. If a system behaves like Cayley-Dickson sign curvature, pair thermography can recover suffix coordinates of omitted points after the birth coordinate. This gives a way to infer hidden coordinate order from multiplication/equality tests alone.

Error correction. A measurement packet now has nested checks: legal heat level, line XOR checksum, Fano-plane XOR checksum, active-line count in $\{0,4,7\}$, and bit-slice zero count in $\{0,6,10,12,14\}$. Violating any one flags corruption, hidden state, nonstationarity, or a different twist model.

AI/tool composition. Boundary-conditioned sweep tests can be made stronger by grouping boundary operations into line and Fano-plane packets. Coherent path sensitivity should obey the checksums; context drift should break them.

Compiler and optimizer safety. Reassociation risk can be localized not only to a pair of boundary operands, but to a whole packet of boundary decompositions. The Fano checksum is a cheap consistency check for regrouping-risk estimates.

10. Round 18 summary

The main equations are:

$$\begin{aligned}b_m(L,t) &= 1 + \sigma_{\{\max(m, \beta_2(L)+1)\}}(t), \\ t_m &= b_m + b_{m+1}, \quad m > \beta_2(L), \quad m < n; \quad t_n = 1 + b_n, \\ \text{XOR}_{\{L \leq U\}} \text{XOR}_{\{t \in L \setminus \{0\}\}} q_n(L,t) &= 2^{n-3} - 1, \\ \text{active-line count in a bit-slice} &\text{ is always } 0, 4, \text{ or } 7.\end{aligned}$$

The new theory direction is: barcode tomography -> thermographic sheaf codes -> coordinate recovery from law heat. The conceptual gain is that conditional law heat can become an invertible, packet-checkable measurement system.

Round 19 Appendix: Derivative Barcode Calculus and Thermographic Checksums

Status. Theorem-candidate layer supported by direct enumeration through $n = 7$ for all Fano planes and by derivation from the Round 18 birth-truncated barcode formula. The claims should still be formalized in a proof assistant before being treated as external mathematical literature.

19.1 Motivation: from high-entropy barcodes to readable codewords

Round 15-18 showed that the outer-pair associator heat produces a high-entropy barcode. For an independent outer pair (x,z) , with $t = x + z$ and line $L =$, define

$$\begin{aligned} H_n(L, t) &= \#\{ y \text{ in } V_n : \text{Phi}_n(x, y, z)=1 \}, \\ q_n(L, t) &= (H_n(L, t) - 4)/8. \end{aligned}$$

The barcode q_n has $n-3$ bits and satisfies line and Fano-plane checksums. This round asks a sharper question: what information is carried by the finite differences of the barcode bits?

19.2 Exceptional finding: the derivative is a birth-gated coordinate probe

Write the barcode in birth-order form

$$q_n(L, t) = \sum_{m=4}^n b_m(L, t) 2^{n-m}.$$

Define the barcode derivative slice

$$D_m(L, t) = b_m(L, t) + b_{m+1}(L, t) \pmod{2}, \quad 4 \leq m < n.$$

Let $\beta_2(L)$ be the first coordinate at which the line L becomes rank-2 visible. Then the derivative has the exact form

$$D_m(L, t) = 1_{\{\beta_2(L) < m\}} * t_m.$$

In words: after a line is born, the derivative of its heat barcode is exactly the m -th coordinate of the omitted point t . Before birth, the derivative is zero. Thus the thermographic barcode is not merely an entropy-rich fingerprint; its discrete derivative is a coordinate probe gated by rank birth.

19.3 Line derivative checksum

For every projective line L with nonzero points a, b, c , the derivative bits obey

$$D_m(L, a) + D_m(L, b) + D_m(L, c) = 0.$$

This follows because either the line has not yet been born, in which case all three derivative bits are zero, or the line has been born, in which case D_m is just the linear coordinate functional $t \rightarrow t_m$ on the three nonzero points of L , and $a+b+c = 0$.

This is a local parity check on the barcode derivative. A single flipped derivative bit on a line is immediately detectable.

19.4 Fano-plane derivative quantization

Let U be a 3-plane, i.e. a Fano plane on its seven nonzero points. For a coordinate m , define the active-line set

$$A_m(U) = \{ \text{lines } L \leq U : \beta_2(L) < m \}.$$

Let $r_{m-1} = \text{rank}(\pi_{m-1}(U))$, the rank of the projection of U onto the first $m-1$ coordinates. Then the number of active lines is forced:

$$\begin{aligned} |A_m(U)| &= 0 \text{ if } r_{m-1} \leq 1, \\ |A_m(U)| &= 4 \text{ if } r_{m-1} = 2, \end{aligned}$$

$$|A_m(U)| = 7 \text{ if } r_{\{m-1\}} = 3.$$

The total number of one-bits in the 21 pointed-line derivative slice is consequently quantized:

$$R_m(U) = \sum_{\{L \leq U\}} \sum_{\{t \text{ in } L - \{0\}\}} D_m(L, t) \text{ in } \{0, 6, 8, 12\}.$$

The refined cases are: if no lines are active, $R_m=0$. If all seven lines are active, $R_m=0$ when the coordinate functional t_m vanishes on U and $R_m=12$ otherwise. If exactly four lines are active, then $R_m=0, 6$, or 8 depending on how the new coordinate functional meets the prefix-kernel point.

19.5 Machine checks

The following table counts derivative slices over all Fano planes in V_n . Only the legal values $\{0, 6, 8, 12\}$ occurred.

n	Fano planes	derivative slices	R=0	R=6	R=8	R=12
5	155	155	22	56	21	56
6	1,395	2,790	417	728	357	1,288
7	11,811	35,433	5,308	7,064	4,077	18,984

These checks used the exact Cayley-Dickson sign recursion and direct enumeration of line barcodes. They also verify the line derivative checksum and the active-line quantization rules.

19.6 Discovery-machine feature selection: value, derivative, checksum

The thermographic feature-selection idea now has three layers:

1. Value layer: $q_n(L, t)$ is a high-entropy conditional heat barcode.
2. Derivative layer: $D_m(L, t)$ is a birth-gated coordinate probe.
3. Checksum layer: line and Fano packets impose legal parity/quantization constraints.

This suggests a stronger scoring system for automated law discovery:

$$\text{Score}(L, S) = \text{TIG}(L, S) + \text{Decoder}(L, S) + \text{Checksum}(L, S) - \text{Cost}(L, S).$$

Here TIG is thermographic information gain, Decoder measures whether the fingerprint can be inverted into structural information, and Checksum measures whether the feature generates legal packet constraints for error detection.

19.7 Most useful new equation

The most useful equation from this round is the birth-gated derivative law:

$$D_m(L, t) = 1_{\{\beta_2(L) < m\}} t_m.$$

It turns the barcode from a descriptive statistic into a readable signal. The value q_n acts like an accumulated suffix integral; D_m acts like its local derivative. This is why barcode thermography is a good discovery-machine feature: it simultaneously has entropy, interpretability, and legal checksums.

19.8 Applications

Black-box coordinate tomography. If multiplication and equality are observable but coordinates are hidden, derivative slices can recover coordinate bits after line birth. Repeating across lines reconstructs partial suffix coordinate structure.

Error-correcting algebra. Legal derivative packets obey line parity and Fano quantization. Illegal values, such as a line derivative sum of one or a Fano derivative weight outside $\{0, 6, 8, 12\}$, signal measurement error, table corruption, hidden-state collapse, nonstationarity, or a different twist.

AI/tool-composition diagnostics. Fix two boundary operations and sweep the middle operation. The heat barcode shows boundary-conditioned path sensitivity; derivative slices reveal when a new feature or context bit first becomes active; packet checksums detect inconsistent or path-dependent tool behavior.

Compiler and optimizer safety. A rewrite may look globally safe but have boundary-conditioned defects. Barcode derivatives identify which feature channels create the defect and whether the defect obeys coherent rank-birth rules.

Discovery-machine law ranking. Prefer law features that have high entropy, a known inverse or derivative, and packet checksums. Such features can discover structure and validate their own measurements.

19.9 Summary

$q_n(L, t)$ = thermographic suffix integral,
 $D_m(L, t)$ = birth-gated coordinate derivative,
 $D_m(L, a) + D_m(L, b) + D_m(L, c) = 0$ on every line,
 $R_m(U)$ in $\{0, 6, 8, 12\}$ on every Fano plane.

The theory direction is now:

barcode tomography -> derivative thermography -> self-checking discovery features.

The central conceptual gain is that conditional law heat can be differentiated. Once differentiated, heat becomes a coordinate-sensitive signal with parity checks, not just a scalar temperature.

Round 20 Appendix: Barcode Distribution, Edge-Scar Measure, and Thermographic Priors

Status. Theorem-candidate layer. The exact counting formulas below are derived from the birth-truncated barcode recurrence and checked against direct enumeration for $n \leq 9$. The entropy-limit statement is computational evidence from the closed count formula, not yet a formal asymptotic proof.

20.1 Motivation: discovery features need priors, not only signatures

Rounds 15-19 showed that outer-pair associator thermography produces a high-entropy barcode $q_n(L, t)$, and that its derivative is a birth-gated coordinate probe. A discovery machine can use this as a feature, but to judge anomalies it also needs the legal distribution: which barcode values are rare, which are common, and why.

This round computes the exact global distribution of the pointed-line barcode. The result is surprising: all barcode values occur, but not uniformly. The nonuniformity is concentrated near the two edges of the barcode interval, creating a nested edge-scar measure.

20.2 Setup: pointed lines and barcode levels

For a projective line L in V_n and a pointed omitted element t in $L - \{0\}$, recall

$$H_n(L, t) = \# \{ y \text{ in } V_n : \Phi_n(x, y, z) = 1 \}, \text{ where } \{x, z, t\} = L - \{0\}, \\ q_n(L, t) = (H_n(L, t) - 4)/8.$$

The legal barcode range is

$$q_n(L, t) \text{ in } \{0, 1, \dots, 2^{n-3} - 1\}.$$

The number of pointed projective lines is

$$P_n = 3 \left[\binom{n}{2} \right]_2 = ((2^n - 1)(2^n - 2))/2.$$

20.3 Exceptional finding: exact barcode distribution

Let $M = 2^{n-3}$. For each q in $\{0, \dots, M-1\}$, define the two-sided edge bands

$$E_m = \{ q : q < 2^{n-m-1} \text{ or } q \geq M - 2^{n-m-1} \}, \quad 4 \leq m < n.$$

Then the exact number of pointed lines having barcode q is

$$C_n(q) = 3 \cdot 2^{n-1} + 9 \cdot 2^{n-3} \\ + \sum_{m=4}^{n-1} 1_{\{q \text{ in } E_m\}} \cdot 3(2^{m-1} - 1)2^{n-m} \\ + 1_{\{q=M-1\}} \cdot 3(2^{n-1} - 1).$$

This is the new barcode prior. It says that early line-birth layers spread uniformly across all barcode values, while later birth layers pile up on nested edge bands. The all-ones barcode $q=M-1$ also receives a terminal late-birth spike.

20.4 Birth-layer derivation

Let $\beta_2(L)=m$ be the first coordinate where L becomes rank-2 visible. The number of pointed lines born at m is

$$R_m(n) = 3(2^{m-1} - 1)4^{n-m}, \quad 2 \leq m \leq n.$$

For $m=2$ and $m=3$, the barcode is uniform over all M values. Their per-level contributions are

$$R_2(n)/M = 3 \cdot 2^{n-1}, \\ R_3(n)/M = 9 \cdot 2^{n-3}.$$

For $4 \leq m < n$, birth layer m only occupies the two extreme bands E_m , uniformly within that support. Its per-level contribution is

$$R_m(n)/|E_m| = 3(2^{m-1} - 1)2^{n-m}.$$

For $m=n$, the line is born so late that the barcode saturates at $q=M-1$, producing the terminal spike.

20.5 Edge-scar interpretation

The distribution is neither uniform nor arbitrary. It is an edge-shell law. Each later birth layer contributes mass to a smaller pair of edge neighborhoods. Thus q values near 0 and $M-1$ are more common than central q values, and $q=M-1$ is most common because it also receives the terminal spike.

This gives a practical anomaly baseline: rare central values are legal but low-prior; illegal values outside the arithmetic progression are impossible; edge-heavy excess beyond the formula suggests corruption, convention drift, hidden state, or nonstationarity.

20.6 Entropy law and thermographic information gain

Even though the distribution is edge-biased, the entropy remains close to maximal. Let

$$H_{\text{bar}}(n) = - \sum_q (C_n(q)/P_n) \log_2(C_n(q)/P_n).$$

The computations from the exact count formula show

$$H_{\text{bar}}(n) = (n-3) - \gamma + o(1),$$

with γ approximately 0.14712 bits.

Thus the barcode loses only about 0.147 bits compared with a perfectly uniform $n-3$ bit barcode. This explains why the feature is so useful for discovery-machine feature selection: it is nearly maximally informative, but not structureless.

n	levels M	entropy H	gap to max	min count	max count
8	32	4.869761	0.130239	672	2,499
9	64	5.862277	0.137723	1,344	5,763
10	128	6.858016	0.141984	2,688	13,059
11	256	7.855648	0.144352	5,376	29,187
12	512	8.854356	0.145644	10,752	64,515
13	1024	9.853660	0.146340	21,504	141,315
14	2048	10.853290	0.146710	43,008	307,203
15	4096	11.853094	0.146906	86,016	663,555
16	8192	12.852991	0.147009	172,032	1,425,411

The gap stabilizes quickly. By $n=16$ it is about 0.147009 bits; by $n=24$ it is about 0.147120 bits.

20.7 Direct enumeration checks

The closed distribution was checked against direct enumeration of pointed lines using the exact Cayley-Dickson sign recursion. The support is full and the formula matches the enumerated counts.

n	pointed lines	barcode levels	support full?	minimum count	maximum count
4	105	2	yes	42	63
5	465	4	yes	84	171
6	1,953	8	yes	168	435
7	8,001	16	yes	336	1,059
8	32,385	32	yes	672	2,499
9	130,305	64	yes	1,344	5,763

20.8 Discovery-machine scoring update

Round 19 proposed scoring law-features by entropy, decodability, checksum structure, and test cost. This round adds a fifth term: prior sharpness.

$$\text{Score}(L, S) = \text{TIG} + \text{Decoder} + \text{Checksum} + \text{PriorSharpness} - \text{Cost}.$$

PriorSharpness measures whether the feature has a known legal distribution. A feature with a distribution law can support statistical anomaly tests, rarity scores, active sampling, and Bayesian model comparison.

For the barcode q , the rarity score is simply

$$\text{Rare}_n(q) = -\log_2(C_n(q)/P_n).$$

This allows the discovery machine to distinguish impossible measurements, merely rare legal measurements, and common coherent measurements.

20.9 Applications

Automated discovery. Candidate laws should be ranked by conditional entropy, decodability, checksum constraints, and whether their legal distribution is known. The outer-pair associator barcode is a model feature: nearly maximal entropy, derivative decoding, packet checksums, and an exact prior.

Error-correcting algebra. The exact distribution supplies soft information. Illegal q-values are hard errors; legal but low-prior q-values are soft warnings; systematic deviations from the edge-shell prior suggest a different operation class.

Black-box coordinate tomography. Central barcode values usually correspond to early-born lines, while edge values mix in later births. Combining q, derivative slices, and the edge prior improves reconstruction of hidden coordinate order.

AI/tool-composition diagnostics. In a tool pipeline, a boundary-conditioned heat barcode can be scored against its expected prior. A stable coherent system should show lawful edge bias; drifting hidden context may produce impossible or statistically distorted barcode histograms.

Compiler and optimizer safety. Boundary-conditioned regrouping tests can be sampled adaptively. The prior tells which boundary pairs are expected to be informative and which anomalies deserve expensive verification.

20.10 Summary

$$\begin{aligned} P_n &= ((2^n-1)(2^n-2))/2, \\ R_m(n) &= 3(2^{m-1}-1)4^{n-m}, \\ C_n(q) &= \text{uniform early mass} + \text{nested edge-band birth mass} + \text{terminal all-ones spike}, \\ H_{\text{bar}}(n) &= (n-3) - 0.14712\dots + o(1). \end{aligned}$$

The new theory direction is:

barcode tomography -> derivative thermography -> thermographic priors.

The central conceptual gain is that a discovery feature should not merely be hot or high-entropy. It should also have a legal distribution. Once a thermographic feature has a prior, it becomes useful for anomaly detection, active sampling, model comparison, and self-validating discovery.

Round 21 Appendix - Apex-Incidence Thermography and Flag Peeling

Status. Theorem candidate with exact finite verification through $n=8$. This appendix continues the law-spectrum/thermographic-feature-selection program. It focuses on a feature that was previously scored as low entropy, but turns out to carry a rare landmark: the newest Cayley-Dickson generator and its projective star.

21.1 Motivation: entropy is not the only value signal

Rounds 15-20 emphasized high-entropy conditional heat features, especially the outer-pair associator barcode. This round adds a complementary principle: low-entropy thermographic features can be high-value if they identify rare anchors, coordinate cuts, or quotient maps. A discovery machine should therefore rank a law-feature by both entropy and landmark value.

The adjacent-pair associator heat looked weak in entropy tables: it has only three heat values. But those three values exactly classify whether the fixed pair is dependent, lies on a line through the newest top-bit generator, or lies on an ordinary line.

21.2 Apex-incidence law

Let $X=2^n$ and let $p=e_n$ be the newest Cayley-Dickson basis label. For fixed adjacent slots define

$$H_{\text{adj}}^{(n)}(a,b) = \#\{ z \text{ in } V_n : \Phi_n(a,b,z)=1 \}.$$

The exact law is

$$H_{\text{adj}}^{(n)}(a,b) = 0 \text{ if } \dim < 2, \quad X - 4 \text{ if } p \text{ in } , \dim=2, \quad X / 2 \text{ otherwise.}$$

The same formula holds for the other adjacent pair, fixing (y,z) and sweeping x . For $n=3$ the last two numerical values collide because $X/2=X-4=4$; for $n \geq 4$ the apex-incidence class is thermographically visible.

21.3 Counts and interpretation

The ordered pair classes have exact sizes

$$\text{dependent pairs: } 3X - 2 \quad \text{apex-line pairs: } 3X - 6 \quad \text{ordinary line pairs: } (X - 2)(X - 4).$$

These add to X^2 . Projectively, the apex-line class is the star of the newest point p : there are $(X-2)/2$ projective lines through p , and each line contributes 6 ordered independent pairs.

This gives a new interpretation of adjacent-pair heat: it is not a broad coordinate barcode, but a rare-incidence detector. It identifies which two-dimensional subspaces contain the newest generator.

n	X	dependent pairs	apex-line pairs	ordinary pairs	heat values
3	8	22	18	24	0, 4, 4
4	16	46	42	168	0, 8, 12
5	32	94	90	840	0, 16, 28
6	64	190	186	3720	0, 32, 60
7	128	382	378	15624	0, 64, 124
8	256	766	762	64008	0, 128, 252

21.4 Flag-peeling theorem candidate

Combine Round 14 one-slot thermography with the new adjacent-pair law. For $n \geq 4$, the scalar 0 and newest generator $p=e_n$ are thermographically identifiable:

$$H_{\Phi}^{(n)}(a) = 0 \text{ if } a=0, \quad X^2/2 - 8 \text{ if } a \text{ is nonzero and } a \neq p, \quad (X-2)(X-4) \text{ if } a=p.$$

Once p is known, adjacent-pair heat recovers the star of p :

$$p \text{ in } \Leftrightarrow H_{\text{adj}}^{(n)}(a,b) = X-4, \text{ for independent } a,b.$$

Therefore the black-box thermographic data reconstructs the quotient-pairing relation

$$a \sim a+p.$$

That is a one-step coordinate-flag peel. It identifies the newest projective point and partitions all other nonzero labels into pairs on lines through that point. This is exactly the first quotient map $V_n \rightarrow V_n/$ at the level of projective incidence.

21.5 Complementarity with the outer-pair barcode

The outer-pair barcode is high entropy and suffix-decodable. The adjacent-pair feature is low entropy but anchor-decodable. They serve different discovery roles:

outer-pair heat: broad barcode, high entropy, many levels, suffix tomography adjacent-pair heat: rare landmark, low entropy, apex-star and quotient pairing one-slot heat: scalar/apex calibration

This suggests a two-channel feature-selection rule: use high-entropy features to reconstruct many coordinates, and low-entropy landmark features to calibrate the coordinate flag and quotient structure.

21.6 Experimental verification

Exact enumeration using the fixed Cayley-Dickson sign recursion found zero mismatches for the adjacent-pair formula for $n=3,4,5,6,7,8$. The same verification was performed for both adjacent slots: fixed (x,y) sweeping z and fixed (y,z) sweeping x . The one-slot apex formula was also rechecked through $n=8$.

Observed class counts matched the closed formulas in every tested dimension. The $n=3$ case is degenerate only in measurement resolution: the apex-line and ordinary-line heat values both equal 4, although the formula still classifies them conceptually.

21.7 Discovery-machine scoring update

The previous thermographic information gain score emphasized entropy per cost. Round 21 adds an anchor term. A practical score should include

$$\text{Score}(L,S) = \text{entropy} + \text{decodability} + \text{checksum strength} + \text{prior sharpness} + \text{landmark value} - \text{cost}.$$

Landmark value is high when a feature isolates a rare structural object: an identity element, newest generator, center, ideal, quotient fiber, orbit representative, or symmetry-breaking scar. The adjacent-pair law is the prototype: its entropy goes to zero, but its structural value remains high because it identifies the top-bit projective star.

21.8 Applications

Black-box coordinate calibration. With no coordinate access, a tester can identify the scalar, the newest generator, and the lines through that generator using only associator failure counts.

Recursive algebra tomography. The apex-star relation gives the pairing $a \sim a+p$, suggesting a recursive quotient-peeling algorithm for reconstructing the Cayley-Dickson coordinate flag from law heat alone.

Error detection. If an independent adjacent pair has heat other than $X/2$ or $X-4$, the table is not coherent under the Cayley-Dickson rank-birth model. If the number of apex-line pairs differs from $3X-6$, the newest-generator thermographic class is corrupted or the sign convention differs.

AI/tool composition. A low-entropy test can still be invaluable if it identifies boundary operations, reset tokens, wrapper operations, or context pivots that define the quotient structure of a workflow. Discovery machines should not discard low-entropy features automatically.

21.9 Round 21 summary

The most useful new equation is

$$H_{\text{adj}}^{(n)}(a,b) = 0 \text{ if } \dim < 2; \quad X-4 \text{ if } e_n \text{ in } ; \quad X/2 \text{ otherwise.}$$

The new theory direction is

thermographic feature selection -> barcode tomography -> anchor-and-barcode tomography.

Main conceptual gain: high-entropy features decode coordinates, while low-entropy thermographic scars can identify the anchors that make those coordinates meaningful.

Round 22 Appendix: Feature-Portfolio Thermography and Discovery-Machine Law Ranking

Status: exploratory theorem candidates and exact finite checks inside the fixed Cayley-Dickson sign convention. This round continues the thermographic feature-selection program begun in Rounds 14-21.

22.1 Motivation

Rounds 15-20 showed that the outer-pair associator heat is a high-entropy barcode. Round 21 showed that adjacent-pair heat is low-entropy but identifies a rare structural anchor: the newest generator and its projective star. This round formalizes the lesson: a discovery machine should not rank candidate laws by scalar temperature or entropy alone. It should rank a portfolio of law-features by what kind of information each feature provides.

Law-feature = (law L , fixed slot set S , statistic $\Theta_{\{L,S\}}$).

A feature can be valuable because it is hot, because it is cold, because it is high entropy, because it decodes coordinates, because it has checksums, or because it identifies a rare landmark. These are different utilities.

22.2 Anchor versus barcode: a separation theorem candidate

Let $X = 2^n$ and $p = e_n$ be the newest Cayley-Dickson generator. The adjacent-pair associator thermograph is

$$H_{\text{adj}}(a,b) = \#\{ z \text{ in } V_n : \Phi_n(a,b,z)=1 \}.$$

Round 21 found the exact three-class law

$$H_{\text{adj}}(a,b) = 0 \text{ if } \dim < 2, \quad X - 4 \text{ if } p \text{ in } , \quad X/2 \text{ otherwise.}$$

Thus adjacent-pair heat is a perfect anchor detector: it tells whether an ordered pair is dependent, spans an apex line, or spans an ordinary line.

By contrast, the outer-pair barcode

$$q_n(L,t) = (\#\{ y : \Phi_n(x,y,z)=1 \} - 4) / 8, \text{ where } t=x+z,$$

has almost $n-3$ bits of entropy, but only weakly detects apex incidence. It is a coordinate-rich barcode, not a landmark detector. This produces a clean separation:

adjacent-pair heat = low entropy, high landmark value; outer-pair heat = high entropy, high coordinate value, low landmark value.

22.3 Exact class counts for the adjacent-pair anchor

Over all ordered pairs (a,b) in V_n^2 , the adjacent-pair feature has exact class counts:

$$\text{dependent pairs} = 3X - 2, \text{ apex-line pairs} = 3X - 6, \text{ ordinary line pairs} = (X - 2)(X - 4).$$

The heat value is therefore a lossless code for the three-class target C_{adj} in this model. Consequently

$$I(H_{\text{adj}} ; C_{\text{adj}}) = H(C_{\text{adj}}).$$

This mutual information can shrink with n because the apex class is rare, but the feature remains structurally priceless: it identifies the coordinate flag anchor needed before barcode decoding becomes meaningful.

22.4 Outer barcode is high entropy but weakly apex-aware

Let A be the event that a pointed projective line (L,t) lies on an apex line, i.e. $p \text{ in } L$. Since the number of projective lines through p is $X/2 - 1$,

$$\Pr(A) = 3/(X-1) \text{ over pointed projective lines.}$$

The outer barcode q_n has near-maximal entropy, but its mutual information with A is small. In particular, even the saturated barcode $q=M-1$ is not an apex certificate. The exact count of saturated pointed lines is

$$C_{\text{max}}(n) = 3(2n - 3)2^{\{n-2\}} + 3, \text{ where } M=2^{\{n-3\}}.$$

The apex contribution to this saturated class is $3(X/2 - 1)$, so

$$\Pr(A | q=M-1) = (3 \cdot 2^{\{n-1\}} - 3) / (3(2n-3)2^{\{n-2\}} + 3) \sim 1/(n-3/2).$$

Thus the saturated barcode is a useful rarity signal, but not a reliable anchor detector. The adjacent-pair thermograph should be used for anchoring; the outer barcode should be used for decoding.

22.5 Computed information table

The following exact enumerations compare the two feature types. Adjacent-pair information is target-perfect for its three-class anchor target; outer-pair information is high-entropy but weak for apex detection.

n	H(adj heat) bits	I(adj; class)	H(outer q) bits	I(q; apex) bits	Pr(apex q=max)
4	1.271600	1.271600	0.970951	0.170951	0.333333
5	0.859019	0.859019	1.933131	0.152918	0.263158
6	0.534229	0.534229	2.902899	0.109441	0.213793
7	0.317231	0.317231	3.882456	0.071751	0.178470
8	0.182987	0.182987	4.869761	0.044746	0.152461

22.6 Landmark-adjusted thermographic information gain

Define a candidate feature $F=(L,S,\Theta)$. For an unsupervised discovery machine, scalar entropy is useful but incomplete. For a known target T , mutual information is better. For structure discovery, checksum and decodability also matter. A practical score should be vector-valued:

$$V(F;T) = (\tau, H(\Theta), I(\Theta;T), \text{Decode}(\Theta), \text{Check}(\Theta), \text{Landmark}(\Theta), \text{Cost}(F)).$$

A single scalar can be formed only after choosing a mission. One possible weighted objective is

$$\text{LTIG}(F;T) = [\alpha H(\Theta) + \beta I(\Theta;T) + \gamma \text{Decode} + \delta \text{Check} + \epsilon \text{Landmark}] / \text{Cost}(F).$$

Here LTIG means Landmark-adjusted Thermographic Information Gain. The key point is that Landmark should not be reduced to entropy. A rare anchor may have low entropy but high strategic value.

22.7 Two-stage discovery-machine policy

The experiments suggest a two-stage thermographic strategy.

Stage 1: anchor calibration. Use low-entropy scar laws such as one-slot and adjacent-pair associator thermography to identify scalar elements, newest generators, quotient pairs, no-op/reset operations, and boundary-sensitive slots.

Stage 2: barcode tomography. Once anchors are fixed, use high-entropy conditional features such as outer-pair associator barcodes and derivative barcodes to recover coordinate-like information, detect legal distributions, and build packet checksums.

anchor features choose the coordinate frame; barcode features fill the coordinate frame.

This resolves the apparent conflict between Round 21 and Rounds 15-20: the best discovery system needs both low-entropy anchors and high-entropy barcodes.

22.8 Applications

Automated law discovery

Instead of ranking candidate identities by global failure temperature alone, a discovery engine should generate many law-features (law plus fixed slots), compute thermographic profiles, and score them by entropy, target information, decodability, legality constraints, landmark value, and test cost.

Black-box algebra tomography

Adjacent-pair heat can identify the newest-generator star; outer-pair heat can then decode suffix-like information. Together they form an anchor-and-barcode reconstruction workflow.

AI and tool-composition diagnostics

A reset/no-op/context-boundary tool may be rare and low-entropy but crucial. First identify anchors and boundary operations. Then run high-entropy sweep tests to detect path sensitivity, context leakage, and nonstationary composition.

Compiler and optimizer safety

Low-entropy thermographic scars can identify operands or slots that should not be moved across rewrite boundaries. High-entropy barcodes can then locate where reorder/regroup rewrites become unsafe.

22.9 Round 22 summary

The main conceptual gain is a feature-selection separation law:

High entropy is not the same as high discovery value. Low-entropy anchors and high-entropy barcodes are complementary.

The most useful formulas are:

$H_{\text{adj}}(a,b) \in \{0, X-4, X/2\}$, with $X-4$ iff p in ; $\Pr(A)=3/(X-1)$ for apex incidence on pointed lines; $\Pr(A \mid q=M-1) = (3 \cdot 2^{\{n-1\}} - 3) / (3(2n-3)2^{\{n-2\}} + 3)$.

The new theory direction is:

thermographic feature selection -> anchor-and-barcode portfolios -> mission-aware law ranking.